LA-UR-22-32122
Accepted Manuscript

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Provided by the author(s) and the Los Alamos National Laboratory (2024-06-19).

To be published in: Automatica

DOI to publisher's version: 10.1016/j.automatica.2023.111481

Permalink to record:
https://permalink.lanl.gov/object/view?what=info:lanl-repo/lareport/LA-UR-22-32122







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Automatica 161 (2024) 111481

Contents lists available at ScienceDirect

Automatica

journal homepage: www.elsevier.com/locate/automatica

Survey paper

100 years of extremum seeking: A survey ☆

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ARTICLE INFO

Article history:
Received 2 April 2021
Received in revised form 19 July 2023
Accepted 29 November 2023
Available online xxxx

Keywords:
Extremum seeking

ABSTRACT

Extremum seeking (ES) is a powerful approach to the optimization and stabilization of unknown dynamic systems and is an active field of research in control theory. This paper surveys recent ES algorithm developments. ES has evolved from its origins as a simple but brilliant engineering invention in the 1922 paper of Leblanc to the breakthrough work of Krstić and Wang in 2000 in which the first general stability analysis was given, to the most recent and numerous analytical results which prove convergence for a wide range of general dynamic systems. This survey provides a thorough overview of the dynamic systems for which ES stabilization and optimization results have been provided, the various analytical methods for proving convergence, and an overview of many real world applications for which ES has been utilized in industry and at scientific laboratories.

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1. Introduction

Many real world control problems require optimizing the performance of time-varying and uncertain dynamic systems based on measurable but analytically unknown output functions. Examples of such systems include maximal power tracking for solar panels and wind farms operating in unpredictable weather conditions, anti-lock brakes which must compensate for unpredictable and changing road surfaces, power management in large computer server farms with time-varying demand, autonomous formation control and collision avoidance in changing environments, and the control and optimization of charged particle beams in particle accelerators.

The problem of optimizing the performance of a dynamic system by tracking the extremum of its output function is known as extremum seeking (ES). ES is a general problem and many methods have been developed over the years including well known and established methods such as nonlinear programming (Teel & Popovic, 2001). Various adaptive ES methods have also been developed which utilize the knowledge of a system model with unknown parameters and optimize system performance by using state rather than objective function measurements (Adetola & Guay, 2006, 2007; DeHaan & Guay, 2005; Favache, Dochain, Perrier, & Guay, 2008; Guay, Dochain, Perrier, & Hudon, 2007; Guay & Zhang, 2003).

In this survey we focus on a perturbation driven method of extremum seeking for optimization and control. This approach is now ∼100 years old, as it was first proposed in the 1922 paper of Leblanc (1922). In his paper, Leblanc reported on a new method for maintaining maximum power transfer from a transmission line to a tram car. Leblanc's paper was a description of an ingenious engineering design, but there was no mathematical analysis of the scheme's dynamics. Eventually, Leblanc's simple method became a popular tool for maximizing or minimizing unknown output functions of dynamic systems.

Various ES approaches utilize a broad range of advanced mathematical and control theory methods including singular perturbation theory, stochastic calculus, differential geometry, nonlinear perturbation and averaging theory, weak convergence in Hilbert spaces, and partial differential equations. Before surveying the many recent cutting edge advances in ES, we start with a brief description which provides an intuition which is the foundation for all ES methods.

Consider an analytically unknown function $h(x)$ whose maximum we are interested in finding. We assume that $h(x)$ converges to its Taylor series about a point $x^\star$

$$h(x) = \sum_{n=0}^{\infty} \frac{h^{(n)}(x^\star)}{n!}(x - x^\star)^n. \tag{1}$$

Assume there is a local maximum at $x = x^\star$ such that $h'(x^\star) = 0$ and $h''(x^\star) < 0$. We consider a small neighborhood of $x^\star$ in which the Taylor series can be accurately approximated as the quadratic function

$$h(x) \approx h(x^\star) + \frac{h''(x^\star)}{2}(x - x^\star)^2. \tag{2}$$

☆ This work was supported by the Los Alamos National Laboratory Laboratory Directed Research and Development (LDRD) Program Directed Research (DR) project "Charged Particle Beam Control and Diagnostics using Adaptive Machine Learning," project number 20220074DR. The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Editor Sonia Martinez.

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https://doi.org/10.1016/j.automatica.2023.111481
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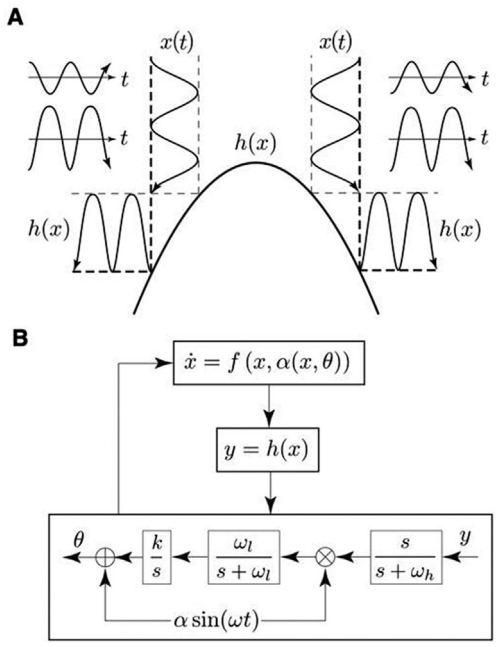


Fig. 1. An intuitive sketch of the mechanism in searching for the maximum of $h(x)$ near an extremum at which $h(x) \approx -x^2$ is also shown (A). Depending on whether $x(t) > x^\star$ or $x(t) < x^\star$ the value of the perturbed function $h(x(t))$ will be in or out of phase with the perturbing signal. The ES setup from Krstić and Wang (2000) is shown for optimization of general nonlinear dynamic systems (B).

To illustrate the ES mechanism we start at a point $x_0 = x^\star + \delta$, $\delta > 0$ and dither our signal with a small perturbation size α. Our dithered signal dynamics are

$$x(t) = x_0 + \alpha \sin(\omega t) = x^\star + \delta + \alpha \sin(\omega t), \tag{3}$$

which we plug into (2) to approximate $h(x)$ as

$$h(x^\star) + \frac{h''(x^\star)}{2}\left(\delta^2 + 2\delta\alpha \sin(\omega t) + \alpha^2 \sin^2(\omega t)\right). \tag{4}$$

If (4) is filtered to remove the non-oscillating terms (for example, this can be done by averaging and then subtracting its average value) and then multiplied by the same dithering frequency $\sin(\omega t)$ the resulting signal is (neglecting phase shifts)

$$\delta\alpha h''(x^\star)\sin^2(\omega t) + h''(x^\star)\alpha^2 \sin^3(\omega t)/2.$$

The signal is next low pass filtered, or simply integrated, so that $\sin^2(\omega t)$ averages to $1/2$ while $\sin^3(\omega t)$ averages to 0. Finally this signal is multiplied by a gain factor k which is used to control the convergence speed of the ES algorithm, resulting in

$$k\frac{\alpha\delta}{2}h''(x^\star) < 0, \tag{5}$$

which acts as a proportional feedback pushing $x(t) > x^\star$ in the negative direction back towards $x^\star$. The same argument holds for a starting point $x(t) < x^\star$. An intuitive sketch of the procedure is shown in Fig. 1A. For dynamic systems of the form

$$\dot{x} = f(x, u), \quad u = \alpha(x, \theta), \quad y = h(x), \tag{6}$$

the method can be used for finding the optimal steady state value of an unknown output function $h(x)$ by adjusting a control parameter θ which parameterizes an existing stabilizing controller, as shown in Fig. 1B. Although the intuition behind finding the optimal of an unknown function $h(x)$ as described above is clear, the analysis of this method and an analytical proof of its convergence for output functions of nonlinear systems such as (6) is a highly challenging problem which required approximately 80 years to solve.

Leblanc's simple and effective optimization scheme gained attention in Russia starting in the 1940s (Kazakevich, 1943, 1944) and in the U.S. the scheme was introduced in the 1950s (Blackman, 1962; Draper & Li, 1951; Tsien, 1954) for optimization of the performance of an internal combustion engine. In the 1950s and 1960s the algorithm was modified and its performance for specific problems was analyzed (Meerkov, 1967; Morosanov, 1957; Obabkov, 1967; Ostrovsky, 1957), but rigorous performance and design analysis had not taken place. In 1971 a Lyapunov based stability analysis of a particular system was performed (Luxat & Lees, 1971), but general stability proofs would not arrive until 30 years later.

In 2000 the first general stability analysis of the extremum seeking method was performed by Krstić and Wang (2000), by utilizing an elegant combination of averaging and singular perturbation techniques. The work of Krstić and Wang sparked renewed interest in the field. Since 2000 the extremum seeking method, as a real-time non-model-based optimization approach, has seen significant theoretical advances (Ariyur & Krstić, 2003; Choi, Krstić, Ariyur, & Lee, 2002; Rotea, 2000), with extension to semi-global convergence (Tan, Nešić, & Mareels, 2006). In 2012 a method of utilizing ES directly as a feedback controller was introduced for the stabilization of open-loop unstable time-varying dynamic systems with unknown control direction and for optimizing their performance based on analytically unknown noisy output functions (Scheinker, 2012; Scheinker & Krstić, 2012a).

Over the past 20 years a large body of work has been published on various ES methods for systems of increasing generality. Although a few reviews of ES have been written in the past (Åström & Wittenmark, 1995a, 1995b; Scheinker & Krstić, 2017; Tan, Moase, Manzie, Nešić, & Mareels, 2010; Zhang & Ordóñez, 2011), this survey attempts to cover the most recent developments in this quickly growing field including both theory and applications.

1.1. Organization

Fig. 2 shows an incomplete high level view of the ES approaches that are surveyed here and how they are related. The survey is broken down into a set of sections which emphasize areas of ES research and somewhat follow along the timeline shown in Fig. 2. We start with classical ES methods and survey various extensions and modifications in Sections 2–5. In Sections 6–11 forms of stabilizing ES and their generalizations are surveyed. Starting with Section 12 the survey jumps back to the classic form of ES, after which areas of overlap between classical and stabilizing ES are shown. The arrangement is described in more detail below.

Section 2 gives an overview of various ES developments in most of which the original form of the ES approach is only slightly modified, the standard system assumptions are made, and the classical approaches to ES proofs of stability are applied via a combination of averaging and singular perturbation theory. Also discussed are extensions, such as passing through local minima, multi unit systems, and systems distributed over networks. Section 3 focuses on ES for games including Nash equilibrium seeking in noncooperative games, cooperative games for groups of agents, multi-population games for multi-constrained optimization problems, and multi-objective ES for finding Pareto sets of optimal solutions. Section 4 is focused on a Newton-based ES, a generalization to higher order derivatives, and multivariable Newton ES, methods which add complexity to the ES scheme, but can also result in faster convergence. Section 5 is focused on a stochastic approach to ES.

In Section 6 a new approach of using high frequency ES as a stabilizing feedback for unknown open-loop unstable systems for

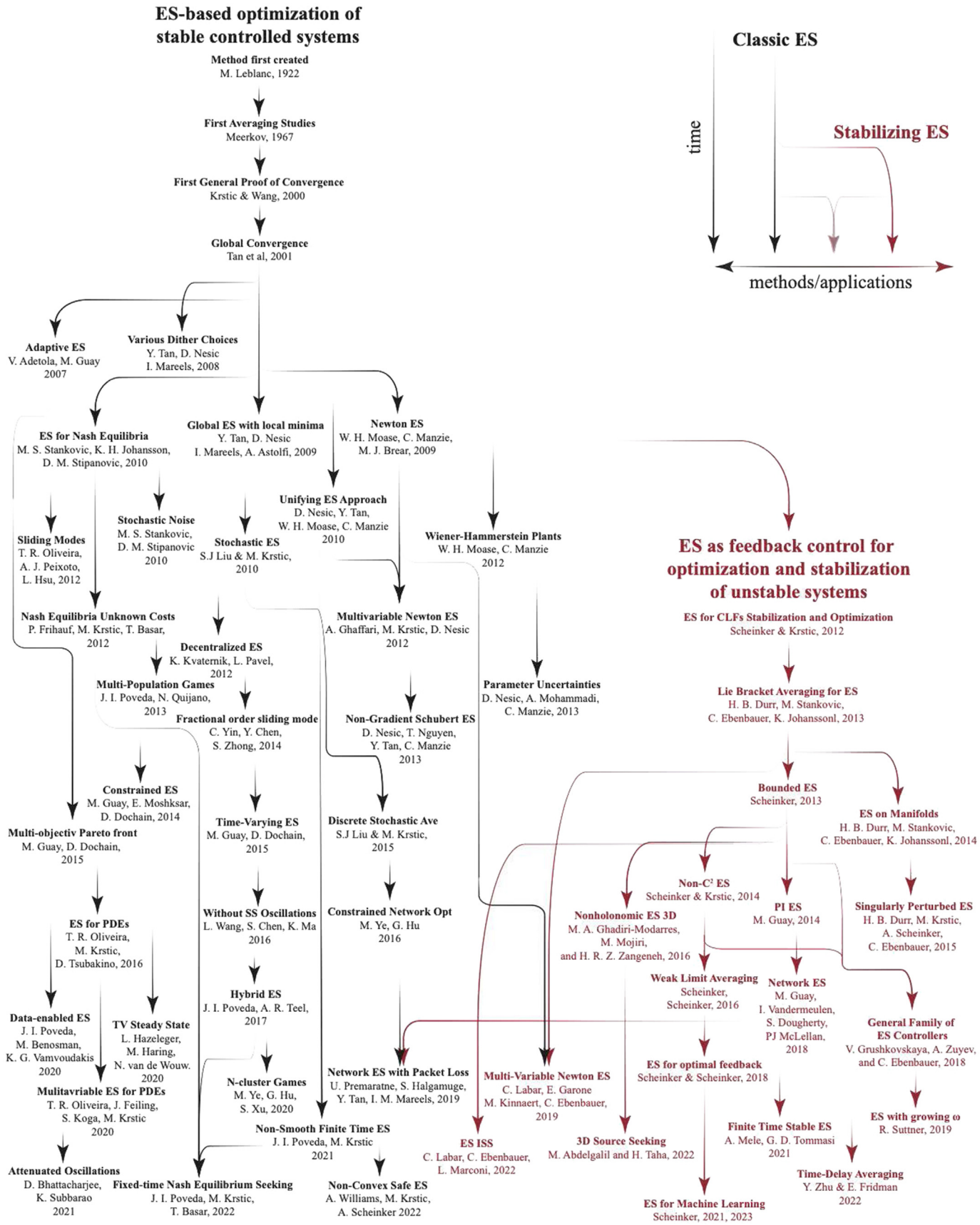


Fig. 2. The inset in the top right corner shows a very high level overview of the evolution of ES as it has split into two major categories, which in recent years have started to merge and exchange ideas and methods. The rest of the figure shows a more detailed view with the major developments and bifurcations in ES research which are discussed in more detail in this survey.

both control and optimization is introduced. Section 7 is focused on Lie bracket averaging, a powerful technique for studying the dynamics of systems in which ES is used directly as a feedback controller. In Section 8, a new bounded form of ES is introduced as well as a weak limit averaging approach which is a generalization of the Lie bracket averaging techniques to non-periodic, discontinuous, and non-differentiable dithers. In Section 9, a general family of ES controllers is introduced which includes as special cases many of the controllers from Sections 6–8. Section 10 describes a proportional integral form of feedback ES that has improved convergence properties. In Section 11, the finite-time stabilization properties of stabilizing ES are surveyed.

In Section 12, ES is extended to the study of PDEs. In Section 13 hybrid ES schemes are described and in Section 14 these techniques are extended to finite time Newton ES. In Section 15, ES with safety guarantees such as collision avoidance and safety barrier functions is surveyed, including both classical and stabilizing ES techniques. Section 16 discusses ES methods for adaptively tuning the low dimensional latent space of generative deep neural networks to enable their application for time-varying systems. Section 17 is a summary of theoretical ES techniques, Section 18 discusses patents, and Section 19 gives a broad overview of applications.

2. Classic ES developments

The original stability analysis for extremum seeking was performed in Krstić and Wang (2000) for the setup shown in Fig. 1. The goal was to maximize the stable steady state output of an analytically unknown output function $h(x)$ of the unknown dynamic system

$$\dot{x} = f(x,u), \quad y = h(x), \tag{7}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}$, $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$, $y : \mathbb{R}^n \to \mathbb{R}$, and the controller u is parameterized by a scalar θ:

$$u = \alpha(x,\theta). \tag{8}$$

The following assumptions were required.

Assumption 1. There exists a smooth function $l : \mathbb{R} \to \mathbb{R}^n$ such that

$$f(x,\alpha(x,\theta)) = 0 \quad \text{if and only if } x = l(\theta). \tag{9}$$

Assumption 2. For each $\theta \in \mathbb{R}$, the equilibrium $x = l(\theta)$ of the system (7), (8) is locally exponentially stable with decay and overshoot constants uniform in θ.

These two assumptions guarantee that the systems is closed loop stable and that given any choice of control parameter θ there is an exponentially stable equilibrium point. One more assumption is made about the existence of a maximum of the output function $h(x)$.

Assumption 3. There exists $\theta^* \in \mathbb{R}$ such that

$$(h \circ l)'(\theta^*) = 0, \quad (h \circ l)''(\theta^*) < 0. \tag{10}$$

A visual description of the consequences of Assumptions 1–3 is shown in Fig. 3 for a 2 dimensional (x_1, x_2) dynamic system which has a stable equilibrium $l(\theta)$ and an output function $y = h(x_1, x_2)$ with a maximum at θ^*.

A proof of convergence follows by considering the closed loop dynamics

$$\dot{x} = f\left(x, \alpha\left(x, \hat{\theta} + a\sin\omega t\right)\right), \tag{11}$$

$$\dot{\hat{\theta}} = k\xi, \tag{12}$$

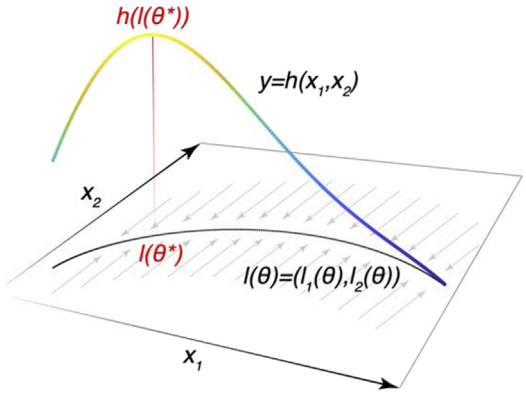


Fig. 3. The consequences of Assumptions 1–3 are described by this image for a 2 dimensional system. The light gray flow lines in the (x_1, x_2) plane represent the exponential stability of the equilibria points parameterized by $l(\theta)$, for which there is a maximum at θ^*. For any given value of θ singular perturbation theory assumes that the system quickly converges to the curve $l(\theta)$ and ES then slowly finds its way along $l(\theta)$ to the maximum.

$$\dot{\xi} = -\omega_l \xi + \omega_l (y - \eta) a \sin\omega t, \tag{13}$$

$$\dot{\eta} = -\omega_h \eta + \omega_h y, \tag{14}$$

in which the dithering frequency ω is arbitrarily small and the gain of the integrator is chosen as $k = \omega\delta K$ where δ is also small and $K > 0$. Defining coordinates

$$\tilde{\theta} = \hat{\theta} - \theta^*, \tag{15}$$

$$\tilde{\eta} = \eta - h(l(\theta^*)), \tag{16}$$

the dynamic ES system is rewritten as two parts with the evolution of one arbitrarily slow relative to the other. Separation of the system into fast and slow parts depends on the choice of $|\delta|, |\omega| \ll 1$. This can be seen by considering the dynamics with time scale $\tau = \omega t$:

$$\frac{dx}{d\tau} = \underbrace{\frac{1}{\omega} f\left(x, \alpha\left(x, \theta^* + \tilde{\theta} + a\sin\tau\right)\right)}_{\text{fast}},$$

$$\begin{bmatrix} \frac{d\tilde{\theta}}{d\tau} \\ \frac{d\xi}{d\tau} \\ \frac{d\tilde{\eta}}{d\tau} \end{bmatrix} = \underbrace{\delta \begin{bmatrix} K'\xi \\ -\omega_L'\xi + \omega_L'\left(h(x) - h(l(\theta^*)) - \tilde{\eta}\right) a\sin\tau \\ -\omega_H'\tilde{\eta} + \omega_H'\left(h(x) - h(l(\theta^*))\right) \end{bmatrix}}_{\text{slow}}.$$

The slow and fast dynamics are then analyzed using a combination of averaging and singular perturbation techniques. As $\omega \to 0$, the fast system is frozen at its equilibrium $x = l(\theta^* + \tilde{\theta} + a\sin\tau)$ which is substituted into the slow system for which averaging analysis proves the existence of an exponentially stable equilibrium (Krstić & Wang, 2000). The final result is summarized by the following theorem.

Theorem 1. *Consider the system (11)–(14) under Assumptions 1–3. There exists a ball of initial conditions around the point $(x, \hat{\theta}, \xi, \eta) = (l(\theta^*), \theta^*, 0, h \circ l(\theta^*))$ and constants $\bar{\omega}$, $\bar{\delta}$, and $\bar{a}$ such that for all $\omega \in (0, \bar{\omega})$, $\delta \in (0, \bar{\delta})$, and $a \in (0, \bar{a})$, the solution $(x(t), \hat{\theta}(t), \xi(t), \eta(t))$ exponentially converges to an $O(\omega + \delta + a)$-neighborhood of that point. Furthermore, $y(t)$ converges to an $O(\omega + \delta + a)$-neighborhood of $h \circ l(\theta^*)$.*

Theorem 1 and the associated Assumptions 1–3 have been utilized in a wide body of work with slight modifications. The approach described above has been studied for source seeking with a nonholonomic unicycle (Zhang, Arnold, Ghods, Siranosian,

& Krstić, 2007), in which the velocity v of a spinning vehicle is tuned based on the ES scheme with equations of motion

$$\dot{x} = v\cos(\omega_0 t), \quad \dot{y} = v\sin(\omega_0 t), \tag{17}$$

for maximizing quadratic functions for which the location of the maximum is unknown. These results were then studied for a force actuated point mass and it was shown that ES could find maxima and track slowly varying trajectories even for moderately unstable systems (Zhang, Siranosian, & Krstić, 2007). Various dither choices for the ES scheme have also been compared in terms of speed of convergence, domain of convergence, and accuracy for sinusoidal, square wave, and triangle wave dithers (Tan, Nešić, & Mareels, 2008). It has been shown that the method described above can handle stochastic additive measurement noise (Stanković & Stipanović, 2010) and an approach to decentralized ES for multi-agent systems has been developed which uses numerical gradient estimation methods rather than sinusoidal perturbations (Kvaternik & Pavel, 2011, 2012a, 2012b). The method has also been applied using high frequency dithers for Weiner–Hammerstein plants (Moase & Manzie, 2012) and for Hammerstein plants (Moase & Manzie, 2011), a sliding mode approach has been developed for systems in normal form (Oliveira, Peixoto, & Hsu, 2012), as well as event-triggered ES (Poveda & Teel, 2017b).

The ability of the ES scheme to pass through local minima has also been studied and it was shown that for a special class of systems if the dithering amplitude α is sufficiently large ES will pass over local minima and eventually settle at a global minimum once $\alpha(t)$ is reduced towards zero based on the dynamics update $\dot{\alpha} = -\omega\delta\epsilon g(\alpha)$, where ω, δ, ϵ, and $g(\alpha)$ must be designed properly (Tan, Nešić, Mareels, & Astolfi, 2009). As a simple example of non-local ES, consider the following dynamic system with an ES-based optimizer whose dithering amplitude $a(t)$ starts at $a(0) = 3.0$ and then exponentially decays:

$$\begin{aligned} \dot{x}_1 &= x_1 + x_2, \quad \dot{x}_2 = x_2 + u, \\ u &= -x_1 + 4x_2 + \theta + a\sin(\omega t), \\ \dot{\theta} &= \omega\delta y(x_1, x_2)\sin(\omega t), \\ \dot{a} &= -\omega\delta\epsilon a, \end{aligned} \tag{18}$$

where the output function being maximized is

$$y = -(x_1 + 3x_2)^4 + \frac{8}{15}(x_1 + 3x_2)^3 + \frac{5}{6}(x_1 + 3x_2)^2 + 10.$$

A simulation of the above system is compared to the same system with a fixed dithering signal of $a = 1$. The output function y has a local maximum at which the small fixed dither size ES gets stuck, whereas the ES with decaying dither amplitude passes through the local maximum and finds the global maximum, as shown in Fig. 4. One limitation of such an approach is for time-varying systems. Once the dither amplitude has decayed, if the optimal point changes, the ES will not respond. An alternative non-gradient approach to ES over compact sets which may contain local minima was developed for systems with known Lipschitz constants by combining ES with a local Schubert search algorithm (Nešić, Nguyen, Tan, & Manzie, 2013).

ES approaches have also been developed for multi unit systems with sampled data (Khong, Nešić, Tan, & Manzie, 2013) and time-varying ES approaches have also been studied in which using adaptive modification of the ES controller during convergence minimizes the impact of the choice of dither signal (Guay, 2014; Guay & Dochain, 2015b), with attenuated steady-state oscillations (Bhattacharjee & Subbarao, 2021). Constrained systems have been handled by adding barrier functions to the cost function (Guay, Moshksar, & Dochain, 2015). A version of ES for nonlinear systems has been developed using fractional-order sliding modes (Yin, Chen, & Zhong, 2014), and a form of ES has been developed without steady state oscillations (Wang, Chen, & Ma, 2016), and for constrained network optimization (Ye & Hu, 2016).

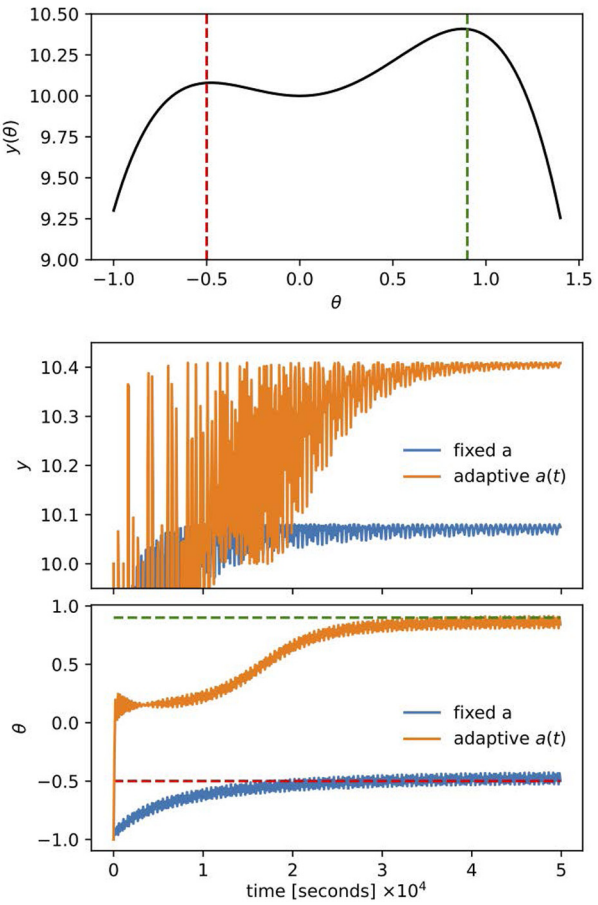


Fig. 4. The top plot shows the function y which has a local and a global maximum. The middle plot shows the evolution of the cost function with and without an adaptive dither size. The bottom plot shows convergence of the parameter θ to its local or global maximizing value.

Recently, data-enabled ES has been studied in which the incorporation of memory to store recorded data enables the use of information-rich datasets during the optimization process, which removes the need for a persistently exciting time-varying excitation dither signal needed by standard ES algorithms (Poveda, Benosman, & Vamvoudakis, 2021). Data-enabled ES is studied for systems of the form

$$\dot{x} = f(x, u), \quad y = h(x, u), \quad \phi(u) = h(l(u), u), \tag{19}$$

where $\phi(u)$ is to be minimized. The analytically unknown function $\phi(u)$ is parameterized as a linear combination of basis functions:

$$\phi(u) = \sum_{i=1}^{p} b_i(u)w_i + \epsilon(u), \tag{20}$$

where $\epsilon(u)$ is the approximation error. The approach is to fit this representation to ϕ over a compact set in order to locate the minimum.

ES for the optimization of periodic time-varying outputs was presented in Haring, van de Wouw, and Nešić (2013) for systems of the form

$$\dot{x} = f(x, \alpha(x, \theta), \theta, w(t)), \quad y = h(x, w(t)), \tag{21}$$

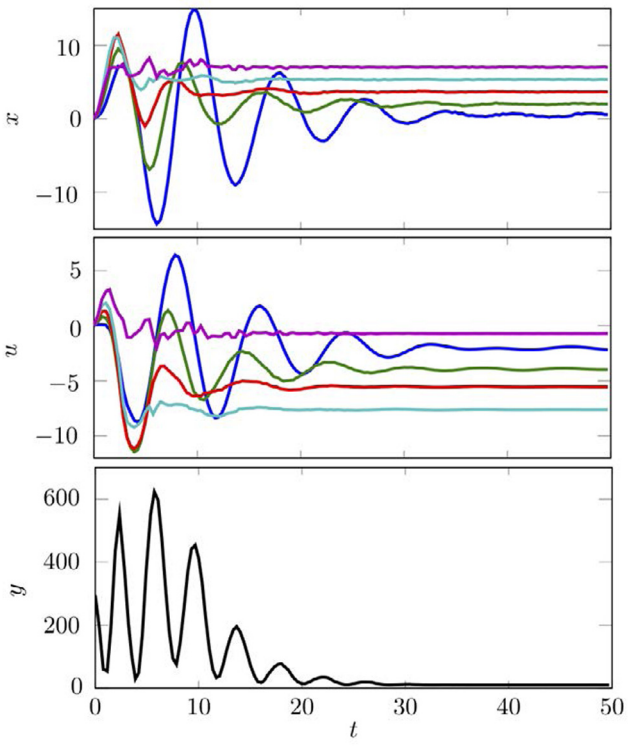


Fig. 5. Convergence of networked ES agents.
Source: Figure adapted from Guay, Vandermeulen, Dougherty, and McLellan (2018).

where $w(t)$ is periodic with known period T_w. Here the standard ES Assumption 1 above is generalized to the existence of a twice differentiable map $M(\theta, w(t))$ such that for any fixed θ there exists a trajectory $\bar{x}_{\theta,w}(t) = M(\theta, w(t))$ to which the system's trajectory naturally converges. For such systems semi-global practically asymptotic stability is proven. In this approach, a performance functional of the form

$$L_p(y_d(t)) = \left(\frac{1}{T_w}\int_{t-T_w}^{t} |y(\tau)|^p \, d\tau\right)^{\frac{1}{p}} \tag{22}$$

is used. The term (22) is multiplied by the perturbing signal $\sin(\omega[t-\phi])$ and passed through a moving average filter over one period of the ES dither:

$$e(t) = \frac{w}{a\pi}\int_{t-\frac{2\pi}{w}}^{t} L_p(y_d(\tau))\sin(\omega[\tau-\phi])d\tau. \tag{23}$$

The result is a better estimate of the unknown function's gradient and faster optimization compared to classical ES because harmonics of the dithering signal are removed. This approach is effective for systems with steady state oscillations such as simple harmonic oscillators:

$$\ddot{x} = -\omega_0^2 x - b(\theta)\dot{x} + w(t), \tag{24}$$

where the goal is to find the optimal $\theta^\star$ which minimizes the unknown damping term $b(\theta)$, $w(t)$ is a periodic external disturbance, and the optimal output is a sinusoidal signal with the maximal amplitude. The benefit of this moving-average approach compared to the standard low pass filter ES approach described above can be seen with example system (24). Addition of the high frequency terms $x_2 = \dot{x}_1 = \dot{x}$ in the cost function prevents the low pass filter version of ES from converging, but is handled by the moving average filtered approach, as shown in Fig. 6.

Another approach to the optimization of time-varying steady state responses of nonlinear systems was taken in Hazeleger, Haring, and van de Wouw (2020) for systems of the form

$$\dot{x} = f(x, u, w(t)), \quad y = g(x, u, w(t)), \tag{25}$$

where y is a measured output function and w represents a disturbance that must be bounded, but is not assumed to be periodic.

There are also approaches that incorporate system knowledge into optimization, which are useful for stable systems with existing controllers and known structure, but unknown parameters, systems of the form

$$\dot{x} = f(x, \theta, u), \quad y = Q(\theta, x), \tag{26}$$

where the function f is known, but the parameters θ and the output function y are unknown (Nesic, Mohammadi, & Manzie, 2012). A multi-parametric ES approach for online estimation of parametric uncertainties of a model has also been developed for systems of the form

$$\dot{x} = f(x) + \Delta f(x, t) + g(x)u, \quad y = h(x), \tag{27}$$

where $\Delta f(x, t)$ is an additive model uncertainty (Benosman, Farahmand, & Xia, 2019).

The classical ES algorithm has also been studied for open loop LQR control of discrete repeatable systems with fixed initial conditions of the form

$$x_{k+1} = A_k x_k + B_k u_k, \quad y_k = C_k x_k, \tag{28}$$

over a finite time horizon N with the goal of finding a sequence of open-loop control steps $\{u_k\}_{k=0}^{N-1}$ that minimize the cost function

$$J = \frac{1}{2}y_N^T Q_N y_n + \frac{1}{2}\sum_{k=0}^{N-1} y_k^T Q_k y_k + R_k u_k^2, \tag{29}$$

where A_k, B_k and the analytic form of J are unknown (Frihauf, Krstić, & Başar, 2013). In this approach the system is re-initialized repeatedly from a fixed initial condition and multi-variable ES tunes a sequence of control steps to minimize J.

An innovative ES-based method has also been developed for 3D environmental source seeking with the navigation of a non-holonomic mobile robot (Matveev, Hoy, & Savkin, 2014).

3. ES for games

For the ES methods described above, extending an approach to multi-parameter systems is easily handled by dithering each parameter with a unique frequency, resulting in a gradient descent in a high dimensional space, with minimal changes in the resulting analysis. Beyond such a high dimensional ES approach, a large body of work has focused on developing multi-parameter ES in which the relationships between various tuned parameters are studied in more complex arrangements such as cooperative games or for Nash equilibrium seeking in noncooperative games with limited game information.

The first such approach was studied in Stanković, Johansson, and Stipanović (2010) for distributed Nash equilibrium seeking in mobile sensor networks in which agents generate actions based only on measurements of noise corrupted local cost functions. In this setup the actions and structural parameters of an agent's and of other players' costs are unknown. Each player utilizes their own local ES loop with a dithering frequency ω_i that is distinct from that used by all of the other agents. This discrete ES setup is shown for a system of 2D players each with coordinates $(x_i, y_i) \in \mathbb{R}^2$ in Fig. 7, where u_{-i} denotes the actions of all other players within a neighboring set $\mathcal{N}_i$. In this setup all neighboring agents in

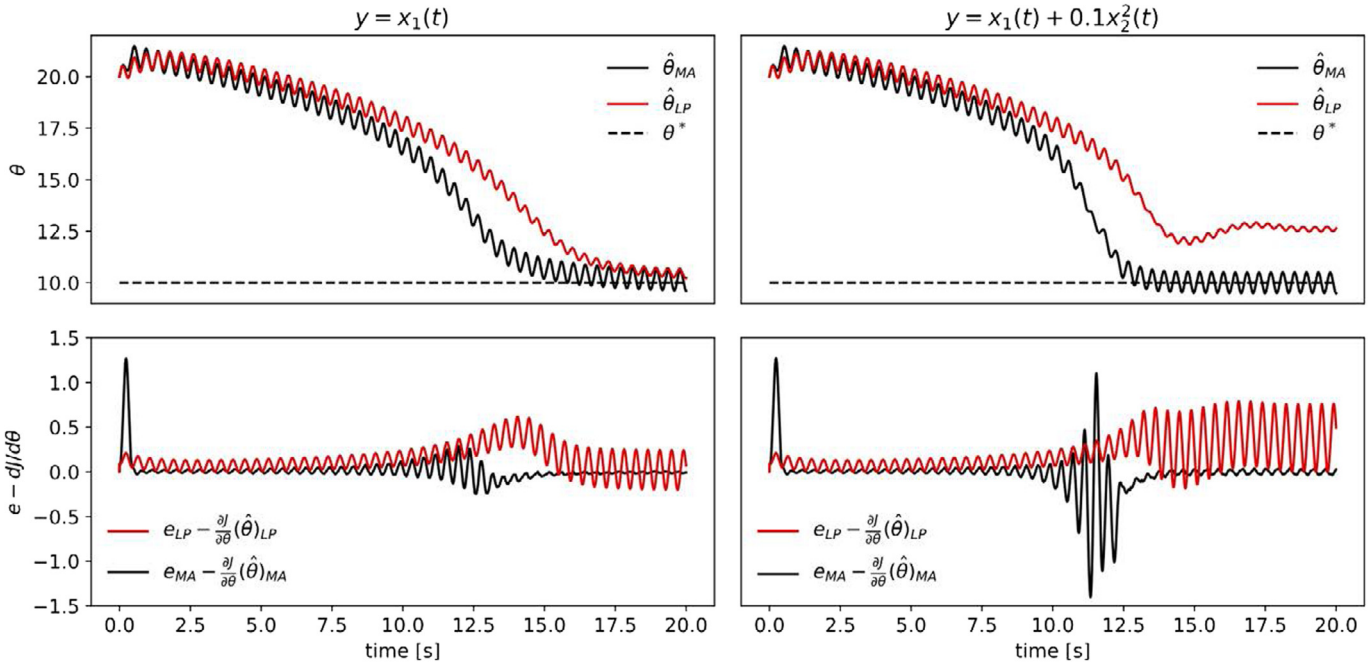


Fig. 6. Left column: Convergence to the optimal value of θ is shown for the classical low pass filtered version of ES $\hat{\theta}_{LP}$ compared to the performance of the moving average estimate $\hat{\theta}_{MA}$. Both converge to the optimal, and the slightly better performance of the moving average approach can be explained by the more accurate estimate of the derivative. Right column: Once a small high frequency term $0.1x_2^2 = 0.1\dot{x}_1$ is added to the cost function the low pass filtered ES can only get to within a neighborhood of the optimal whereas the moving average approach filters out the high frequency disturbances much more efficiently.

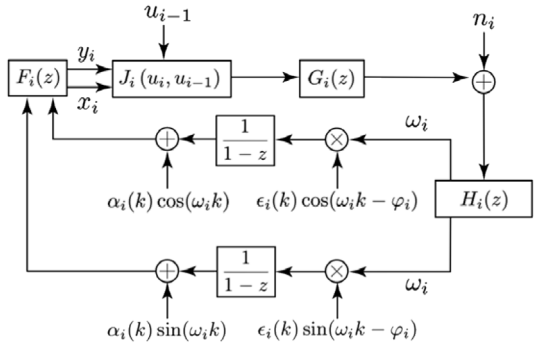


Fig. 7. Nash equilibrium seeking discrete time ES scheme.

$\mathcal{N}_i$ have different dithering frequencies $\omega_j \neq \omega_i$, the $G_i(z)$ are low-pass filters, the $H_i(z)$ are high-pass filters, and n is measurement noise. The overall setup dynamics are given by

$$w_i(k) = H_i(z)\left[G_i(z)\left[J_i(u_i(k), u_{-i}(k))\right] + n_i(k)\right], \tag{30}$$

$$u_i(k) = u_0^i(k) + F_i(z)\left[\frac{1}{1-z}[\xi_i(k)]\right], \tag{31}$$

$$\xi_i(k) = \epsilon_i(k)C_i(k)w_i(k), \tag{32}$$

and the system almost surely converges to the Nash equilibrium at u^* where

$$\nabla_1 J_1(u^*)^T, \ldots, \nabla_N J_N(u^*)^T = 0. \tag{33}$$

A similar approach was later studied for continuous systems in Frihauf, Krstić, and Basar (2011). A fixed-time Nash equilibrium seeking approach has also been studied (Poveda, Krstić, & Başar, 2022), as well as Nash equilibrium seeking in N-cluster noncooperative games (Ye, Hu, & Xu, 2020).

ES methods have also been developed for cooperative games for groups of agents, such as distributed control of wind turbines for maximizing wind farm power capture (Menon & Baras, 2014). In this approach each agent indexed by i of a group of n agents has access to its own private utility function $f_i(u)$ which can depend on the collective actions taken by all agents $u = (u_1, \ldots, u_n)$. The utility functions are assumed to be analytically unknown and each agent only has access to its own utility function. The objective of the multi-agent system is to optimize a welfare function which is the sum of all of the individual utility functions:

$$W(u) = \sum_{i=1}^{n} f_i(u). \tag{34}$$

The agents run a dynamic average consensus algorithm so that the consensus output tracks $W(u)$. Information is exchanged between agents over an undirected, connected communication graph and each agent runs part of an ES loop so that the action of agent i is on-average given by

$$\dot{u}_i \approx -\frac{\partial W(u)}{\partial u_i}, \tag{35}$$

resulting in convergence to within an arbitrarily small neighborhood of a local minimizer of $W(u)$.

A class of extremum seeking controllers has also been developed for multi-population games, for solving multi-constrained optimization problems on multi-agent dynamical systems (Poveda & Quijano, 2013). In this setup, clusters of agents interact via different cost functions which depend on the states of all other agents. The agents of any single cluster attempt to simultaneously maximize their common cost function, which is analytically unknown and only accessible via measurements. A numerical example of such a problem is presented as two connected microgrids, with each one attempting to satisfy its own power demand by utilizing local generators.

The work considers sets of populations $\mathcal{P} = \{1, \ldots, M\}$ and of social strategies $\mathcal{H} = \{1, \ldots, N\}$, where the set of strategies used by population $j \in \mathcal{P}$ is the set $\mathcal{H}_j \subset \mathcal{H}$. At any given time

t, $p_{ij}(t) \geq 0$ denotes the number of individuals playing the ith strategy in the jth population. The amount of entities associated to the jth population is then given by $\sum_{i\in\mathcal{H}_j} p_{ij}(t) = P_j(t)$ and the number of entities associated to the ith strategy is given by $\sum_{j\in\mathcal{P}_i} p_{ij}(t) = S_i(t)$. The total amount of entities in the entire society is then $\sum_{j\in\mathcal{P}}\sum_{i\in\mathcal{H}_j} p_{ij} = \mathcal{T}$. The authors consider two types of society states: (1) normalized with respect to $\mathcal{P}_j$, for every population $j \in \mathcal{P}$; and (2) normalized with respect to S_i, for each strategy $i \in \mathcal{H}$.

For the first case, the states are defined as $y_{ij}(t) = \frac{p_{ij}(t)}{\mathcal{P}_j} Y_j$, which defines the proportion of entities playing the ith strategy in the jth population, where $Y_j \in \mathbb{R}_{>0}$ and where $\mathcal{P}_j$ is assumed to be constant for every $j \in \mathcal{P}$. With this setup the society state for the jth population is defined $y_j(t)$ and the overall society state is given by $[y_1(t)^T, \ldots, y_M(t)^T]^T$. The authors consider state dynamics of the form

$$\dot{y}_{ij} = \beta_j y_{ij}\left(f_{ij}(y) - \frac{1}{Y_j}\sum_{i\in\mathcal{H}_j} y_{ij} f_{ij}(y)\right), \tag{36}$$

where $\beta_j > 0$ and the function f_{ij} represents the payoff perceived by the portion of entities of the jth population implementing the ith strategy in the society. By utilizing interesting differential geometric arguments, the authors show how the dynamics (36) can be related to a gradient system with respect to a potential function J_j for each population $j \in \mathcal{P}$, and one of the ES problems they consider is for a system of Ω agents and the goal is to maximize $J_i(x, \theta)$ where x represents agent states and each agent's internal dynamics are given by $\dot{\theta}_{ij} = g_{ij}(x, \theta)$, $\theta \in \mathbb{R}^{\Omega}$. The authors propose ES dynamics for the ith agent of the jth cluster in the $\tau = \bar{\omega}t$ time scale given by

$$\bar{\omega}\frac{\partial \theta_{ij}}{\partial \tau} = g_{ij}(\hat{x} + a\mu(\tau), \theta), \tag{37}$$

$$\frac{\partial \hat{x}_{ij}}{\partial \tau} = 2\epsilon\hat{x}_{ij}\left(\frac{J_j(x,\theta)\mu_{ij}(\tau)}{X_j}\sum_{k\in\mathcal{H}_j}\hat{x}_{kj} + C\cdots \right.$$
$$\left. \cdots - \frac{1}{X_j}\sum_{k\in\mathcal{H}_j}\hat{x}_{kj}\left[J_j(\theta)\mu_{kj}(\tau) + C\right]\right), \tag{38}$$

where $\hat{x}_{ij} = x_{ij} - a\mu_{ij}(t)$ with $\mu_{ij}(t)$ the dither signals defined as $\sin(\bar{\omega}\omega_{ij})$ with distinct ω_{ij} and $k = 2\epsilon\bar{\omega}$ is the adaptation gain for all agents. The authors prove that by appropriately tuning the parameters $\bar{\omega}$, ϵ, a, and C, the system can be made semi-globally practically asymptotically stable.

As a final example, we consider a real-time multi-objective ES controller for uncertain nonlinear systems which is guaranteed to converge to points of the Pareto set of optimal solutions (Guay & Dochain, 2015a). This form of ES is designed for engineering problems in which multiple competing objectives must be handled for various design goals. Nonlinear systems are studied of the form

$$\frac{dx}{dt} = f(x,u), \quad y = h(x), \tag{39}$$

where $x \in \mathbb{R}^n$ is a vector of state variables, the control input u takes values in a compact convex set $\mathcal{U} \subset \mathbb{R}^p$, and $y = h(x) \in \mathbb{R}^m$ is a vector valued unknown function to be minimized. The usual ES assumptions are made, such as $f(x,u)$ and $h(x)$ being smooth functions and there being an equilibrium (steady-state) map given by the n-dimensional vector $\pi(u)$ such that $f(\pi(u), u) = 0$ and the equilibrium cost function is given by $y = h(\pi(u)) = l(u)$ with the equilibrium manifold $x = \pi(u)$ being locally exponentially stable uniformly in u, $\forall u \in \mathcal{U}$.

Given a vector of weights, λ, the authors show that computing elements of the Pareto curve can be achieved by solving the minimization problem

$$\min_{\lambda} \left\|\lambda^T \nabla_u l(u)\right\|^2, \tag{40}$$

subject to the constraints

$$\lambda_i \geq 0,\ i = 1, \ldots, m, \quad \sum_{i=1}^{m}\lambda_i = 1, \tag{41}$$

and that this problem is equivalent to the unconstrained optimization problem:

$$\min_{\beta} \left\|\lambda(\beta)^T \nabla_u l(u)\right\|^2, \tag{42}$$

where $\lambda_m(\beta) = \prod_{i=1}^{m-1}\cos(\beta_i)^2$ is a parameterization of the λ_i, which can be solved by a gradient descent algorithm to yield a minimum squared norm element of $\lambda^T\nabla u$ as a function of β, based on which the Pareto optimal solution can be found following a simple gradient update of the input u. These two gradient descent dynamics are then combined as the following controller:

$$\dot{\beta} = -k\frac{\partial\lambda(\beta)^T}{\partial\beta}\nabla_u l(u)\nabla_u l(u)^T\lambda(\beta), \tag{43}$$

$$\dot{u} = -\epsilon_2 k\nabla_u l(u)^T\lambda(\beta),\ \epsilon_2 > 0. \tag{44}$$

In practice, it is often desirable to define a particular optimal solution associated with specific compromises between multiple objectives. Assuming such a solution is known and defined as y^*, the above controller is then modified as

$$\dot{\beta} = -Mk\frac{\partial\lambda(\beta)^T}{\partial\beta}\nabla_u l(u)\nabla_u l(u)^T\lambda(\beta), \tag{45}$$

$$\dot{u} = -Mk\nabla_u l(u)^T\lambda(\beta) - k\nabla_u l(u)^T\left[l(u) - y^*\right], \tag{46}$$

with $M > 0$.

The ES part is now developed for an analytically unknown function $l(u)$ for which time-varying estimates of the unknown gradients are developed. Considering the dynamics

$$\frac{dy}{dt} = \frac{\partial l}{\partial u}^T\frac{du}{dt}, \tag{47}$$

for each output function y_i, the unknown gradient is defined as $\theta_i(t) = \frac{\partial l_i}{\partial u}$ so that

$$\frac{dy}{dt} = \theta_i(t)^T\frac{du}{dt}, \tag{48}$$

and estimates $\hat{y}_i$ of y_i are then defined by the dynamics

$$\frac{d\hat{y}_i}{dt} = \dot{u}\hat{\theta}_i + Ke_i + c_i^T\dot{\hat{\theta}}_i, \tag{49}$$

where $K > 0$ is a design variable, $e_i = y_i - \hat{y}_i$ is the estimation error, and the time-varying vector $c_i(t)$ has dynamics

$$\frac{dc_i}{dt}^T = -Kc_i^T + \dot{u}^T. \tag{50}$$

A parameter estimation update approach is then developed which guarantees convergence of the parameter estimates.

For a $p \times m$ matrix of parameter estimates $\hat{\Theta} = \left[\hat{\theta}_1, \ldots, \hat{\theta}_m\right]^T$ and the vector of weightings of Pareto solutions $\lambda = [\lambda_1, \ldots, \lambda_m]^T$, the proposed ES controller is

$$\frac{du}{dt} = -\epsilon_2\left(k\sum_{i=1}^{m}\left[\lambda_i + \epsilon_1(y_i - y_i^*)\right]\hat{\theta}_i + d(t)\right)$$
$$= \epsilon_2\left(-k\hat{\Theta}\left[\lambda + \epsilon_1(y - y^*)\right] + d(t)\right), \tag{51}$$

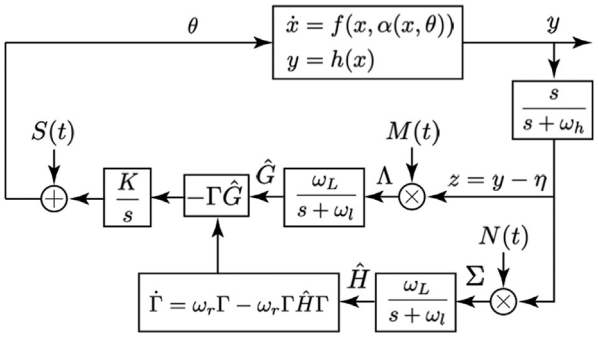


Fig. 8. Newton-based ES for multivariable systems, which requires an initial condition of $\Gamma(0) < 0$.

$$\frac{d\beta}{dt} = -k\left(\sum_{i=1}^{m}\frac{\partial\lambda_i(\beta)}{\partial\beta}\hat{\theta}_i\right)\left(\sum_{i=1}^{m}\lambda_i^T\hat{\theta}_i\right) = -k\frac{\partial\lambda^T}{\partial\beta}\hat{\Theta}^T\hat{\Theta}\lambda, \tag{52}$$

where $d(t)$ is a vector of bounded dither signals, $k > 0$ is a chosen gain, and $\epsilon_1 > 0$ is a time-scale separation parameter.

4. Newton-based ES

Another major extension of ES was the development of a Newton-based approach. Averaging analysis shows that extremum seeking methods converge by continuously following the unknown gradients of the functions being maximized or minimized. This makes the algorithm convergence speed locally dependent on a function's curvature. Convergence speed being proportional to curvature is clearly demonstrated by the following example from Moase, Manzie, and Brear (2009) in which an ES setup with a tuned parameter θ is being adjusted to minimize the output function y with averaged ES dynamics

$$\dot{\hat{\theta}} = -k\hat{y}', \tag{53}$$

where $\hat{y}'$ is an estimate of $dy/d\theta$ near a local minimum at θ^* where $y'(\theta^*) = y'_* = 0$. We can Taylor expand y' and y'' near θ^* as

$$y'(\theta - \theta^*) = (\theta - \theta^*)y''_* + \mathcal{O}(\theta^2), \tag{54}$$
$$y''(\theta - \theta^*) = y''_* + \mathcal{O}(\theta), \tag{55}$$
$$\frac{y'(\theta - \theta^*)}{y''(\theta - \theta^*)} \sim (\theta - \theta^*). \tag{56}$$

Therefore by introducing a local ES convergence proportional to a Newton rate of y'/y'' removes the curvature dependence from the convergence rate. Such an approach to Newton-like ES was developed in Moase et al. (2009) and Moase, Manzie, and Brear (2010) and generalized to higher order derivatives in Nesić, Tan, Moase, and Manzie (2010).

The price of using a Newton method to improve convergence rate is an increase in complexity. In order to apply a Newton-like ES scheme, in addition to the typically used estimates of y' there is a requirement to estimate y''. The method introduced in Moase et al. (2009, 2010) is to consider a second order Taylor expansion of the plant output y with an ES dither signal $\theta = \theta_0 + a\sin(\omega_\theta t)$

$$y = y_0 + y'_0 a\sin(\omega_\theta t) + \frac{1}{2}y''_0 a^2\sin(\omega_\theta t) + h(\theta_0, a, t), \tag{57}$$

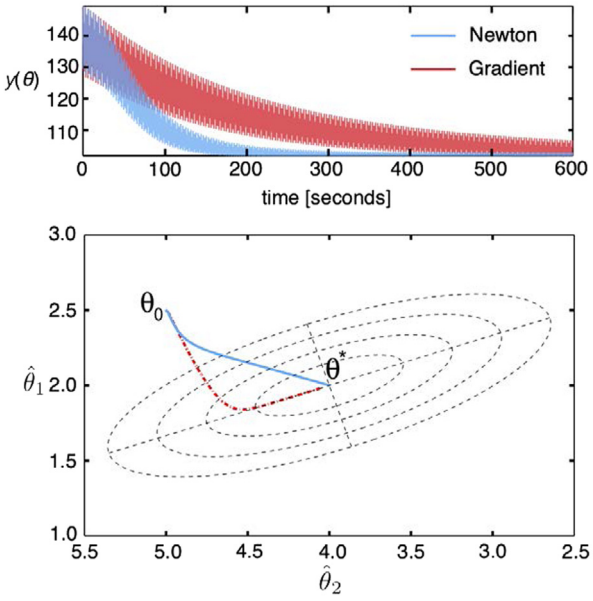


Fig. 9. Example of convergence via the Newton-based ES method for a two dimensional system compared to the standard ES approach.
Source: Figure adapted from the simulation results section of Ghaffari, Krstić, and Nešić (2012).

where $h(\theta_0, a, t)$ contains terms of third order and higher in a. This plant output is represented as a dynamic system in matrix form as

$$y = Cx + h(\theta_0, a, t), \tag{58}$$
$$\frac{dx}{dt} = \omega_\theta Ax + \frac{\partial x}{\partial\theta_0}\frac{\partial\theta_0}{\partial t} + \frac{\partial x}{\partial a}\frac{\partial a}{\partial t}, \tag{59}$$

where $C = (1\ 1\ 0\ 0\ -1/4)$,

$$x = \begin{bmatrix} y_0 + \frac{1}{4}a^2y''_0 \\ y'_0 a\mathcal{S}_1 \\ y'_0 a\mathcal{C}_1 \\ y''_0 a^2\mathcal{S}_2 \\ y''_0 a^2\mathcal{C}_2 \end{bmatrix},\quad A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}, \tag{60}$$

$\mathcal{S}_n = \sin(n\omega_\theta t)$ and $\mathcal{C}_n = \cos(n\omega_\theta t)$. System (59) is then dynamically estimated by the state-space observer

$$\hat{y} = C\hat{x}, \tag{61}$$
$$\frac{d\hat{x}}{dt} = \omega_\theta A\hat{x} + \omega_\theta L(y - \hat{y}), \tag{62}$$

with $L \in \mathbb{R}^5$ a gain vector and the estimates of y'_0 and y''_0 are given by

$$a\hat{y'_0} = C'\hat{x},\quad C' = (0\ \mathcal{S}_1\ \mathcal{C}_1\ 0\ 0), \tag{63}$$
$$a^2\hat{y''_0} = C''\hat{x},\quad C''(0\ 0\ 0\ \mathcal{S}_2\ \mathcal{C}_2). \tag{64}$$

See Fig. 9.
The Newton-based approach was later extended to multi-input-single-output (MISO) nonlinear dynamic systems of the form

$$\dot{x} = f(x, u),\quad y = h(x), \tag{65}$$

where $x \in \mathbb{R}^m$, $u \in \mathbb{R}^n$, and $y \in \mathbb{R}$ (Ghaffari et al., 2012). The multivariable ES approach requires typical ES assumptions of a known smooth control law $u = \alpha(x, \theta)$ such that the closed loop system $\dot{x} = f(x, \alpha(x, \theta))$ has a set of equilibria parameterized by a smooth function $l(\theta)$, that for each value of θ the equilibrium point $x = l(\theta)$ is locally exponentially stable uniformly in θ, and

that there exists $\theta^* \in \mathbb{R}^n$ such that $h(l(\theta^*)) = 0$ and where the second derivative Hessian matrix H is negative definite.

The multivariable Newton ES work utilizes the approach of Nesić et al. (2010) to find an estimate $\hat{H}$ of the Hessian matrix H and then use a Riccati equation to find its inverse. Considering the filter dynamics

$$\dot{\mathcal{H}} = -\omega_r \mathcal{H} + \omega_r \hat{H}, \tag{66}$$

whose state converges to $\hat{H}$, if we denote $\Gamma = \mathcal{H}^{-1}$, then the $\dot{\Gamma} = -\Gamma \dot{\mathcal{H}} \Gamma$ dynamics can be written as

$$\dot{\Gamma} = \omega_r \Gamma - \omega_r \Gamma \hat{H} \Gamma, \tag{67}$$

with equilibria $\Gamma^* = 0_{n\times n}$ and $\Gamma^* = \hat{H}^{-1}$, whose 0 equilibrium is unstable for $\omega_r > 0$. This dynamic estimate of the Hessian inverse is robust even for $\hat{H}$ which are close to singular or whose inverse is indefinite. The approach is then to estimate $\hat{H}$, and estimate its inverse Γ, which can be used to remove the influence of the Hessian from the ES dynamics whose rate of convergence can then be specified by a gain matrix K.

The overall multivariable Newton ES setup is shown in Fig. 8 with dynamics

$$\begin{aligned}
\dot{x} &= f\left(x, \alpha\left(x, \theta^* + \tilde{\theta} + S(t)\right)\right)\\
\dot{\tilde{\theta}} &= -K\left(\tilde{\Gamma} + H^{-1}\right)\hat{G}\\
\dot{\hat{G}} &= -\omega_l \hat{G} + \omega_l\left(y - h\circ l(\theta^*) - \tilde{\eta}\right) M(t)\\
\dot{\tilde{\Gamma}} &= \omega_r\left(\tilde{\Gamma} + H^{-1}\right)\left(I - \left(\tilde{H} + H\right)\left(\tilde{\Gamma} + H^{-1}\right)\right)\\
\dot{\tilde{H}} &= -\omega_l \tilde{H} - \omega_l H + \omega_l\left(y - h\circ l(\theta^*) - \tilde{\eta}\right) N(t)\\
\dot{\tilde{\eta}} &= -\omega_h \tilde{\eta} + \omega_h (y - h\circ l(\theta^*)),
\end{aligned} \tag{68}$$

where $\tilde{\theta} = \hat{\theta} - \theta^*$, $\theta = \hat{\theta} + S(t)$, $\tilde{\eta} = \eta - h\circ l(\theta^*)$, $\tilde{\Gamma} = \Gamma - H^{-1}$, $\tilde{H} = \hat{H} - H$, and

$$\dot{\eta} = -\omega_h \eta + \omega_h y. \tag{69}$$

5. Stochastic ES

ES methods have also been developed for stochastic systems by first proving general stochastic averaging results, which are then applied to an ES approach which utilizes stochastic dithers (Liu & Krstić, 2010, 2012). One of the main motivations for a stochastic ES method is that there is a chance of escaping from local minima due to the possibility of a large random jump, which may then result in a trajectory that locates the global minimum (see Fig. 10).

In Liu and Krstić (2010, 2012), the authors consider a system of the form

$$\frac{dX_t^\epsilon}{dt} = a\left(X_t^\epsilon, Y_{t/\epsilon}\right), \tag{70}$$

where X_t^ϵ is a continuous Markov process on probability space $(\Omega, \mathcal{F}, P)$, $a(x, y)$ is a continuous function of (x, y) which for any $x \in \mathbb{R}^n$ is a bounded function of y, and satisfies a Lipschitz condition in $x \in \mathbb{R}^n$ such that for any compact set $D \subset \mathbb{R}^n$ there exists a constant k_D such that for all y and any $x_1, x_2 \in D$,

$$|a(x_1, y) - a(x_2, y)| < k_D |x_1 - x_2|. \tag{71}$$

For such systems, one of the stochastic ES excitation signals proposed by the authors is the Brownian motion on the unit circle $(Y_t, t \geq 0)$

$$Y_t = e^{jW_t} = [\cos(W_t), \sin(W_t)]^T, \tag{72}$$

which is the solution to the following system of stochastic differential equations

$$dY_t = -\frac{1}{2} Y_t dt + B Y_t dW_t, \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \tag{73}$$

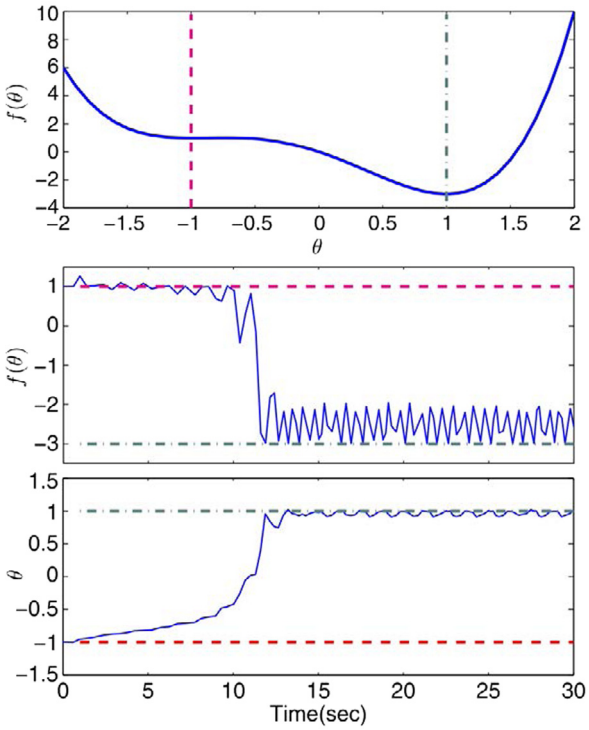


Fig. 10. Example of stochastic ES escaping from a local minimum. *Source:* Figure adapted from Liu and Krstić (2012).

where W_t is a standard 1D Brownian motion. Using the fact that Brownian motion on the unit circle is exponentially ergodic with invariant distribution μ given by the uniform measure $T = \left\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 1\right\}$, such that $\mu(S) = l(S)/2\pi$ for any set $S \subset T$ where $l(S)$ is the Lebesgue measure of S, the authors show that utilizing dither (72) results in an averaged dynamics of system (70) of the form

$$\frac{d\bar{X}_t}{dt} = \bar{a}(\bar{X}_t), \quad \bar{a}(x) = \int_{S_Y} a(x, y)\mu(dy). \tag{74}$$

In order to study stability, the authors rewrite system (70) as

$$\frac{dX_t^\epsilon}{dt} = \bar{a}(X_t^\epsilon) + R(X^{\epsilon t}, Y_{t/\epsilon}), \; X_0^\epsilon = x, \tag{75}$$

where $R(X^{\epsilon t}, Y_{t/\epsilon}) = a\left(X_t^\epsilon, Y_{t/\epsilon}\right) - \bar{a}(X_t^\epsilon)$, such that (75) can be viewed as a random perturbation of the averaged system (74), with assumption that $\bar{a}(0) = 0$, $\bar{X}_t \equiv 0$ is a stable solution of (74). The authors then develop an approximation theorem which provides a closeness of trajectories condition for their system, ensuring that the perturbed and averaged system trajectories will remain within any $\delta > 0$ for arbitrarily long time for arbitrarily small ϵ almost surely and in probability as stated in their Theorem 1:

For any $\delta > 0$

$$\liminf_{\epsilon\to 0} \left\{t \geq 0 : |X_t^\epsilon - \bar{X}_t| > \delta\right\} = +\infty, \text{ a.s} \tag{76}$$

and there exists a function $T(\epsilon) : (0, \epsilon_0) \to \mathbb{N}$ such that for any $\delta > 0$

$$\lim_{\epsilon\to 0} P\left\{\sup_{0\leq t\leq T(\epsilon)} |X_t^\epsilon - \bar{X}_t| > \delta\right\} = 0, \tag{77}$$

where

$$\lim_{\epsilon\to 0} T(\epsilon) = +\infty. \tag{78}$$

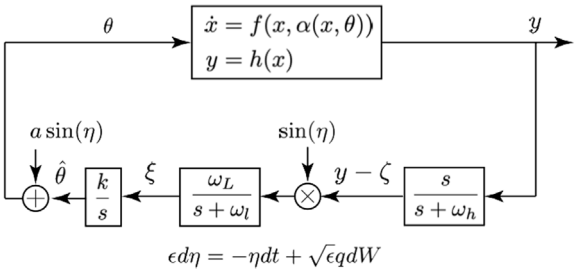


Fig. 11. Stochastic ES setup using white noise for a dynamic system.

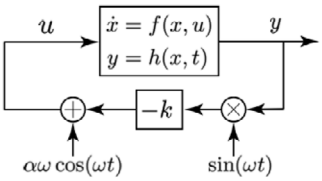


Fig. 12. ES as an optimizer or as a stabilizing feedback setup with the system dynamics being utilized as the natural integrator of the ES input as developed in Scheinker (2012).

See Fig. 11.

Utilizing this stochastic closeness of trajectories property the authors are able to relate various forms of stability of the averaged system (74) with the original system (70), the strongest of which is an exponential stability result which states that if the equilibrium $\bar{X}_t \equiv 0$ of (74) is exponentially stable, then it is weakly exponentially stable under random perturbations $R(\cdot, Y_{t/\epsilon})$, implying there exist constants $r > 0$, $c > 0$, and $\gamma > 0$ such that for any initial condition with norm bounded by r, and for any $\delta > 0$, the solution of (70) satisfies

$$\liminf_{\epsilon \to 0} \left\{ t \geq 0 : |X_t^\epsilon| > c|x|e^{-\gamma t} + \delta \right\} = +\infty, \text{ a.s.} \tag{79}$$

as well as the boundedness result

$$\lim_{\epsilon \to 0} P\left\{ |X_t^\epsilon| \leq c|x|e^{-\gamma t} + \delta, \forall t \in [0, T(\epsilon)] \right\} = 1. \tag{80}$$

With these stochastic averaging results in hand the authors next go on to develop their stochastic ES controller for general nonlinear systems of the form

$$\dot{x} = f(x, u), \quad y = h(x), \tag{81}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}$ is the input, and $y \in \mathbb{R}$ is an analytically unknown output function. A smooth control law of the form $u = \alpha(x, \theta)$ is assumed, parameterized by a scalar parameter θ. The stochastic ES update law for finding the optimal value θ^* of θ is then designed as

$$\theta = a\sin(\eta) + \hat{\theta}, \tag{82}$$

$$\frac{d\hat{\theta}}{dt} = k\xi, \tag{83}$$

$$\frac{d\xi}{dt} = -w_1\xi + w_1(y - \zeta)\sin(\eta), \tag{84}$$

$$\frac{d\zeta}{dt} = -w_2\zeta + w_2 y, \tag{85}$$

$$\epsilon d\eta = -\eta dt + \sqrt{\epsilon} q dW, \tag{86}$$

a stochastic analog of system (11)–(14), where $k > 0$, $w_1 > 0$, $w_2 > 0, \epsilon > 0$, and $q > 0$ are design parameters and $(W(t), t \geq 0)$ is a 1D Brownian motion.

Subsequently, stochastic ES methods have also been developed for discrete implementations (Liu & Krstić, 2015; Manzie & Krstić, 2009), Newton-based stochastic ES (Liu & Krstić, 2014), and stochastic ES for dynamic maps with delays (Rušiti, Evangelisti, Oliveira, Gerdts, & Krstić, 2018).

6. ES as stabilizing feedback for systems with unknown time-varying control directions

In 2012 the idea was proposed to use extremum seeking as a stabilizing feedback directly for unknown, unstable, and time-varying systems and to minimize their analytically unknown output functions (Scheinker, 2012; Scheinker & Krstić, 2012b). This approach to developing ES-based stabilizing feedbacks was married to the theories of stabilization and control Lyapunov functions (clf) (Freeman & Kokotovic, 1996; Krstić & Deng, 1998; Krstić, Kokotovic, & Kanellakopoulos, 1995; Sepulchre, Jankovic, & Kokotovic, 2012; Sontag, 1989). The marriage of clf techniques with high frequency ES studies has allowed for the stabilization of unknown systems (Scheinker, 2012; Scheinker & Krstić, 2012a, 2012b), such as systems with unknown control directions (Mudgett & Morse, 1985; Nussbaum, 1983), which unlike other approaches (Ge, Yang, & Lee, 2008), allows the unknown control gain matrix $B(t)$ to repeatedly pass through zero. The original idea of using ES as a stabilizing feedback was developed for an open-loop unstable system

$$\dot{x} = ax + bu, \quad y = h(x, t), \tag{87}$$

where $a > 0$, b, and h are unknown, as shown in Fig. 12. In the case of stabilization, the proposed feedback controller was defined as

$$u = \alpha\omega\cos(\omega t) - k\sin(\omega t)V(x), \tag{88}$$

where $y = V(x)$ is a Lyapunov function for (87). For $y = x^2$, the closed loop dynamics are

$$\dot{x} = ax + b\alpha\omega\cos(\omega t) - bk\sin(\omega t)x^2, \tag{89}$$

where α can be thought of as the persistent dithering amplitude, k the feedback gain, and ω must be sufficiently large to guarantee stability, which is a departure from the classical ES methods described above in which the ES dithering frequency was required to be small. It is clear from (89) that even if $x(t)$ is near zero a persistent dithering term of the form $b\alpha\sin(\omega t)$ will remain. To study stability we change variables to

$$y(t) = x(t) - b\alpha\sin(\omega t), \tag{90}$$

where we have subtracted off the persistent oscillating term. By considering the time scale $\tau = \omega t$, the system dynamics can be rewritten as

$$\frac{dy}{d\tau} = \frac{1}{\omega}\left(ay + ab\alpha\sin(\tau) - y^2 bk\sin(\tau) - 2b^2 k\alpha y\sin^2(\tau) - b^3 k\alpha^2\sin^3(\tau)\right), \tag{91}$$

which, for large ω, can be related to the averaged system

$$\frac{d\bar{y}}{d\tau} = \left(a - k\alpha b^2\right)\bar{y}. \tag{92}$$

The strength of this ES-based feedback approach is immediately evident from (92) as the unknown control direction b is squared in the average dynamics. Therefore stabilization is achieved independently of the sign of b simply by ensuring that $k\alpha > a/b^2$, which ensures that the origin of the average system (92) is exponentially stable no matter what the sign of b is. The trajectory $x(t)$ then exponentially converges to an arbitrarily small neighborhood of the origin proportional to α and $1/\omega$, with a solution of the form

$$x(t) \to b\alpha\sin(\omega t) + O(1/\omega). \tag{93}$$

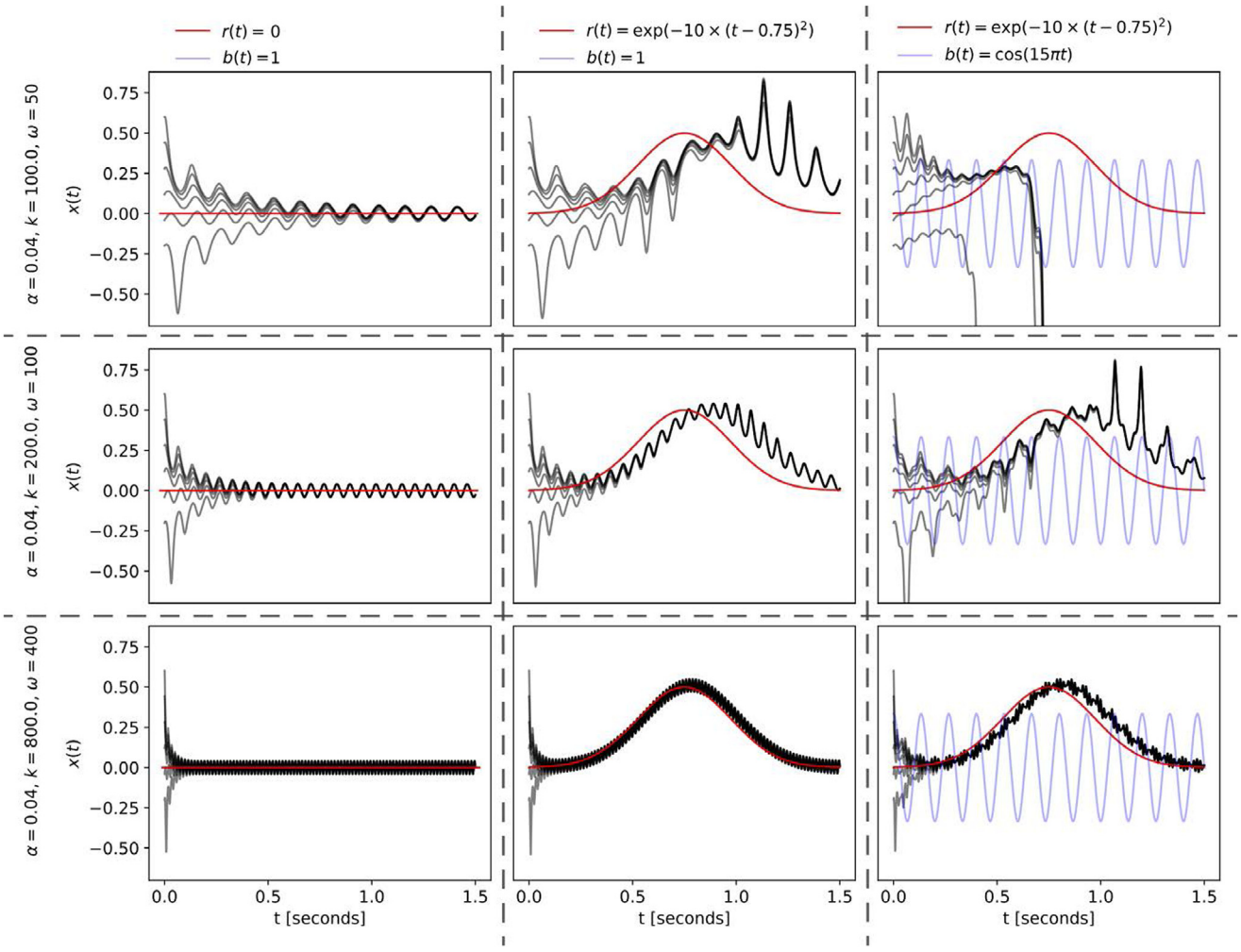


Fig. 13. Convergence of $x(t)$ to a small neighborhood of a tracked trajectory $r(t)$ is shown for various initial conditions for an unstable system (96) with $a(t) \equiv 1$. The columns show results for various forms of $r(t)$ and $b(t)$ while the rows show results for various values of k, ω. The ability to overcome the instability while tracking a time-varying trajectory in the face of a quickly time-varying control direction $b(t)$ improves as the dithering frequency ω and ES gain k are increased. The blue curve in the right column shows a scaled down version of $b(t)$ to demonstrate its rate of time variation relative to the system dynamics. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This result is summarized by the following theorem.

Theorem 2. *Consider system (87), (88) and a compact set $K \subset \mathbb{R}$, if $k\alpha > a/b^2$ then there exists $\omega^* > 0$ such that for all $\omega > \omega^*$ and for all $x(0) \in K$*

$$x(t) = e^{\left(a - k\alpha b^2\right)t} x(0) + b\alpha \sin(\omega t) + O\left(1/\omega\right) \quad \forall t. \tag{94}$$

For the more general time-varying system

$$\dot{x} = a(t)x + b(t)u, \tag{95}$$

$$u = \alpha\omega \cos(\omega t) - k \sin(\omega t) V(x, t), \tag{96}$$

a similar analysis as the one performed above results in an averaged system of the form

$$\frac{d\bar{y}}{d\tau} = a(t)\bar{y} - \frac{k\alpha}{2} b^2(t) \frac{\partial V(\bar{y}, t)}{\partial \bar{y}}, \tag{97}$$

which simultaneously minimizes $V(x, t)$ and stabilizes the location of its minimum. The function $V(x, t)$ can either be a chosen Lyapunov-type function or an analytically unknown output function to be minimized. We simulate the dynamics (95), (96) for $a(t) \equiv 1$ and several choice of $b(t) = 1$ and $b(t) = \cos(wt)$ in order to minimize $V(x, t)$ of several forms including the simple time-invariant $V = x^2$ and also the time-varying $V(x, t) = (x(t) - r(t))^2$ where $r(t) = \exp(-10(t - 0.75)^2)$. In Fig. 13 for each choice of $b(t)$ and $r(t)$ we show how increasing k and ω improves convergence.

ES-based feedback controllers have been developed for stabilizing a wide range of time-varying vector-valued linear and nonlinear systems (Scheinker & Krstić, 2012a), and their convergence proofs utilized a Lie bracket averaging approach, which is summarized in the next section. The implications of using ES as a feedback controller to minimize Lyapunov-like functions, resulting in stabilization, are summarized in Fig. 14. For example, consider the possibly open loop unstable vector valued nonlinear system

$$\dot{x} = f(x, t) + g(x, t)u, \quad y = V(x, t), \tag{98}$$

where $x \in \mathbb{R}^n$ and $V(x, t)$ is either a function chosen by the user to achieve stabilization (a Lyapunov-type function) or an analytically unknown measurable output function of the system to be minimized as in classical ES applications. In either case, applying the ES-based feedback controller

$$u = \sqrt{\omega}\left[\alpha \cos(\omega t) - k \sin(\omega t) V(x, t)\right], \tag{99}$$

results in average closed-loop dynamics of the form

$$\dot{\bar{x}} = f(\bar{x}, t) - \frac{k\alpha}{2} g(\bar{x}, t) g^T(\bar{x}, t) \nabla_{\bar{x}} V(\bar{x}, t)$$

$$= f(\bar{x},t) - \frac{k\alpha}{2} g(\bar{x},t)\left(L_g V(\bar{x},t)\right)^T, \tag{100}$$

where $L_g V = \nabla_{\bar{x}} V g$. Note that in the average dynamics the control direction matrix is now of the form $g(\bar{x},t)g^T(\bar{x},t) \geq 0$ and the system can be stabilized with unknown and time-varying control direction.

The form of Eq. (100) motivates two assumptions about a system for which an ES-based stabilizing feedback is being designed.

Assumption 4 (*Strong $L_g V$-Stabilizability*)**.** There exists a positive definite, radially unbounded, continuously differentiable function $V : \mathbb{R}^n \to \mathbb{R}^+$ and $\beta > 0$ such that

$$L_f V - \beta \left(L_g V\right)^2 < 0, \ \forall x \neq 0. \tag{101}$$

It is well known that a continuous time-invariant system of the form (98) is stabilizable by feedback continuous at the origin and smooth away from the origin if and only if there exists a control Lyapunov function (clf) with a suitable "small control property" (scp) (Sontag, 1989), defined as:

Definition 1. A positive definite radially unbounded function W has small control property (scp) relative to a system of the form (98) if

$$L_g W = 0 \implies L_f W < 0,$$
$$L_f W + L_g W \alpha_c < 0, \ \forall x \neq 0,$$

for some continuous function α_c.

In addition, if we also make the following boundedness assumption the above clf results are extended to time-varying systems of the form (98) (Albertini & Sontag, 1997).

Assumption 5. For any compact set K and $r_1, r_2 > 0$ there exists $M_{r_1,r_2} > 0$ such that

$$\sup_{\substack{|x|_K \leq r_1 \\ d(u,0)\leq r_2}} |f(x,t) + g(x,t)u| \leq M_{r_1,r_2}, \ \forall t \in \mathbb{R}, \tag{102}$$

where $|x|_K$ denotes the distance of a point from a compact set K.

While Assumption 5 is trivially satisfied by any time-invariant system with continuous dynamics, Assumption 4 is more restrictive than mere stabilizability. For example, for the system

$$\dot{x} = x^3 + x^2 u, \tag{103}$$

which is stabilizable by a simple smooth feedback $u = -2x$, no function V exists that satisfies (101) for some $\beta > 0$, and yet $W = x^2/2$ is a clf with an scp.

However, Assumption 4 is satisfied for any stabilizable system whose clf W satisfies not only the clf condition $L_g W = 0 \implies L_f W < 0$ but also a *strong small control property* (sscp) that

$$\lim_{\epsilon \to 0} \max_{\substack{L_f W(x)>0 \\ \text{and } |x|=\epsilon}} \frac{L_f W(x)}{(L_g W(x))^2} < \infty. \tag{104}$$

For a system satisfying (104) it can be shown that Assumption 4 is satisfied for any $\beta \geq 1$ by a new clf V constructed as

$$V = \int_0^W \rho(r) dr, \tag{105}$$

where

$$\rho(r) = 1 + 2 \sup_{x:V(x)\leq r} \frac{L_f V + \sqrt{\left(L_f V\right)^2 + \left(L_g V\right)^4}}{\left(L_g V\right)^2}. \tag{106}$$

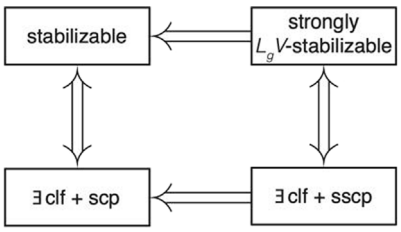


Fig. 14. The existence of a clf with sscp is equivalent to the system being strong $L_g V$-stabilizable, which implies that the system is stabilizable. Because stabilizability is equivalent to the existence of a clf with scp, as shown above the existence of a clf with sscp implies existence of a clf with scp.
Source: Figure adapted from Scheinker and Krstić (2012a).

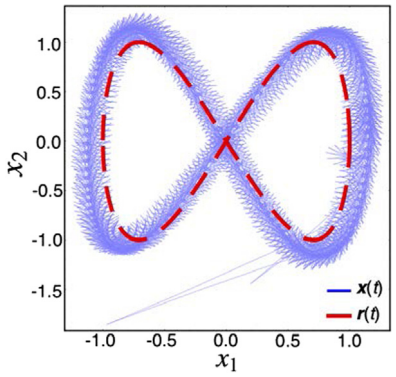


Fig. 15. After an initial transient, the simultaneous stabilization and tracking of a time-varying trajectory by utilizing ES as a stabilizing feedback controller is shown.

Therefore a system is strongly $L_g V$-stabilizable if it has a clf with sscp and the converse is also true by noting that (101) implies that $L_g V = 0$ implies that $L_f V < 0$ whenever $x \neq 0$ and therefore

$$\lim_{\epsilon \to 0} \max_{\substack{L_f V(x)>0 \\ \text{and } |x|=\epsilon}} \frac{L_f V(x)}{(L_g V(x))^2} < \beta < \infty. \tag{107}$$

The relationships between stabilizability, Assumption 4, the existence of a clf, and the scp and sscp properties are summarized in Fig. 14.

As a simple example, we consider the 2D open-loop unstable system with unknown and quickly time-varying control direction vector $b(t) = (\cos(20t), \sin(20t))$,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1.1 & 1.2 \\ -1.1 & 1\dot{x}_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \cos(20t) \\ \sin(20t) \end{bmatrix} u, \tag{108}$$

note that the system is unstable with poles $1.05 \pm 1.15i$. In order to track the trajectory

$$r(t) = [\cos(t/4), \sin(t/2)]^T, \tag{109}$$

we define the error $e(t) = x(t) - r(t)$ and utilize the stabilizing ES feedback controller

$$u = \alpha\sqrt{\omega}\cos(\omega t) - k\sqrt{\omega}\sin(\omega t)|e(t)|^2. \tag{110}$$

The trajectories of this system are shown in Fig. 15.

This stabilizing ES approach is able to simultaneously stabilize open-loop unstable systems and optimize their analytically unknown output functions (Scheinker, 2013b).

7. Lie bracket averaging for ES

For the study of high frequency feedback ES dynamics, an interesting connection between ES and Lie bracket averaging was

developed in Dürr, Stanković , Ebenbauer, and Johansson (2013) by building on averaging techniques (Gurvits, 1992; Gurvits & Li, 1992; Kurzweil & Jarník, 1988; Sussmann, 1992; Sussmann & Liu, 1991) and semiglobal practical stability theorems (Moreau & Aeyels, 2000).

In Dürr, Stanković , Ebenbauer, and Johansson (2013) the authors considered stable T-periodic systems of the form

$$\dot{x} = \sum_{i=0}^{m} b_i(x,t)\sqrt{\omega}u_i(t,\omega t),\ \ u_0 \equiv 1/\sqrt{\omega}, \tag{111}$$

and showed that the trajectory $x(t)$ of (111) can be related to the trajectory of the Lie bracket averaged system

$$\dot{z} = \sum_{i=0, j=i+1} \left[b_i(z,t), b_j(z,t)\right]\beta_{ij}, \tag{112}$$

where

$$\beta_{ij} = \frac{1}{T}\int_{\sigma}^{T} u_j(t,\sigma)\int_{0}^{\sigma} u_i(t,\tau)d\tau d\sigma, \tag{113}$$

and the terms $[b_i, b_j]$ are Lie brackets defined as

$$[b_i, b_j] = \frac{db_i}{dx}b_j - \frac{db_j}{dx}b_i. \tag{114}$$

The relationship between $z(t)$ and $x(t)$ is that they can be made arbitrarily close for arbitrarily long times by a sufficiently large value of ω.

Theorem 3. *For any bounded set K for which all trajectories of (111) have solutions bounded by some $A > 0$, for any $T_f > 0$ and any $\delta > 0$ there exists ω^* such that for all $\omega > \omega^*$ if $z(t_0) = x(t_0)$ the following holds:*

$$|x(t) - z(t)| < \delta,\ \ \forall t \in [t_0, t_0 + T_f]. \tag{115}$$

The approach in Dürr, Stanković , Ebenbauer, and Johansson (2013) was designed for a system of 2D agents for Nash equilibrium seeking with the coordinates of agent $i \in 1, \ldots, N$ given by (x_i, y_i) and dynamics

$$\begin{aligned}
\dot{x}_i &= c_i(r_i - e_i h_i)\sqrt{\omega_i}\sin(a_i\omega_i t - \varphi_i) \\
&\quad + \alpha_i\sqrt{\omega_i}\cos(a_i\omega_i t) \\
\dot{y}_i &= -c_i(r_i - e_i h_i)\sqrt{\omega_i}\cos(a_i\omega_i t - \varphi_i) \\
&\quad + \alpha_i\sqrt{\omega_i}\sin(a_i\omega_i t) \\
\dot{e}_i &= -e_i h_i + r_i,
\end{aligned} \tag{116}$$

where $c_i > 0, h_i > 0$ are design parameters including gains and filter coefficients which remove DC offsets, the r_i are utility functions, and e_i are filtered versions of the r_i after passing through a high-pass filter represented in state space as $s/(s + h_i)$, and the potential function to be maximized, $Q(x, y)$, satisfies $\nabla_{x_i,y_i}Q(x,y) = \nabla_{x_i,y_i}r_i(x,y)$.

The Lie bracket averaged system (112) of (116) is given by

$$\begin{aligned}
\dot{\bar{x}}_i &= \frac{1}{2}\left(c_i\alpha_i\nabla_{\bar{x}_i}r_i(\bar{x},\bar{y})\cos(\varphi_i) + c_i\alpha_i\nabla_{\bar{y}_i}r_i(\bar{x},\bar{y})\sin(\varphi_i)\right. \\
&\quad \left. - c_i^2\nabla_{\bar{y}_i}r_i(\bar{x},\bar{y})\cos(\varphi_i)(r_i(\bar{x},\bar{y}) - \bar{e}_i h_i)\right), \\
\dot{\bar{y}}_i &= \frac{1}{2}\left(c_i\alpha_i\nabla_{\bar{y}_i}r_i(\bar{x},\bar{y})\cos(\varphi_i) - c_i\alpha_i\nabla_{\bar{x}_i}r_i(\bar{x},\bar{y})\sin(\varphi_i)\right. \\
&\quad \left. + c_i^2\nabla_{\bar{x}_i}r_i(\bar{x},\bar{y})\cos(\varphi_i)(r_i(\bar{x},\bar{y}) - \bar{e}_i h_i)\right), \\
\dot{\bar{e}}_i &= -\bar{e}_i h_i + r_i,
\end{aligned} \tag{117}$$

which can be shown to converge to the maximum of Q by a Lyapunov analysis with $V(\bar{x},\bar{y}) = Q(x^*, y^*) - Q(\bar{x},\bar{y})$. Simulation results showing the proximity of agent dynamics to their Lie bracket averages for $\omega = 100$ are shown in Fig. 16.

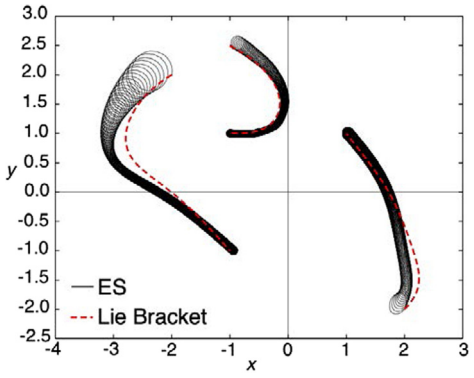


Fig. 16. Trajectories of agents relative to the Lie bracket averages of their trajectories are shown.
Source: Figure adapted from Dürr, Stanković , Ebenbauer, and Johansson (2013).

The Lie bracket averaging-based approach to studying ES dynamics is a powerful method which has been applied to a large variety of systems including ES on submanifolds (Dürr, Stanković, Johansson, & Ebenbauer, 2014), with singular perturbations (Dürr, Krstić, Scheinker, & Ebenbauer, 2015, 2017), for gain tuning (Benosman, 2016), for formation control (Suttner & Sun, 2018), obstacle avoidance (Grushkovskaya, Zuyev, & Ebenbauer, 2018b), multi-agent ES (Poveda, Benosman, & Teel, 2017), for a unicycle model (Grushkovskaya, Michalowsky, Zuyev, May, & Ebenbauer, 2018), for nonlinear systems (Grushkovskaya & Ebenbauer, 2021), for systems with a dead zone (Shao, Teel, Tan, Liu, & Wang, 2019), for iterative learning control (Khong, Nešić, & Krstić, 2016), for maximizing map sensitivity (Mills & Krstić, 2018), a Newton-based second-order Lie bracket approach has been developed (Labar, Garone, Kinnaert, & Ebenbauer, 2019b), and for non-holonomic systems (Suttner, 2020). In Labar, Ebenbauer, and Marconi (2022) input-to-state like stability properties are proven for a class of Lie bracket-averaged ES systems, in Zhu and Fridman (2022) a time-delayed approach to averaging was developed for studying stabilizing ES dynamics, and in Michalowsky, Gharesifard, and Ebenbauer (2018) Lie bracket approximations are used to design an approach to distributed optimization.

8. Bounded ES and weak limit averaging

In Scheinker (2013a) a new bounded form of ES was developed in which the feedback control output has analytically guaranteed bounds despite acting on analytically unknown, noisy, and time-varying functions. This bounded form of ES was developed for systems of the form

$$\dot{x} = f(x,t) + g(x,t)u, \quad y = h(x,t) + n(t), \tag{118}$$

where $f(x,t)$ and $g(x,t)$ are unknown, the system may be open-loop unstable, and y is a noise-corrupted measurement of an analytically unknown function $h(x,t)$. The ES feedback controller in this case is given by

$$u = \sqrt{\alpha\omega}\cos(\omega t + ky), \tag{119}$$

and the closed-loop average dynamics are

$$\dot{\bar{x}} = f(\bar{x},t) - \frac{k\alpha}{2}g(\bar{x},t)g^T(\bar{x},t)\nabla_{\bar{x}}h(\bar{x},t). \tag{120}$$

In the case of optimization, $h(x,t)$ is an analytically unknown cost function which is minimized. In the case of stabilization, where $h(x,t)$ may be chosen, by picking a Lyapunov-type function (such as $h = x^Tx$) stabilization is achieved. This simple bounded ES setup is shown in Fig. 17. Although there are no bounds assumed

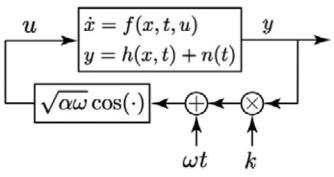


Fig. 17. Bounded ES as an optimizer or as a stabilizing feedback.

on the optimized function $h(x,t)$, the control effort of bounded ES has analytically guaranteed bounds

$$|u| = \left|\sqrt{\alpha\omega}\cos(\omega t + ky)\right| \leq \sqrt{\alpha\omega}, \tag{121}$$

which makes the algorithm especially well suited for in-hardware applications for sensitive systems in which tight control is required over parameter settings while optimizing an analytically unknown and time-varying cost function. Recently a group of researchers confirmed that this exact bonded ES mechanism is utilized by sea urchin sperm when finding the egg (Abdelgalil, Aboelkassem, & Taha, 2022).

In Scheinker and Scheinker (2016) the authors extended bounded ES to a much more general class systems for ES-based stabilization of vector-valued nonlinear time-varying systems which are not periodic and whose ES controllers are not periodic, not differentiable, and not even required to be continuous by utilizing the general weak limit averaging approach from the field of functional analysis (Conway, 1990). We start with a brief reminder and some definitions. *Riemann–Lebesgue Lemma*: If $f \in L^1(K)$, then

$$\lim_{\omega\to\infty}\int_K f(x)e^{-i\omega x}dx = 0. \tag{122}$$

A sequence of functions $\{f_k\} \subset L^2[0,1]$ is said to converge weakly to f in $L^2[0,1]$, denoted $f_k \rightharpoonup f$, if

$$\lim_{k\to\infty}\int_0^1 f_k(\tau)g(\tau)d\tau = \int_0^1 f(\tau)g(\tau)d\tau, \quad \forall g \in L^2[0,1].$$

For example, by application of the Riemann–Lebesgue Lemma,

$$\cos(\omega t)\sin(\omega t) \rightharpoonup 0, \tag{123}$$

$$\cos^2(\omega t) \rightharpoonup \frac{1}{2}, \quad \sin^2(\omega t) \rightharpoonup \frac{1}{2}, \tag{124}$$

$$\cos(\omega_1 t)\cos(\omega_2 t) \rightharpoonup 0, \quad \forall\, \omega_1 \neq \omega_2, \tag{125}$$

$$\sin(\omega_1 t)\sin(\omega_2 t) \rightharpoonup 0, \quad \forall\, \omega_1 \neq \omega_2. \tag{126}$$

In Scheinker and Scheinker (2016) the authors consider a general class of vector-valued, time-varying, nonlinear systems

$$\dot{x} = f(x,t) + g(x,t)u(y,t), \qquad y = \psi(x,t), \tag{127}$$

where $x \in \mathbb{R}^n$, and the functions $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$, $g : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^{n\times n}$, $\psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ are unknown.

It is assumed that f and g are twice continuously differentiable with respect to x and assume that the value y of $\psi(x,t)$ is available for measurement. The controller u is given by

$$u(y,t) = \sum_{i=1}^{m} k_i(y,t)h_{i,\omega}(t), \quad k_i : \mathbb{R}\times\mathbb{R} \to \mathbb{R}^n, \tag{128}$$

where the functions $k_i(y,t)$ are continuously differentiable and the functions $h_{i,\omega}(t)$ are piece-wise continuous. System (127), (128) has the following equivalent closed-loop form

$$\dot{x}(t) = f(x,t) + \sum_{i=1}^{m} b_i(x,t)h_{i,\omega}(t), \tag{129}$$

$$b_i(x,t) = g(x,t)k_i\left(\psi(x,t),t\right), \tag{130}$$

where the functions defined as

$$H_{i,\omega}(t) = \int_{t_0}^{t} h_{i,\omega}(\tau)d\tau \tag{131}$$

satisfy the uniform limits and weak limits

$$\lim_{\omega\to\infty} H_{i,\omega}(t) = 0, \qquad h_{i,\omega}(t)H_{j,\omega}(t) \rightharpoonup \lambda_{i,j}(t). \tag{132}$$

The dynamics (129) can be approximated arbitrarily accurately with weak limit average dynamics

$$\dot{\bar{x}} = f(\bar{x},t) - \sum_{i\neq j=1}^{m} \lambda_{i,j}(t)\frac{\partial b_i(\bar{x},t)}{\partial \bar{x}}b_j(\bar{x},t), \quad \bar{x}(0) = x(0). \tag{133}$$

The stability result is given by the following theorem.

Theorem 4. *For any compact set $K \subset \mathbb{R}^n$, any $t_0, T \in \mathbb{R}_{\geq 0}$, and any $\delta > 0$, there exists $\omega^\star$ such that for each $\omega > \omega^\star$, the trajectories of (129) and (133), satisfy $\|x(t) - \bar{x}(t)\| < \delta$ for all $t \in [t_0, t_0 + T]$. Therefore, by Moreau and Aeyels (2000), uniform asymptotic stability of (133) over K implies that (129) is $\frac{1}{\omega}$-SPUAS.*

The statement of the theorem utilizes the following definitions of stability.

Definition 2. Considering $\epsilon = \frac{1}{\omega}$ and denoting a trajectory of (127), (128) with $x(t_0) = x_0$ as $\phi^\epsilon(t,t_0,x_0)$, an equilibrium point of (127), (128) is said to be ϵ-SPUAS if it satisfies the following three conditions:

- *Uniform Stability:* For every $c_2 \in (0,\infty)$ there exists $c_1 \in (0,\infty)$ and $\hat{\epsilon} \in (0,\infty)$ such that for all $t_0 \in \mathbb{R}$ and for all $x_0 \in \mathbb{R}^n$ with $\|x_0\| < c_1$ and for all $\epsilon \in (0,\hat{\epsilon})$,

$$\|\phi^\epsilon(t,t_0,x_0)\| < c_2 \quad \forall t \in [t_0,\infty). \tag{134}$$

- *Uniform Boundedness:* For every $c_1 \in (0,\infty)$ there exists $c_2 \in (0,\infty)$ and $\hat{\epsilon} \in (0,\infty)$ such that for all $t_0 \in \mathbb{R}$ and for all $x_0 \in \mathbb{R}^n$ with $\|x_0\| < c_1$ and for all $\epsilon \in (0,\hat{\epsilon})$,

$$\|\phi^\epsilon(t,t_0,x_0)\| < c_2 \quad \forall t \in [t_0,\infty). \tag{135}$$

- *Global Uniform Attractivity:* For all $c_1, c_2 \in (0,\infty)$ there exists $\bar{T} \in (0,\infty)$ and $\hat{\epsilon} \in (0,\infty)$ such that for all $t_0 \in \mathbb{R}$ and for all $x_0 \in \mathbb{R}^n$ with $\|x_0\| < c_1$ and for all $\epsilon \in (0,\hat{\epsilon})$,

$$\|\phi^\epsilon(t,t_0,x_0)\| < c_2 \quad \forall t \in [t_0+\bar{T},\infty). \tag{136}$$

The weak limit-based ES approach has been studied for discontinuous dithers (Scheinker & Scheinker, 2016), bounded ES (Scheinker & Krstić, 2014a), inverted pendulum stabilization (Michalowsky & Ebenbauer, 2013), for systems not affine in control (Scheinker & Scheinker, 2018), and for generating optimal feedback controllers for unknown systems (Scheinker & Scheinker, 2021).

As a simple example of the generality of the weak limit averaging approach, we consider a 2D setting in which we are seeking the minimum of an analytically unknown output function $y = h(x,t)$ for the open-loop unstable system

$$\dot{x}_1 = x_1 + 0.75x_2 + u_1 \tag{137}$$

$$\dot{x}_2 = 0.5x_1 + 2x_2 + u_2, \tag{138}$$

whose open-loop eigenvalues are $\lambda = \{2.29, 0.71\} > 0$. Instead of utilizing two distinct frequencies, we can utilize sin() and cos() functions with a single frequency, or phase-shifted square waves, or triangle waves, or square waves with dead time, as these all satisfy the above requirements of being orthogonal in $L^2[0,T]$. If the dithering amplitudes are chosen so that all of the signals

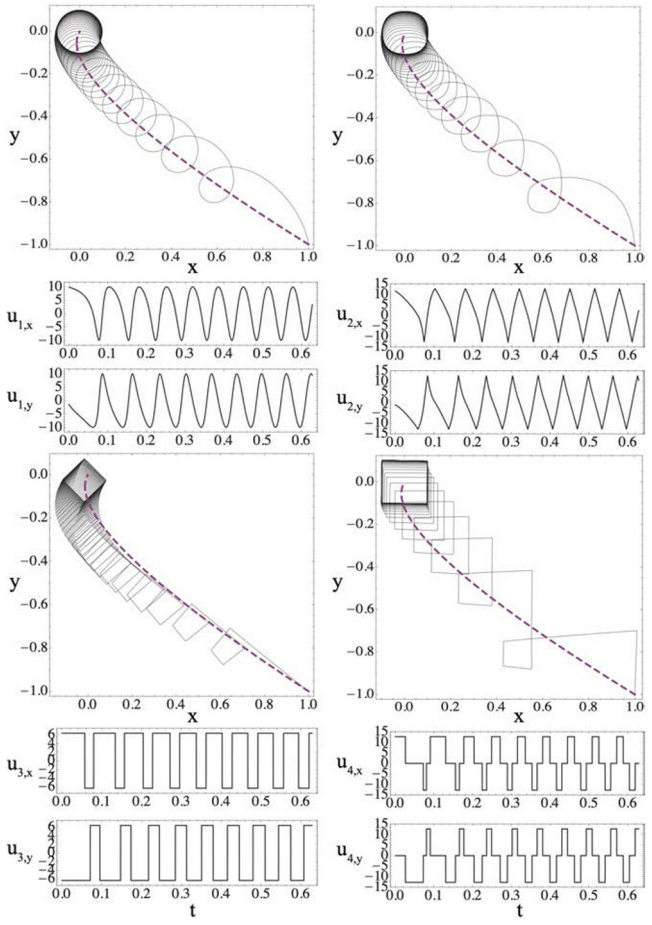


Fig. 18. A family of various dithering signals with identical weak limit average dynamics is shown.
Source: Figure adapted from Scheinker and Scheinker (2016).

have equal average power, then the average dynamics for all of the dithering types are identical, as shown in Fig. 18. This is a useful result because some analog systems can more easily implement continuous signals such as trigonometric functions or triangle waves, while digital systems are usually switching in a discontinuous way with dead time in between.

To get an intuitive understanding for the simple control law (119), we consider a system without outside influence, where we only worry about control according to $\dot{x}_1 = u_1$, $\dot{x}_2 = u_2$, as is typically considered for ES applied for optimization of some unknown function $y = J(x,t)$. In this case if we apply sin() and cos() controllers, the system dynamics are

$$\dot{x}_1 = \sqrt{\alpha\omega}\cos(\omega t + kJ) = v\cos(\theta), \tag{139}$$
$$\dot{x}_2 = \sqrt{\alpha\omega}\sin(\omega t + kJ) = v\sin(\theta), \tag{140}$$

which is simply a unicycle in a 2D environment traveling at constant velocity $v = \sqrt{\alpha\omega}$ with ES being used to control the heading angle $\theta(t) = \omega t + kJ$, as shown in Fig. 19.

An intuition behind the convergence comes from considering the rate of change of this heading angle $\theta(t)$ which is given by

$$\begin{aligned}\frac{d\theta}{dt} &= \omega + k\frac{dJ}{dt} = \omega + k\frac{\partial J}{\partial x_1}\dot{x}_1 + k\frac{\partial J}{\partial x_2}\dot{x}_2\\ &= \omega + k\sqrt{\alpha\omega}\left[\cos(\theta)\frac{\partial J}{\partial x_1} + \sin(\theta)\frac{\partial J}{\partial x_2}\right]\\ &= \omega + \underbrace{kv\,[\cos(\theta),\sin(\theta)]\cdot\nabla_x J}_{\text{angular velocity modulation}},\end{aligned} \tag{141}$$

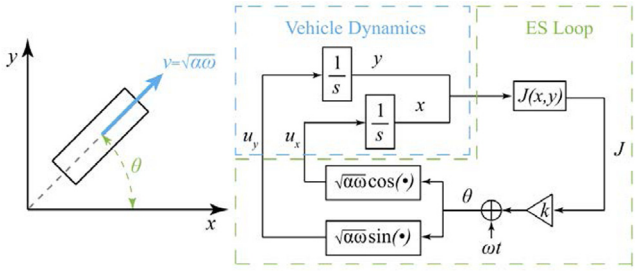


Fig. 19. Bounded ES for 2D unicycle control setup.

which is the dot product of the vehicle's velocity with the gradient of the unknown function J. Therefore, when the vehicle is heading towards the minimum, along a negative gradient of J the rotation slows down and the vehicle spends more time approaching the minimum. If instead the vehicle is moving away from the minimum, the gradient is positive, and the rotation speeds up, and the vehicle spends less time moving away. This sort of behavior is clearly seen in Fig. 18. In Ghadiri-Modarres, Mojiri, and Zangeneh (2016a), this scheme was modified in order to adjust the forward velocity, while rotating the vehicle at a constant angular velocity. In Grushkovskaya and Ebenbauer (2016), this bounded ES controller was applied for multi-agent coordination with Lagrangian measurements.

This simple 2D vehicle control ES setup has since been extended for 3D vehicle control in several papers. In Ghadiri-Modarres, Mojiri, and Zangeneh (2016b) the authors considered a 3D unicycle and proposed the following generalization of the 2D controller described above

$$\dot{x}_1 = V_0\cos(\phi)\cos(\theta), \tag{142}$$
$$\dot{x}_2 = V_0\cos(\phi)\sin(\theta), \tag{143}$$
$$\dot{x}_3 = V_0\sin(\phi), \tag{144}$$
$$\dot{\phi} = \omega_\phi - kh\eta, \tag{145}$$
$$\dot{\theta} = \omega_\theta - kh\eta, \tag{146}$$
$$\eta = \left[\frac{s}{s+h}\right]J(x_1,x_2,x_3,t), \tag{147}$$

where ϕ and θ are the pitch and yaw angles of the vehicle, respectively. The setup and a simulation result are shown in Fig. 20.

In Abdelgalil and Taha (2022) the authors considered a 3D unicycle and proposed another generalization of the 2D controller described above in which they utilized differential geometry techniques and recursive second order averaging for an ES setup with two time scales for a general class of systems of the form

$$\dot{x} = \sqrt{\omega}f_1(x,t,\sqrt{\omega}t,\omega t) + f_2(x,t,\sqrt{\omega}t,\omega t), \tag{148}$$

of which a 3D unicycle is a special case with dynamics

$$\dot{x}_1 = (R\sqrt{2\omega}e_1)_1, \tag{149}$$
$$\dot{x}_2 = (R\sqrt{2\omega}e_1)_2, \tag{150}$$
$$\dot{x}_3 = (R\sqrt{2\omega}e_1)_3, \tag{151}$$
$$\dot{R} = R\Omega, \quad \Omega = \varphi e_1 + \theta e_3, \tag{152}$$
$$\mu\dot{z} = J(x) - z, \tag{153}$$
$$\theta = \omega - \dot{z}, \tag{154}$$
$$\varphi = 2\alpha\sqrt{2\omega}\sin(\omega t - z + \pi/4), \tag{155}$$

where e_i are the standard unit basis vectors, φ and θ are the roll and yaw angles of the rigid body, $R \in SO(3)$ is a rotation matrix,

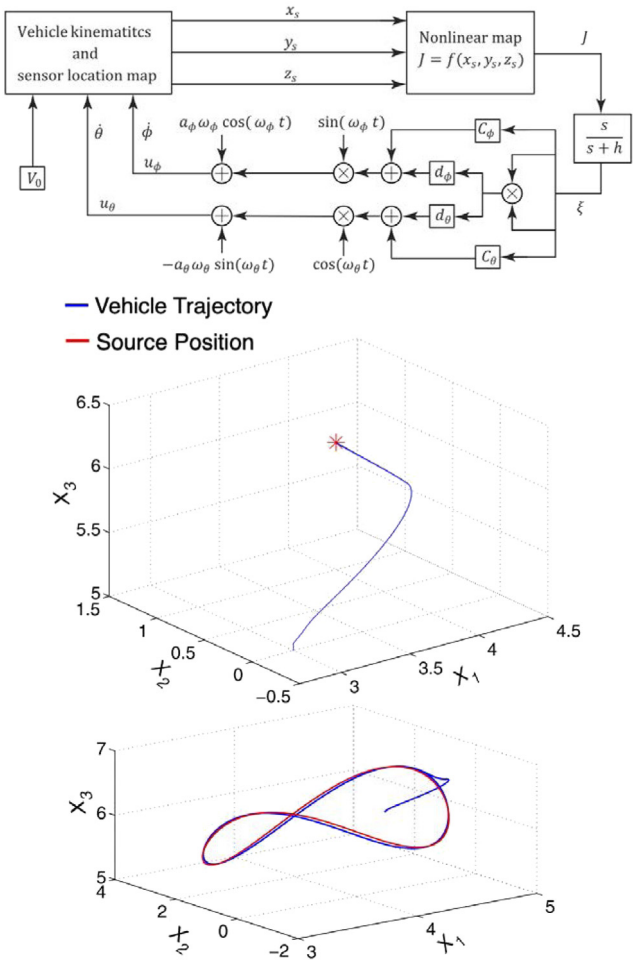


Fig. 20. Bounded ES for 3D unicycle control setup and simulation example. *Source:* Figures adapted from Ghadiri-Modarres et al. (2016a).

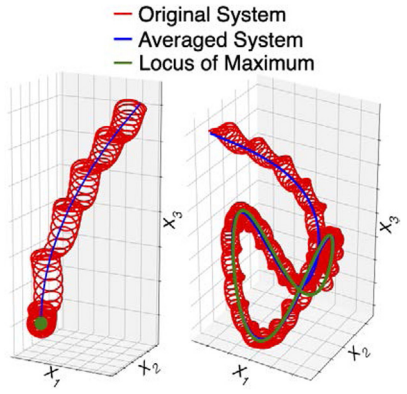


Fig. 21. Bounded ES for 3D unicycle control simulation example. *Source:* Figure adapted from Abdelgalil and Taha (2022).

and $(R\sqrt{2\omega}e_1)_i$ denotes the i component of the vector $R\sqrt{2\omega}e_1$. A simulation result for static source localization as well as trajectory tracking is shown in Fig. 21.

9. A general family of ES controllers

In Scheinker and Krstić (2014b) Lie bracket averaging theory was extended to non-differentiable systems and applied to a general class of ES controllers of the form

$$u_r = \alpha\sqrt{\omega}\cos(\omega t)|x|^r - \frac{k}{1-r}\sqrt{\omega}\sin(\omega t)|x|^{2-r}, \tag{156}$$

for $x \in \mathbb{R}^n$. The analysis of these non-differentiable controllers required the extension of Lie bracket averaging and practical stability results to functions not differentiable over entire sets.

The main advantage of these non-C^2 controllers is that for $r \neq 0, 2$ the entire high frequency oscillating control feedback disappears as equilibrium is reached. The average dynamics of a system of the form

$$\dot{x} = f(x,t) + g(x,t)u, \tag{157}$$

with u as in (156) are given by

$$\dot{\bar{x}} = f(\bar{x},t) - k\alpha g(\bar{x},t)g(\bar{x},t)^T\bar{x}, \tag{158}$$

which, remarkably, are independent of the choice of r.

The studies in Scheinker (2013a) and Scheinker and Krstić (2014a, 2014b) led to a search for a larger class of generating vector fields for stabilizing ES controllers. Another form of dithering functions was introduced in Suttner and Dashkovskiy (2017) with guaranteed exponential stability for systems whose output functions have known minimum values. In Grushkovskaya, Zuyev, and Ebenbauer (2018a) for systems of the form

$$\dot{x} = F_1(J(x))\sqrt{\omega}u_1(\omega t) + F_2(J(x))\sqrt{\omega}u_2(\omega t), \tag{159}$$

it was shown that the results of Suttner and Dashkovskiy (2017) and Scheinker and Krstić (2014b) could be described by a general family of ES controllers defined as

$$F_2(x) = -F_1(x)\int \frac{F_0(x)}{F_1^2(x)}dx, \tag{160}$$

for which the Lie bracket average dynamics of (159) are given by

$$\dot{\bar{x}} = -\beta_{2,1}\nabla J(\bar{x})F_0(J(\bar{x})), \tag{161}$$

where the constant $\beta_{2,1}$ is defined according to the T-periodic dither signals u_1, u_2 as

$$\beta_{2,1} = \int_0^T\int_0^\sigma u_1(\sigma)u_2(\theta)d\theta d\sigma. \tag{162}$$

This general family of ES controllers can be used to generate feedback with nice properties such as minimal overshoot and with vanishing control efforts that allow the system to settle at equilibrium, as shown in Fig. 22.

While the standard ES methods rely on the existence of some small enough ω^* such that the averaging and singular perturbation results hold for all $\omega \in (0, \omega^*)$, the Lie bracket averaging approach relies on the existence of a large enough ω^* such that for all $\omega > \omega^*$ the averaging results hold, with the actual required value of ω^* depending on the specific system being considered. In Suttner (2019) a form of ES was introduced in which the dither signal's frequency continuously increases thereby guaranteeing a large enough ω relative to any system dynamics. The authors considered systems of the form

$$\dot{x} = g_0(t,x) + \sum_{k=1}^m u_k g_k(t,x), \quad y = \psi(x), \tag{163}$$

and utilize controllers

$$u_k = \frac{\sqrt{\omega_k}}{(z-y)^2}\sin\left(\omega_k\Omega + \frac{1}{z-y}\right), \tag{164}$$

$$\dot{z} = -(z-y), \quad \dot{\Omega} = \frac{1}{(z-y)^5}, \tag{165}$$

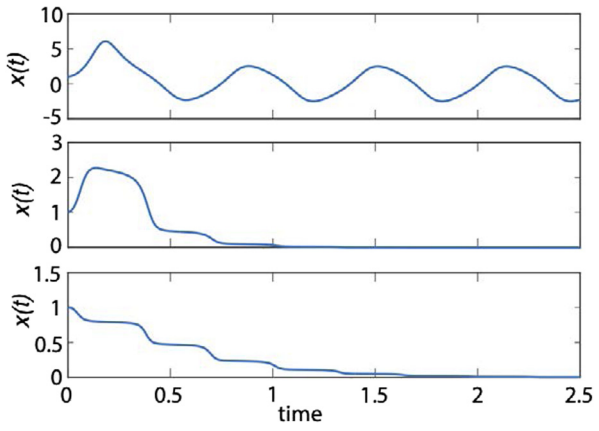


Fig. 22. Based on the choice of controller parameters, a controller can be designed whose effort vanishes allowing the trajectory to settle at the origin.
Source: Figure adapted from Grushkovskaya et al. (2018a).

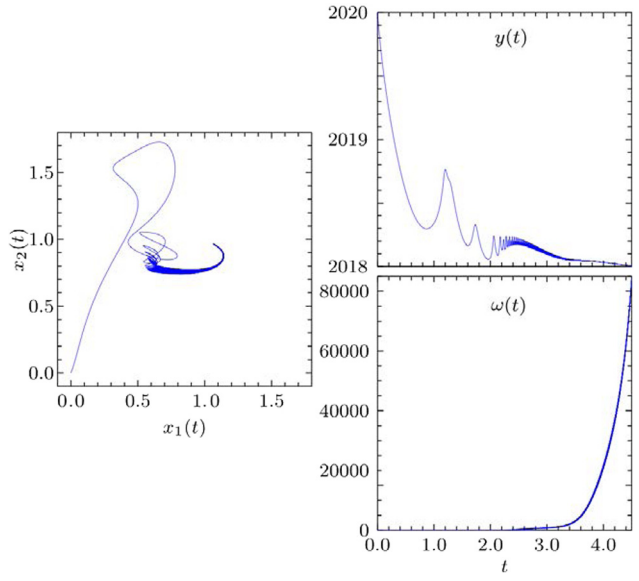


Fig. 23. Simulation example of ES asymptotically converging as the dithering frequency grows.
Source: Figure adapted from Suttner (2019).

which, under certain assumptions, guaranteed asymptotic convergence to the optimal value $y(t) \to y^*$ as $t \to \infty$.

The method is demonstrated by simulating system (163) with

$$g_0(t) = (\sin(2t), \cos(t)) \tag{166}$$

$$g_1(x_1, x_2) = (\cos(x_1 + x_2), \sin(x_1 + x_2)), \tag{167}$$

$$g_2(x_1, x_2) = (-\sin(x_1 + x_2), \cos(x_1 + x_2)), \tag{168}$$

$$\psi(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 1)^2 + 2018, \tag{169}$$

and the simulation results are shown in Fig. 23.

10. Proportional integral ES

Two forms of proportional integral approaches to stabilizing ES have also been developed (Guay, 2016; Guay & Dochain, 2017). In these approaches, the standard ES controller which achieves optimization is treated as an integral term in a PI-like control, with the addition of a proportional terms whose purpose is faster convergence and stabilization. Overall the two time-scale approach can result in improved convergence properties.

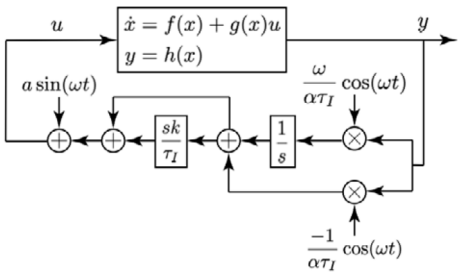


Fig. 24. PI ES as an optimizer and stabilizing feedback.

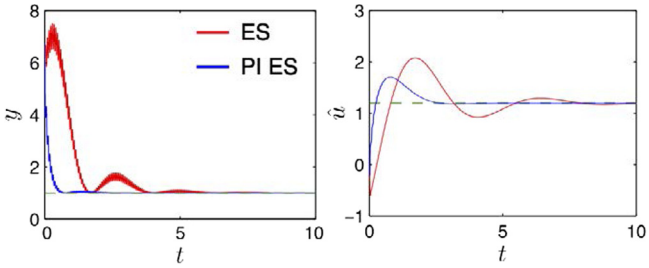


Fig. 25. PI-ES convergence relative to ES.
Source: Figure adapted from Guay (2016).

In Guay (2016), the derivative $\dot{y}$ with respect to time of the unknown cost function being minimized is assumed to be available for measurement and the system dynamics and controller are given by

$$\dot{x} = f(x) + g(x)u, \quad y = h(x) \tag{170}$$

$$\dot{\hat{u}} = -\frac{1}{\alpha\tau_l}\dot{y}\sin(\omega t) \tag{171}$$

$$\begin{aligned} u &= \hat{u} - \frac{k}{\alpha}\dot{y}\sin(\omega t) + a\sin(\omega t) \\ &= \hat{u} + \frac{k}{\tau_l}\dot{\hat{u}} + a\sin(\omega t). \end{aligned} \tag{172}$$

In the above, the term $-k/\alpha\dot{y}\sin(\omega t) + a\sin(\omega t)$ takes on the role of a fast proportional feedback, while $\hat{u}$ takes the role of a proportional feedback, which can be seen more directly by integrating by parts and re-rewriting $\hat{u}$ as (with $\hat{u}(0) = 0$)

$$\hat{u}(t) = -\frac{1}{\alpha\tau_l}y(t)\sin(\omega t) + \frac{\omega}{\alpha\tau_l}\int_0^t y(\tau)\cos(\omega\tau)d\tau. \tag{173}$$

We can then express the closed loop PI-ES system (170)–(172) as shown in Fig. 24.

One example that clearly shows the benefit of the PI-ES result is for the system

$$\dot{x} = -x + u, \quad y = 1 + 4(x - 1.2)^2, \tag{174}$$

where the ES feedback has to fight the origin-stabilizing $-x$ term in order to bring $x(t)$ to its optimal value of 1.2. Simulation results comparing the PI scheme to a standard ES approach are shown in Fig. 25 where we clearly see faster settling with much less overshoot.

In Guay and Dochain (2017) the authors extended their PI-ES results to an approach which does not require direct access to the derivative of the cost function $\dot{y}$, it is instead dynamically approximated. For a dynamic system of the form

$$\dot{x} = f(x) + g(x)u, \quad y = h(x),$$

the $\dot{y}$ dynamics are

$$\dot{y} = L_f h + L_g h u, \tag{175}$$

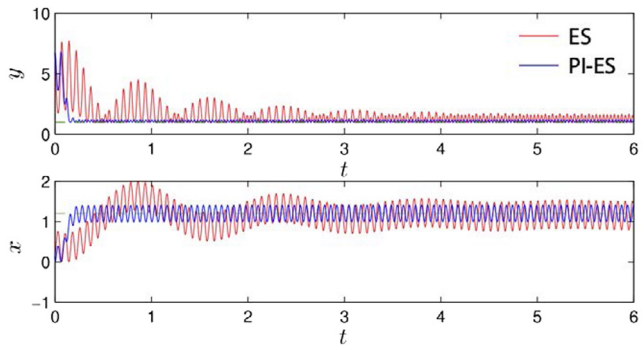


Fig. 26. PI-ES convergence relative to ES. *Source:* Figure adapted from Guay and Dochain (2017).

and the derivative $\dot{y}$ can be parameterized as

$$\dot{y} = \theta_0 + \theta_1(u - \hat{u}) = \phi^T u, \tag{176}$$

where $\phi = [1, (u-\hat{u})^T]^T$ and $\theta = [\theta_0, \theta_1^T]^T$. Considering the error signals $e = y - \hat{y}$ and $\tilde{\theta} = \theta - \hat{\theta}$ the following predictor is set up to estimate $\dot{y}$ as $\hat{y}$

$$\dot{\hat{y}} = \phi^T\hat{\theta} + Ke + c^T\dot{\hat{\theta}}, \tag{177}$$
$$\dot{c} = -Kc^T + \phi^T, \tag{178}$$
$$\dot{e} = \dot{u}^T\tilde{\theta} - Ke - c^T\dot{\hat{\theta}}, \tag{179}$$

where $e(0) = y(0) - \hat{y}(0)$. The full details of the full predictor setup are described in Guay and Dochain (2017), the resulting controller is given by

$$u = -k_g\hat{\theta}_1 + \hat{u} + d(t), \tag{180}$$
$$\dot{\hat{u}} = -\frac{1}{\tau_l}\hat{\theta}_1, \tag{181}$$

where the dither $d(t)$ satisfies the persistency of excitation condition that there exist $\alpha_2, T > 0$ such that

$$\int_t^{t+T} d(\tau)d^T(\tau)d\tau > \alpha_2 I, \quad \forall t. \tag{182}$$

Convergence of this estimated derivative approach for the same system as above is shown in Fig. 26.

The PI-ES method has been studied for application to systems that are distributed over networks (Guay et al., 2018), where p agents are considered, each with dynamics

$$\dot{x}_i = f_i(x, \eta) + g_i(x, \eta)u_i \tag{183}$$
$$y_i = h_i(x), \tag{184}$$

where $\eta \in \mathbb{R}^q$ is a vector of unknown parameters. The overall goal is to minimize the total cost

$$J = \sum_{i=1}^{p} h_i(x) \tag{185}$$

while each agent only has access to its local cost $h_i(x)$ and the costs of its immediate neighbors $h_{i-1}(x)$ and $h_{i+1}(x)$, as shown in Fig. 27. An example of the convergence of 5 networked ES agents is shown in Fig. 5. The problem is solved by utilizing a dynamic average approach to estimate the average of the total cost, for which the PI-ES method is then applied (Guay, 2016; Guay & Dochain, 2017). Network-based control systems with multiple independent agents must be robust to variations in information transmission rates and availability. Therefore, there has also been development of ES for network systems with sporadic packet loss,

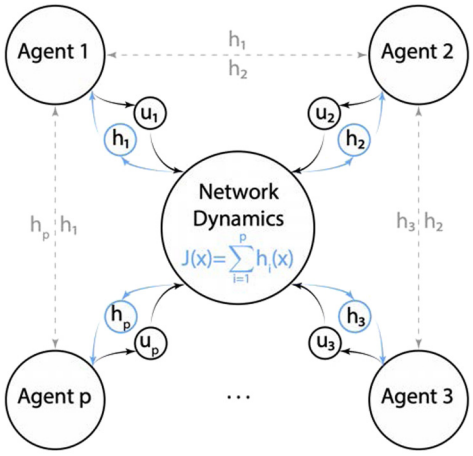


Fig. 27. A group of agents interact with a network, with each agent x_i having its own controller u_i and access only to its own local cost function $h_i(x)$ as well as those of its neighbors, with the goal of minimizing the unknown global cost function $J(x)$.
Source: Figure adapted from Guay et al. (2018).

where the ES algorithm only requires notifications of changes registered at sensors, rather than a continuous full sensor information stream of data (Premaratne, Halgamuge, Tan, & Mareels, 2019).

11. Finite time stabilizing ES

The non-C^2 ES controller described above, in (156), which has the advantageous property that the control effort disappears as the stabilized equilibrium is reached has the additional property of finite time stabilization which can be prescribed over compact sets.

For example, considering the open-loop unstable system

$$\dot{x} = x + u, \tag{186}$$

and two choices of controllers u_r as in (156)

$$u_{r=0} = \alpha\sqrt{\omega}\cos(\omega t) - k\sqrt{\omega}\sin(\omega t)x^2, \tag{187}$$
$$u_{r=\frac{1}{2}} = \alpha\sqrt{\omega}\cos(\omega t)|x|^{\frac{1}{2}} - 2k\sqrt{\omega}\sin(\omega t)|x|^{\frac{3}{2}}, \tag{188}$$

give identical average dynamics

$$\dot{\bar{x}} = (1 - k\alpha)\bar{x} \implies \bar{x}(t) = e^{(1-k\alpha)t}\bar{x}(0). \tag{189}$$

The advantage of the choice $r = 0.5$ is that we now have a feedback controller with vanishing oscillations, allowing the system to settle at equilibrium in a finite time once $x(t)$ reaches 0. We can estimate this finite settling time t_s as it should occur when the amplitude of oscillation is approximately of the same magnitude as $x(t)$ so that the trajectory of $x(t)$ will swing through 0 and stop. To do so, we consider one cycle of ES oscillation, $t \in [t_s, t_s + \Delta t]$ where $\Delta t = \frac{2\pi}{\omega}$, during which $x(t) \approx \bar{x}(t_s)$. During this time let us consider only the dominant first term of (188) which can be integrated to get the oscillation

$$t \in [t_s, t_s + \Delta t], \quad x(t) \approx \bar{x}(t_s) + \frac{\alpha}{\sqrt{\omega}}\cos(\omega t)\sqrt{|\bar{x}(t_s)|}. \tag{190}$$

This term can only reach zero if the oscillation is of the same magnitude as $\bar{x}$, such that

$$\sqrt{|\bar{x}(t_s)|} \approx \frac{\alpha}{\sqrt{\omega}} \implies |x(0)|e^{(1-k\alpha)t_s/2} \approx \frac{\alpha}{\sqrt{\omega}}$$
$$\implies t_s \approx \frac{\ln\left(\frac{2\alpha}{\sqrt{\omega}|x(0)|}\right)}{1 - k\alpha}. \tag{191}$$

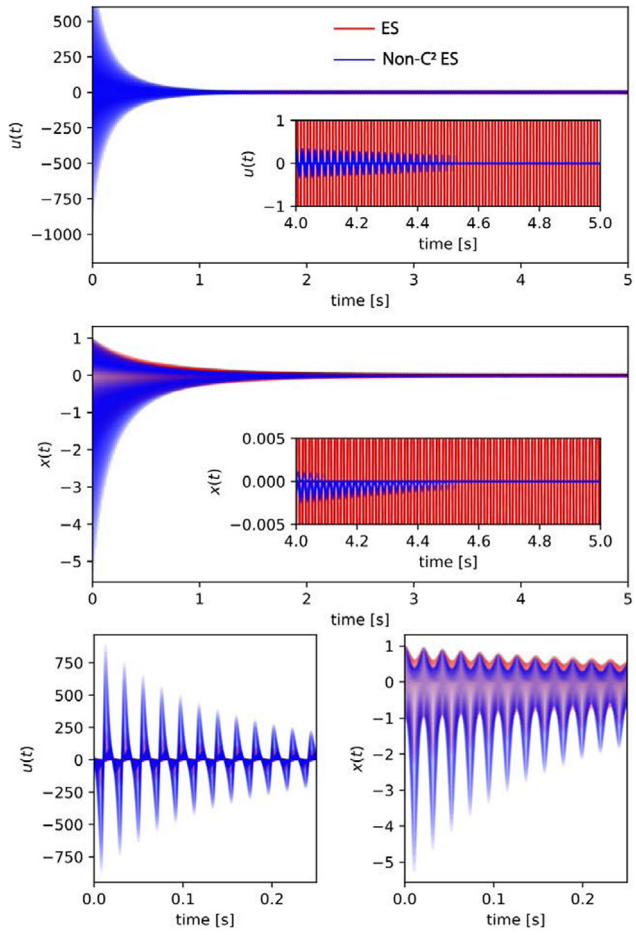


Fig. 28. This figure shows the evolution of the trajectories and the control effort of system (186) using controllers developed in Scheinker and Krstić (2014b). The bottom plots zoom in on the first 0.25 s of evolution and the plot inserts zoom in vertically to show the finite time settling.

We consider 100 random starting conditions uniformly spread between $x(0) \in [-1, 1]$ with $\omega = 300$, $\alpha = 0.5$, and $k = 5$. From (191) these trajectories should have a settling time of $t_s \leq 4.7$ seconds and Fig. 28 confirms that all trajectories have settled by ≈4.5 seconds.

Another finite time ES method based on a stabilizing ES feedback was recently developed in Mele, Tommasi, and Pironti (2021), in which the authors develop a finite time ES method for open-loop unstable dynamic systems. The authors consider the following definition of finite time stabilizability (Amato et al., 2014):

Definition 3. The system

$$\dot{x} = A(t)x(t), \quad x(t_0) = x_0, \tag{192}$$

is said to be finite time stabilizable (FTS) with respect to $(t_0, T, \Gamma(\cdot), R)$ iff

$$x_0^T R x_0 \leq 1 \implies x^T(t)\Gamma(t)x(t) < 1, \quad t \in [t_0, t_0 + T].$$

The FTS condition can be extended to a closed-loop system by the following definition

Definition 4.

$$\dot{x} = A(t)x(t) + B(t)u(t), \quad u(t) = K(t)x(t), \tag{193}$$

is said to be FTS with respect to $(t_0, T, \Gamma(\cdot), R)$ if there exist $L(t), Q(t)$ such that

$$\dot{Q}(t) \succ Q(t)A^T(t) + A(t)Q(t) + L^T(t)B^T(t) + B(t)L(t)$$
$$Q(t) \prec \Gamma^{-1}(t), \quad t \in [t_0, t_0 + T]$$
$$Q(t_0) \succ R^{-1}.$$

When a system is FTS for $Q(t)$, $L(t)$ then the FT stabilizing controller gain is given by

$$K(t) = L(t)Q^{-1}(t). \tag{194}$$

The authors consider open-loop unstable time-varying systems and stabilizing feedback controllers of the form

$$\dot{x} = A(t)x + B(t)u, \tag{195}$$
$$u = \alpha\sqrt{\omega}\cos(\omega t) - k\sqrt{\omega}\sin(\omega t)x^T\Pi(t)x, \tag{196}$$

with Lie bracket-averaged dynamics (for large ω)

$$\dot{\bar{x}} = A(t)\bar{x} - k\alpha B(t)B^T(t)\Pi(t)\bar{x}. \tag{197}$$

System (197) is in the form of system (193) and the idea is then to design $\Pi(t)$ so that the feedback control gain $-k\alpha B(t)B^T(t)\Pi(t)$ renders the system FTS. Once this is accomplished a large enough ω must be found such that the trajectories of (195), (196) and (197) are sufficiently close together such that the original highly oscillatory system satisfies the FTS conditions.

Towards this goal, the authors first prove the following Lemma which allows them to relate the FTS condition of one system to the other based on the distance between the original $(x(t))$ and averaged trajectories $(\bar{x}(t))$.

Lemma 1. *For $x(t), \bar{x}(t) \in \mathbb{R}^n$, for a given $\Delta > 0$, assume*

$$|x(t) - \bar{x}(t)| < \Delta. \tag{198}$$

Consider two symmetric, positive-definite matrix valued functions of time $\Gamma(t)$ and $\bar{\Gamma}(t)$ such that $\forall t \in [t_0, t_0 + T]$

$$\bar{\Gamma}(t) = \frac{1}{r^2(t)}\Gamma(t). \tag{199}$$

If $\bar{x}^T(t)\bar{\Gamma}(t)\bar{x}(t) < 1$ for the time interval $t \in [t_0, t_0 + T$, then on the same time interval

$$x^T(t)\Gamma(t)x(t) < 1, \quad \forall r(t) \leq 1 - \frac{\Delta}{\min_i \gamma_i(t)}, \tag{200}$$

where $\{1/\gamma_i^2(t)\}$ are the eigenvalues of $\Gamma(t)$ at time t.

The main result of the paper is then given as

Theorem 5. *For any $\Delta > 0$, choose ω sufficiently large such that the trajectories of (195)–(197) satisfy the bounds $|x(t) - \bar{x}(t)| < \Delta$ for $t \in [t_0, t_0 + T]$. If there exists some $Q(t)$, k, α, and $\Pi(t)$ such that*

$$\dot{Q}(t) \succ Q(t)A(t) + A^T(t)Q(t) - k\alpha Q(t)\Pi(t)B(t)B^T(t) - k\alpha B(t)B^T(t)\Pi(t)Q(t), \ \forall t \in [t_0, t_0 + T], \tag{201}$$
$$Q(t) \succ \bar{\Gamma}^{-1}(t), \quad \forall t \in [t_0, t_0 + T], \tag{202}$$
$$Q(t_0) \prec R^{-1}, \tag{203}$$

where $\bar{\Gamma}(t)$ is the matrix-valued function of time defined in Lemma 1 with

$$r(t) \leq 1 - \frac{\Delta}{\min_i \gamma_i(t)}, \tag{204}$$

then system (195), (196) is FTS with respect to $(t_0, T, R, \Gamma(t))$ for the same values of k, α, and $\Pi(t)$.

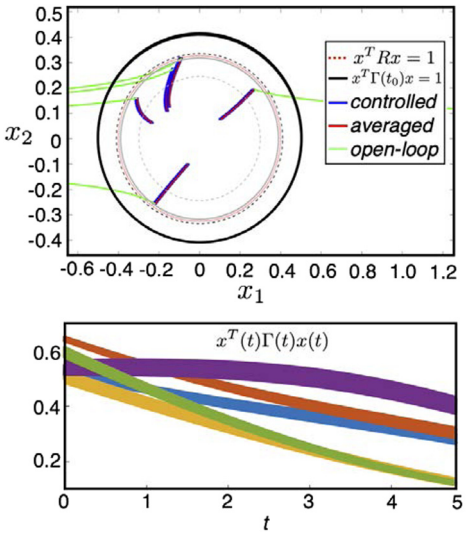


Fig. 29. Simulation results for FTS ES for five different initial conditions. *Source:* Figure adapted from Mele et al. (2021).

Furthermore, following an interesting analysis the authors give an estimate on the required magnitude of ω in order to guarantee the bound $|x(t) - \bar{x}(t)| < \Delta$, for a system with constant $B(t) = B$. The bound is given as

$$\omega \geq \left[\frac{2(\alpha + k)|B|\eta}{\Delta}\right]^2, \tag{205}$$

where

$$\eta = \max_t \left\| e^{\int_{t_0}^t |A(\tau) - k\alpha BB^T \Gamma(\tau)| d\tau} \right\|. \tag{206}$$

This intuitive result shows that the square root of the minimal dithering frequency is inversely proportional to the maximum separation of trajectories Δ and is directly proportional to the terms that influence the amplitude of the perturbation injected into the system (α, k, $|B|$) as well as the scale of instability $A(t)$ of the system.

The FTS ES result is demonstrated with several simulations, including stabilization of the time-varying unstable system

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \left(1 + \frac{t}{10}\right)\begin{bmatrix}0.5 & -0.1 \\ 0 & -0.15\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} u(t), \tag{207}$$

where $u(t)$ is given as (196) and

$$R = \begin{bmatrix}6.25 & 0 \\ 0 & 9.375\end{bmatrix}, \quad \Gamma(t) = \Gamma_0 e^{t/10}, \quad \Gamma_0 = \begin{bmatrix}4 & 0 \\ 0 & 6\end{bmatrix}. \tag{208}$$

The simulation results are shown in Fig. 29 for 5 random initial conditions.

12. ES for PDEs

In order to develop ES methods for systems with delays, the ES analysis was extended to systems governed by partial differential equations (PDE), in which the delay is modeled as a PDE (Oliveira, Krstić, & Tsubakino, 2016).

In Oliveira et al. (2016) the authors consider static maps with a tuned vector of parameters $\theta = (\theta_1, \ldots, \theta_n)$, whose influence on the output measurement $y = Q(\theta)$ has distinct delays in each channel i given by $(D_1, \ldots, D_n) > 0$. The delayed relationship can be summarized as

$$y(t) = Q(\theta(t - D)) = e^{-Ds}[Q(\theta(t))], \tag{209}$$

where $D = \text{diag}\{D_1, \ldots, D_n\}$. Considering the vector of dynamics of the delayed tuned parameters which are delayed functions of inputs U_i, given by

$$\dot{\theta}(t - D) = \begin{bmatrix}U_1(t - D_1) \\ U_2(t - D_2) \\ \vdots \\ U_n(t - D_n)\end{bmatrix}, \tag{210}$$

the delays can be represented as continuous time dynamics by the transport equation PDE according to

$$\dot{\theta}_i(t - D_i) = u_i(0, t), \tag{211}$$
$$\partial_t u_i(x, t) = \partial_x u_i(x, t), \quad x \in [0, D_i], \tag{212}$$
$$u_i(D_i, t) = U_i(t), \quad i = 1, 2, \ldots, n, \tag{213}$$

with solution

$$u_i(x, t) = U_i(t + x - D_i). \tag{214}$$

The system is then analyzed by utilizing averaging theory, Lyapunov–Krasovskii functional analysis, and backstepping transformations in order to derive the controllers

$$u_i(D_i, t) = \frac{c_i}{s + c_i}\left\{-k_i\left[z_i(t) + \int_0^{D_i} u_i(\sigma, t) d\sigma\right]\right\}, \tag{215}$$

where $c_i, k_i > 0$, and

$$z(t) = \Gamma(t)G(t), \tag{216}$$

where $\Gamma(t)$ is the solution to a Riccati differential equation, as described above for Newton-based ES,

$$\dot{\Gamma} = \omega_r \Gamma - \omega_r \Gamma \hat{H} \Gamma \tag{217}$$

which gives an estimate of the inverse of the cost function's Hessian matrix. The matrix $G(t)$ is related to the dithering signals applied to the measured output

$$G(t) = M(t)y(t), \tag{218}$$
$$M(t) = \left[\frac{2}{a_1}\sin(\omega_1 t), \ldots, \frac{2}{a_n}\sin(\omega_n t)\right]^T. \tag{219}$$

The setup is shown in Fig. 30, where

$$N(t) = [a_1 \sin(\omega_1(t + D_1)), \ldots, a_n \sin(\omega_n(t + D_n))]^T. \tag{220}$$

The ES PDE method described above was then extended for multivariate ES of static maps with actuator dynamics governed by diffusion PDEs (Oliveira, Feiling, Koga, & Krstić, 2020). For each control vector θ_i the authors assumed a distinct heat equation so that the propagated actuator vector $\Theta(t) \in \mathbb{R}^n$ has dynamics

$$\Theta(t) = \alpha_i(0, t), \quad \forall i \in \{1, 2, \ldots, n\} \tag{221}$$
$$\partial_t \alpha_i(x, t) = \partial_{xx}\alpha_i(x, t), \quad x \in (0, D_i) \tag{222}$$
$$\partial_t \alpha_i(0, t) = 0 \tag{223}$$
$$\alpha_i(D_i, t) = \theta_i(t) \tag{224}$$

where $\alpha_i : [0, D_i] \times \mathbb{R}_+ \to \mathbb{R}$ and each domain length D_i is known.

System (221)–(224) can be solved as

$$\alpha_i(x, t) = \mathcal{L}^{-1}\left[\frac{\cosh(x\sqrt{s})}{\cosh(D_i\sqrt{s})}\right] * \theta_i(t), \tag{225}$$

where $\mathcal{L}^{-1}(\cdot)$ denotes the inverse Laplace transformation and $*$ is the convolution operator.

This relationship is used to define the *diffusion operator* $\mathcal{D} = \text{diag}\{D_1, \ldots, D_n\}$ with

$$\mathcal{D}_i[\varphi(t)] = \mathcal{L}^{-1}\left[\frac{1}{\cosh(D_i\sqrt{s})}\right] * \varphi(t), \text{ s.t. } \Theta(t) = \mathcal{D}[\theta(t)]. \tag{226}$$

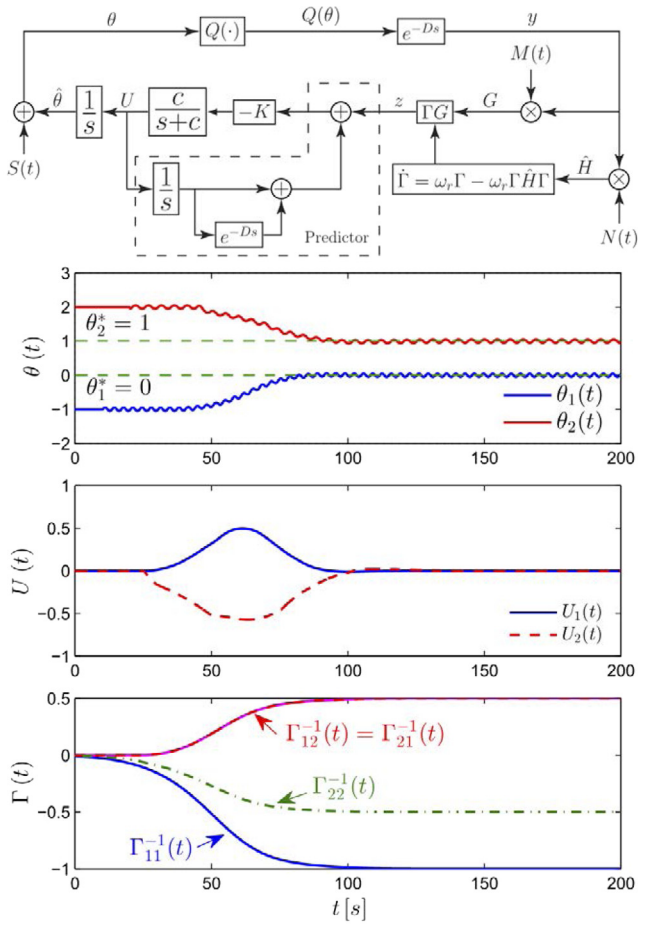


Fig. 30. Top: Newton ES setup for systems with delay. Bottom: Simulation convergence example.
Source: Figures adapted from Oliveira et al. (2016).

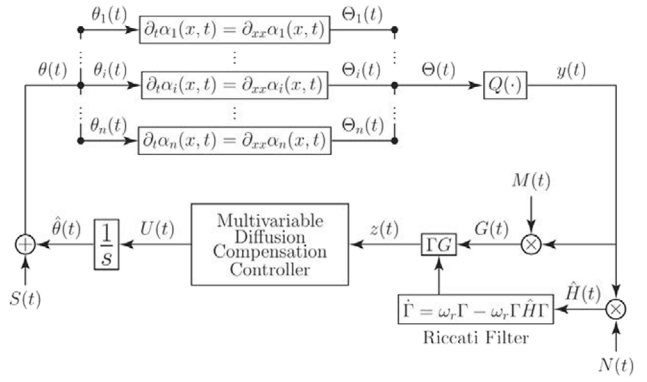


Fig. 31. Newton ES for multivariable PDEs setup.
Source: Figure adapted from Oliveira et al. (2020).

The authors then define the relationship between the scalar measurements y of the unknown static map $Q(\theta)$, with propagated input (221) as

$$y(t) = Q(\mathcal{D}[\theta(t)]) = Q(\Theta(t)),\ \ \Theta(t) = [\Theta_1(t), \ldots, \Theta_n(t)]^T. \quad (227)$$

The unknown static map is assumed to be locally quadratic in the neighborhood of the extremum point y^*, i.e.,

$$y(t) = Q(\Theta(t)) = y^* + \frac{1}{2}(\Theta(t) - \Theta^*)^T H(\Theta(t) - \Theta^*), \quad (228)$$

where the Hessian matrix $H = H^T < 0$ is unknown. A Newton ES method is then developed for optimization of this system, in an approach similar to the one described above for systems with delays. The overall closed-loop system dynamics are given as

$$\dot{\theta}_i = u_i(0, t) \quad (229)$$

$$\partial_t u_i(x, t) = \partial_{xx} u_i(x, t),\ \ x \in (0, D_i) \quad (230)$$

$$\partial_x u_i(0, t) = 0 \quad (231)$$

$$u_i(D_i, t) = -\frac{k_i c_i}{s + c_i} z_i(t) - \frac{k_i c_i}{s + c_i} \int_0^{D_i} (D_i - \sigma) u_i(\sigma, t) d\sigma, \quad (232)$$

and the overall setup shown in Fig. 31.

PDE ES methods have also been developed for wave PDE compensation (Oliveira & Krstić, 2021), ES for systems with delays (Oliveira et al., 2016; Rušiti, Oliveira, Krstić, & Gerdts, 2021), Newton-based ES for systems with time-varying delays (Rušiti et al., 2021), and static maps with actuation dynamics governed by diffusion PDEs (Feiling, Koga, Krstić, & Oliveira, 2018).

13. Hybrid ES with dynamic inclusions

For some systems, introducing the added flexibility of hybrid controllers which switch between various feedback laws can be advantageous and for others it is required due to the structure of the system dynamics. Robust hybrid controllers have been developed for autonomous vehicles operating under unknown or adversarial environments, simultaneously avoiding obstacles and source seeking (Poveda, Benosman, Teel, & Sanfelice, 2021), hybrid ES controllers have been developed for optimization with acceleration and resets (Poveda & Li, 2021), and a hybrid ES approach has been developed for global practical optimization on smooth compact manifolds (Strizic, Poveda, & Teel, 2017).

A general framework has been developed to extend the application of ES to hybrid systems with combined continuous and discrete-time dynamics (Poveda et al., 2018; Poveda & Teel, 2017a). In Poveda and Teel (2017a) controllers were developed for set-valued hybrid dynamical systems (HDS) of the form

$$\dot{x} \in F(x), \qquad x \in C, \quad (233)$$

$$x^+ \in G(x), \qquad x \in D, \quad (234)$$

where $\dot{x} = dx/dt$, x^+ corresponds to values of x after an instantaneous jump, and $F(x)$ and $G(x)$ are the flow map and jump map, respectively. In the approach the dynamic systems involved are not limited to those governed by Lipschitz differential equations, they are extended to systems described by differential inclusions, which are commonly used to analyze discontinuous and switched systems. Because such systems are not guaranteed to have unique solutions the authors find optimal operating set points relative to all of the dynamics that emerge from compact sets of initial conditions. Such a study can be especially useful for real-world applications in which noise, disturbances, and uncertainty require a study of entire sets of possible initial conditions and trajectories rather than single equilibrium point. Such systems emerge frequently in real-world applications in optimization and control systems in the form of switched systems, logic and event-based controllers, continuous-time systems with weakly jumping parameters, supervisory-based control systems, and multi-agent systems with switching communication topologies. The HDS system (233), (234) is represented by the notation $\mathcal{H} := \{C, F, D, G\}$.

Before the main results are presented, a few definitions are required. A set-valued mapping M is outer semi-continuous (OSC) at $x \in \mathbb{R}^m$ if for all sequences $x_i \to x$ and $y_i \in M(x_i)$ such that

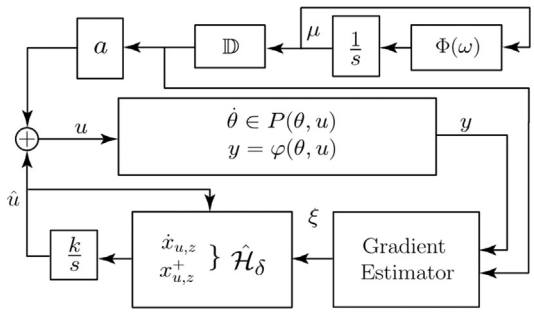


Fig. 32. Hybrid ES setup. *Source:* Figure adapted from Poveda and Teel (2017a).

$y_i \to y$, we have that $y \in M(x)$. A set-valued mapping M is locally bounded (LB) at $x \in \mathbb{R}^m$ if there exists a neighborhood U_x of x such that $M(U_x) \subset \mathbb{R}^n$ is bounded. Given a set $\mathcal{X} \subset \mathbb{R}^m$ the mapping M is said to be OSC and LB relative to $\mathcal{X}$ if the set-valued mapping from $\mathbb{R}^m$ to $\mathbb{R}^n$ defined by $M(x)$ for $x \in \mathcal{X}$ and $\emptyset$ for $x \notin \mathcal{X}$ is OSC and LB at each $x \in \mathcal{X}$. Given a hybrid system $\mathcal{H}$ as described above, and $\mathcal{A} \subset \mathbb{R}^n$ a compact set. The set $\mathcal{A}$ is said to be uniformly globally asymptotically stable (UGAS) for $\mathcal{H}$ if there exists a $\mathcal{KL}$ function β such that any maximal solution x of $\mathcal{H}$ satisfies

$$|x(t,j)|_{\mathcal{A}} \leq \beta(|x(0,0)|_{\mathcal{A}}, t+j), \ \forall (t,j) \in \text{dom}(x). \tag{235}$$

The authors consider a dynamical system with state $\theta \in \mathbb{R}^m$, with input $u \in \mathbb{R}^n$, and output $y \in \mathbb{R}$, modeled by a constrained differential inclusion

$$\dot{\theta} \in P(\theta, u), \quad (\theta, u) \in \Lambda_\theta \times \mathbb{U}, \quad J = \varphi(\theta, u), \tag{236}$$

where P is a set-valued mapping, φ is an output function, and u evolves in the closed set $\mathbb{U} = \hat{\mathbb{U}} + \mathbb{B}$, where $\hat{\mathbb{U}} \subset \mathbb{R}^n$. The following regularity assumptions are made.

Assumption 6. The set $\hat{\mathbb{U}}$ is closed. The set-valued mapping $P(\cdot,\cdot)$ is OSC and LB relative to $\mathbb{R}^m \times \mathbb{U}$, $\mathbb{R}^m \times \mathbb{U} \subset \text{dom}(P)$, and $P(\theta, u)$ is convex for every $(\theta, u) \in \mathbb{R}^m \times \mathbb{U}$. The mapping $\varphi(\cdot,\cdot)$ is continuous with respect to both arguments.

Assumption 7. There exists a set-valued mapping H that is OSC and LB relative to $\mathbb{U}$, such that for each $\rho > 0$ the compact set $\mathbb{M}_\rho := \{(\theta, u) : \theta \in H(u), u \in \mathbb{U} \cap \rho\mathbb{B}\}$ is UGAS for the restricted HDS $\mathcal{H}_{p_\rho} := \{\Lambda_\theta \times (\mathbb{U} \cap \rho\mathbb{B}), P \times 0, \emptyset, \emptyset\}$ with state $[\theta^T, u^T]^T$.

Assumption 8. The response map $J : \mathbb{R}^n \to \mathbb{R}$ is analytic, and there exists an optimal nonempty compact set $\mathcal{O} \subset \hat{\mathbb{U}}$, such that $J(u^*) > J(u)$ for all $u^* \in \mathcal{O}$ and all $u \in \hat{\mathbb{U}} \setminus \mathcal{O}$.

A class of time-invariant gradient-based hybrid extremum seeking controllers is considered which are comprised of dynamic gradient estimators with states $(\xi, \mu) \in \mathbb{R}^n \times \mathbb{R}^{2n}$, and a hybrid learning mechanism with overall state $x_{u,z} := (\hat{u}, \hat{z}) \in \mathbb{R}^n \times \mathbb{R}^r$ which regulate the input of the plant to the optimal set $\mathcal{O}$ by extracting the gradient information of J online based only on output measurements. The hybrid ES setup is shown in Fig. 32, where the gradient estimator for the response map J has dynamics

$$\dot{\xi} = -\omega_L \left(\xi - \frac{2}{a} y \cdot \mathbb{D}\mu \right), \tag{237}$$

$$\dot{\mu} = \Phi(\omega) \cdot \mu, \quad (\xi, \mu) \in \Lambda_\xi \times \mathbb{S}^n, \tag{238}$$

where $\mu = [\mu_1^T, \ldots, \mu_n^T]^T \in \mathbb{S}^n$, $\mu_i = [\mu_{i,1}, \mu_{i,2}]^T \in \mathbb{S}^1$, for all $i \in \{1, \ldots, n\}$, ξ evolves in a compact set $\Lambda_\xi = \lambda_\xi \mathbb{B} \subset \mathbb{R}^n$, where $\lambda_\xi > 0$ is selected sufficiently large to encompass all of the complete solutions of interest, matrix $\mathbb{D} = [e_1, \mathbf{0}, e_2, \mathbf{0}, \ldots, e_i, \ldots, \mathbf{0}, e_n, \mathbf{0}]$ is a block matrix where $\mathbf{0} \in \mathbb{R}^n$ is a column vector of zeros, and $e_i \in \mathbb{R}^n$ is the unit basis vector in dimension i. The ω-parameterized block matrix $\Phi : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ is given by

$$\Phi(\omega) = \begin{bmatrix} \Omega(\omega_1) & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \Omega(\omega_2) & \ldots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \Omega(\omega_n) \end{bmatrix}, \tag{239}$$

where the block components $\Omega(\omega_i)$ are given by

$$\Omega(\omega_i) = \begin{bmatrix} 0 & \omega_i \\ -\omega_i & 0 \end{bmatrix}, \quad \forall i \in \{1, \ldots, n\}, \tag{240}$$

where $\omega = [\omega_1, \ldots, \omega_n]^T$, $\omega_i = \epsilon \cdot \kappa_i$, $\omega_L = \epsilon \cdot \bar{\omega}$, $(\epsilon, \bar{\omega}) \in \mathbb{R}^2_{>0}$, κ_i is a rational number such that $\kappa_i \neq \kappa_j$ and $(i, j) \in \{1, \ldots, n\}$. $u = \hat{u} + a\mathbb{D}\mu$ where $a \in (0, 1)$ is a small tunable parameter, and $\hat{u} \in \mathbb{R}^n$ is an auxiliary state. The continuous-time dynamics associated to the linear oscillator $\dot{\mu} = \Phi(\omega) \cdot \mu$ generates a vector of periodic signals with odd components given by $\mu_{i,1}(t) = \cos(\omega_i t)\mu_{2i-1}(0) + \sin(\omega_i t)\mu_{2i}(0)$, for each $i \in \{1, \ldots, n\}$. These signals are injected into the closed-loop system to extract the gradient information of J.

14. Finite time Newton ES

In Poveda and Krstić (2021) the methods developed above for hybrid ES-based optimization were utilized to develop ES methods which could optimize (in a semi-global practical sense) the analytically unknown output function of a static map in a prescribed finite time. Because the systems considered are non-Lipschitz continuous, the authors consider systems of the form

$$x \in C, \quad \dot{x} = F(x), \tag{241}$$

where $x \in \mathbb{R}^n$ is the state, $C \subset \text{dom}(F)$ is a closed flow set, and F is a set-valued mapping that is OSC, LB, and convex-valued with respect to C. A solution to (241) is an absolutely continuous function $x : \text{dom}(x) \to \mathbb{R}^n$ that satisfies: $x(0) \in C$; $x(t) \in C$ for all $t \in \text{dom}(x)$; and $\dot{x}(t) \in F(x(t))$ for almost all $t \in \text{dom}(x)$. The authors develop their results for dynamic systems of the form

$$\dot{x} = f(x, u), \quad y = h(x, u), \tag{242}$$

where f is locally Lipschitz, h is continuously differentiable, and the operational space S of the states of the plant is closed and bounded. The following assumptions are required.

Assumption 9. There exists a continuous function $l_x : \mathbb{R}^n \to \mathbb{R}^p$, such that for each compact set $K_u \subset \mathbb{R}^n$ the constrained dynamical system $(x, u) \in S \times K_u$, $\dot{x} = f(x, u)$, $\dot{u} = 0$ renders UGAS the compact set $M_{K_u} = \{(x, u) \in S \times K_u : x = l_x(u)\}$.

Assumption 10. Considering the response map defined as $\phi(u) = h(l_x(u), u)$, the function $u \to \phi(u)$ is twice continuously differentiable, and the set

$$\mathcal{A}_\phi = \underset{u \in \mathbb{R}^n}{\text{argmin}}\, \phi(u),$$

is nonempty and compact.

As described in the previous section, the approach is then to regulate the input u towards the set $\mathcal{A}_\phi$ by using only output measurements y without analytic knowledge of f, h, or ϕ.

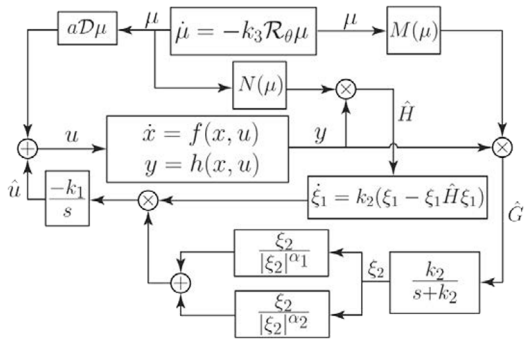


Fig. 33. Finite time Newton ES setup. *Source:* Figure adapted from Poveda and Krstić (2021).

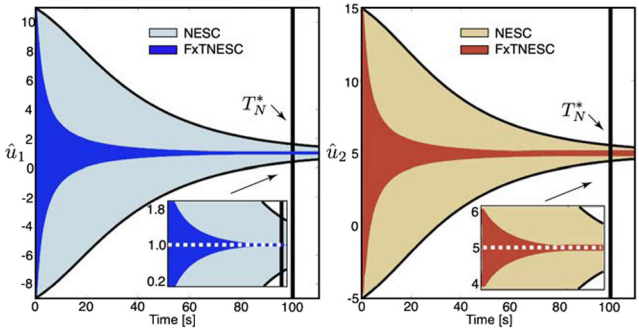


Fig. 34. Finite time Newton ES simulation example. *Source:* Figure adapted from Poveda and Krstić (2021).

The Newton-like fixed-time ES controller is then defined as (in what follows $\dot{x} = dx/d\tau$ where $\tau = \epsilon_0 t$)

$$\dot{\hat{u}} = -kF_u(\hat{u},\xi) \tag{243}$$

$$\dot{\xi}_1 = -\frac{1}{\epsilon_2}\left(\xi_1 F_H(y,\tilde{u})\xi_1 - \xi_1\right) \tag{244}$$

$$\dot{\xi}_2 = -\frac{1}{\epsilon_2}\left(\xi_2 - F_G(y,\tilde{\mu})\right) \tag{245}$$

$$\dot{\mu} = -\frac{2\pi}{\epsilon_1}\mathcal{R}_\theta\tilde{\mu} \tag{246}$$

$$\dot{x} = \frac{1}{\epsilon_0}f(x,\hat{u}+a\tilde{\mu}) \tag{247}$$

$$\hat{u} = u - a\mathcal{D}\mu, \tag{248}$$

where $\mathcal{D} \in \mathbb{R}^{n\times 2n}$ is the matrix that maps a vector $x = [x_1, x_2, \ldots, x_{2n}]^T \in \mathbb{R}^{2n}$ to a vector $\tilde{x} = [x_1, x_3, \ldots, x_{2n-1}]^T$ that has only the odd components of x, and the learning dynamics are defined as

$$F_u(\hat{u},\xi) = \xi_1\left(\frac{\xi_2}{|\xi_2|^{\alpha_1}} + \frac{\xi_2}{|\xi_2|^{\alpha_2}}\right), \tag{249}$$

with $F_u = 0$ whenever $\xi_2 = 0$ and the parameters (α_1, α_2) are selected to guarantee continuity of F_u. For rules determining the settings of all of the constants involved one should refer to the paper. A simulation result for finite time Newton ES is shown in Fig. 34.

The finite time Newton ES setup is shown in Fig. 33. Several other finite time ES optimization methods have now been developed including finite-time ES for unknown maps (Guay & Benosman, 2021), fixed-time Newton-like ES (Poveda & Krstić, 2020b), and fixed-time gradient-based ES (Poveda & Krstić, 2020a).

15. Safe ES

Constrained optimization in unknown environments is often performed through the use of Gaussian Processes (GPs), where a measure of safety can be considered statistically. Safe optimization of quadrotor controllers is described in Berkenkamp, Schoellig, and Krause (2016), and a general method for doing safe optimization via GPs is given by Sui, Gotovos, Burdick, and Krause (2015).

Several approaches have been developed for using ES to solve constrained optimization problems, such as in Guay et al. (2015). Their modified ES design also involves estimating the gradient of the unknown functions, and they perform their analysis without using classical averaging and singular perturbation theory. In Liao, Manzie, Chapman, and Alpcan (2019) ES is blended with "boundary tracing". The formalization here is based on concepts taken from nonlinear programming techniques. ES based algorithms are presented in Poveda and Quijano (2015), which can handle both equality and inequality constraints, inspired by evolutionary game theory. A robust coordinated hybrid source seeking approach with obstacle avoidance has also been developed (Poveda, Benosman, Teel, & Sanfelice, 2021).

A very simple approach to constrained ES was developed for systems of angular velocity-actuated unicycles in Scheinker (2017) in which autonomous agents are attempting to minimize an analytically unknown cost function J based only on its noise-corrupted measurement $\hat{J} = J + n = J_0 + J_c + n$ where J_0 is an unknown source function which the agents are attempting to minimize and J_c is an analytically unknown collision avoidance function based on the proximity to neighboring agents. For example each agent might generate such a function within its neighborhood based on a Gaussian with maximum centered at the agent's location. Physically such a signal could be a temperature level or electric field strength which naturally decays with distance from the source. To consider a group of N agents, we define their coordinates as $x = (x_1, \ldots, x_N)$ where $x_i = (x_{i1}, x_{i2})$ are the 2D coordinates of each agent i. In this case the dynamics are given by

$$\dot{x}_{i1} = \sqrt{\alpha\omega}\cos(\theta), \quad \dot{x}_{i2} = \sqrt{\alpha_i\omega_i}\sin(\theta_i), \tag{250}$$

$$\dot{\theta}_i = \omega_i + k_i\omega_i^2\left(\hat{J}_i(x_i,t) - \eta_i\right), \tag{251}$$

$$\dot{\eta}_i = -\omega_i^2\left(\eta_i - \hat{J}_i(x,t)\right), \tag{252}$$

$$\hat{J}_i = \underbrace{J_{i,0}(x,t)}_{\text{minimization}} + \underbrace{J_{i,c}(x,t)}_{\text{collision avoidance}} + \underbrace{n_i(t)}_{\text{noise}}. \tag{253}$$

which is summarized for a single 2D system, for simplicity, in Fig. 35.

For example, given a two agent system, the goal might be for each agent to minimize its own analytically unknown cost function while avoiding collision with the other agent, with functions $\hat{J}_i$ given by

$$\hat{J}_1 = \underbrace{(x_{11}-1)^2 + x_{12}^2}_{\text{minimization}} + \underbrace{k_p e^{-k_a\left((x_{11}-x21)^2+(x_{12}-x_{22})^2\right)}}_{\text{collision avoidance}} + \underbrace{n_1(t)}_{\text{noise}},$$

$$\hat{J}_2 = \underbrace{(x_{21}+1)^2 + x_{22}^2}_{\text{minimization}} + \underbrace{k_p e^{-k_a\left((x_{11}-x21)^2+(x_{12}-x_{22})^2\right)}}_{\text{collision avoidance}} + \underbrace{n_2(t)}_{\text{noise}},$$

The simulation of such a system with ($k_p > 0$) and without ($k_p = 0$) collision avoidance activated is shown in Fig. 35A, B.

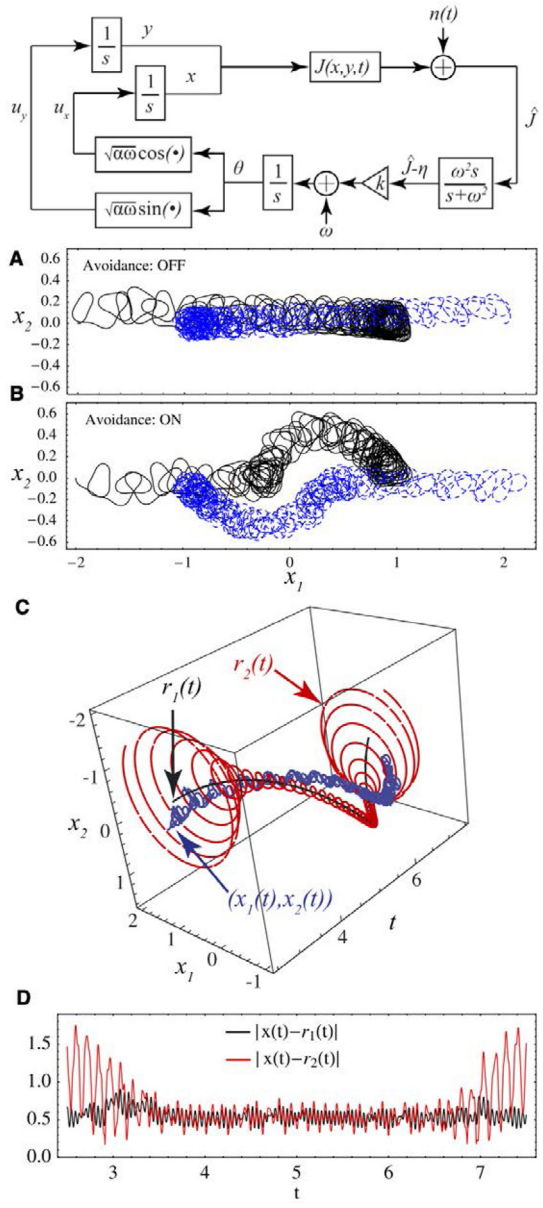


Fig. 35. Example of collision avoidance. The agents are seen to collide with collision avoidance off (A) and automatically change their trajectories to safely pass when it is on (B). The blue agent $(x_1(t), x_2(t))$ can be seen closely tracking $r_1(t)$ without collision, and automatically moving away in order to avoid collision with a third agent $r_2(t)$ when it moves in close to $r_1(t)$ (C). Non-zero distances are retained between the controlled and the two other agents at all times (D). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Adapted from Scheinker (2017).

With avoidance on, the trajectories avoid each other as they make their way to their respective minimizing values.

A more complex example is given by considering a 3 agent system in which the controlled agent $x(t) = (x_1(t), x_2(t))$ must track a second agent $r_1(t) = (\cos(t), \sin(t))$ with unknown position, whose tracking is based only on a measure of some distance-related function while avoiding collision with the tracked agent as well as avoiding collision with a third unknown agent $r_2(t)$ for which a distance-related measurement is available as well. In this case the controlled agent's dynamics are open-loop unstable, given by

$$\dot{x}_1 = \frac{x_1^2(t)}{4} + \sqrt{\alpha\omega}\cos(\omega t), \tag{254}$$

$$\dot{x}_2 = \frac{x_2^2(t)}{4} + \sqrt{\alpha\omega}\sin(\omega t), \tag{255}$$

and the avoidance/minimization function is given as

$$\hat{J}(x,t) = \underbrace{k_0|x(t)-r_1(t)|^2}_{\text{minimization}} + \underbrace{k_1 e^{-\sigma|x(t)-r_1(t)|^2}}_{\text{collision avoidance}} + \underbrace{k_2 e^{-\sigma|x(t)-r_2(t)|^2}}_{\text{collision avoidance}} + \underbrace{n(t)}_{\text{noise}}. \tag{256}$$

A demonstration of such safe tracking is shown in Fig. 35C, D where the controlled agent $(x_1(t), x_2(t))$ is seen to temporarily move away from the tracked agent $r_1(t)$ when the third agent $r_2(t)$ spirals in close to $r_1(t)$, with close tracking resuming after $r_2(t)$ has moved far away again. A plot of the distances between the agents shows that collisions were avoided at all times.

Another constrained ES approach is presented in Labar, Garone, Kinnaert, and Ebenbauer (2019a), which is described using the Lie bracket formalism. They approximate the solution of the constrained optimization problem by using so called "saddle point dynamics" which introduce an additional Lagrange-multiplier state. The authors consider the following constrained optimization problem

$$\min_{x\in\mathbb{R}^n} h(x) \quad \text{s.t.} \quad g(x) \le 0, \tag{257}$$

where $h(x)$ is strictly convex and C^2 and the constraint $g(x)$ is convex and also C^2. The goal is to steer $x(t)$ towards the solution of (257) while ensuring

$$g(x(t)) \le \beta^{-1}, \quad \forall t \ge t_0, \tag{258}$$

where $\beta \in \mathbb{R}_{>0}$ is a user-defined value. Towards this goal the authors consider the set $\Omega = \{x \in \mathbb{R}^n : -\beta g(x) + 1 > 0\}$, for which they utilize a modified barrier function $L(x,\lambda) : \Omega \times \mathbb{R}_{>0} \to \mathbb{R}$ defined as

$$L(x,\lambda) = h(x) - \lambda\beta^{-1}\log(-\beta g(x) + 1). \tag{259}$$

Intuitively, $L(x,\lambda)$ tends to infinity as $g(x)$ approaches β^{-1}. $L(x,\lambda)$ can also be seen as the Lagrangian of the constrained optimization problem

$$\min_{x\in\mathbb{R}^n} h(x) \quad \text{s.t.} \quad \beta^{-1}\log(-\beta g(x) + 1) \ge 0, \tag{260}$$

which is equivalent to (257). The solution of (260) can be achieved by the following dynamics, as proposed in Dürr, Zeng, and Ebenbauer (2013),

$$\dot{x} = -\rho_x \nabla_x L(x,\lambda), \tag{261}$$

$$\dot{\lambda} = \rho_\lambda \lambda \nabla_\lambda L(x,\lambda) = -\rho_\lambda \lambda \beta^{-1}\log(-\beta g(x) + 1), \tag{262}$$

where $\rho_x > 0$ and $\rho_\lambda > 0$, with the following result.

Theorem 6. *Let $\Omega = \{x \in \mathbb{R}^n : -\beta g(x) + 1 > 0\}$. Let $S_L \subset \Omega \times \mathbb{R}_{\ge 0}$ be the set of all saddle points for $L(x,\lambda)$. Assume that S_L is non-empty and compact. Then S_L is regionally uniformly asymptotically stable in $\Omega \times \mathbb{R}_{>0}$ for system (261), (262).*

Next, taking into account that the set of admissible values is a priori unknown to a user and $\log(-\beta g(x) + 1)$ is not defined for $g(x) \ge \beta^{-1}$, the authors consider a modified version of the minimization dynamics (261), (262), considering instead

$$\dot{x} = -\rho_x \left(\frac{-\beta g(x) + 1}{1 + (-\beta g(x) + 1)}\right)^2 \nabla_x L(x,\lambda), \tag{263}$$

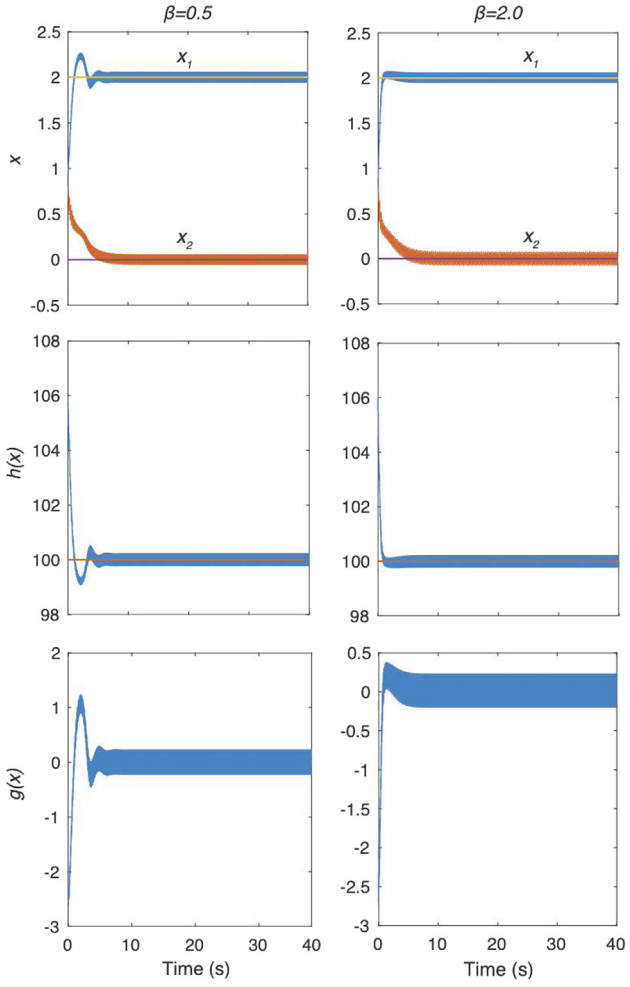


Fig. 36. ES trajectories for constrained optimization.
Source: Figure adapted from Labar et al. (2019a).

$$\dot{\lambda} = \rho_\lambda \lambda \left(\frac{-\beta g(x)+1}{1+(-\beta g(x)+1)} \right)^2 \nabla_\lambda L(x,\lambda), \tag{264}$$

with $\rho_x > 0$ and $\rho_\lambda > 0$. Intuitively, considering the new dynamics (263), (264), no matter what the value of the estimated $\nabla_x L(x,\lambda)$ is, if $g(x_0) < \beta^{-1}$, then $g(x(t)) < \beta^{-1}$, $\forall t \geq t_0$. This approach ensures that the $\log(-\beta g(x)+1)$ term in $\nabla_\lambda L(x,\lambda) = -\beta^{-1}\log(-\beta g(x)+1)$ is always well defined. These results are summarized as

Theorem 7. *Let $\Omega = \{x \in \mathbb{R}^n : -\beta g(x)+1 > 0\}$. Let $S_L \subset \Omega \times \mathbb{R}_{\geq 0}$ be the set of all saddle points for $L(x,\lambda)$. Assume that S_L is non-empty and compact. Then S_L is regionally uniformly asymptotically stable in $\Omega \times \mathbb{R}_{>0}$ for system (263), (264).*

Finally, the authors generalize this approach for a system with an analytically unknown optimization function $h(x)$ and analytically unknown barrier function $g(x)$, relying only on their online measurements. For these unknown functions the gradient $\nabla_x L(x,\lambda)$ is also unknown and so a bounded ES-based version is developed using the bounded ES approach as in (119), as developed in Scheinker (2013a) and Scheinker and Krstić (2014a). This bounded ES version of (263), (264) is utilized so that the gradient can be estimated online in real-time. The bounded ES-based dynamics are

$$\dot{x}_i = \frac{-\beta g(x)+1}{1+(-\beta g(x)+1)} \sqrt{2\rho_x \omega k_i} \cos\left(\mu k_i \omega t + L(x,\lambda)\right),$$

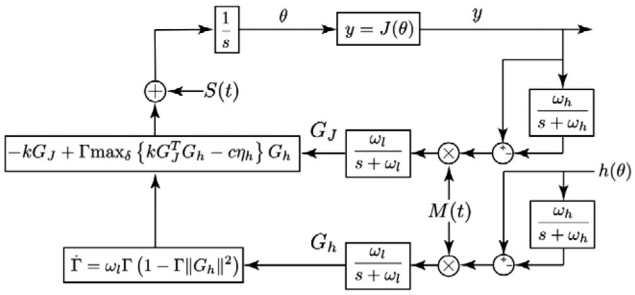


Fig. 37. Safe ES setup for the method developed in Williams et al. (2022).

$$\dot{\lambda} = -\mu\rho_\lambda \lambda \left(\frac{-\beta g(x)+1}{1+(-\beta g(x)+1)} \right)^2 \frac{\log(-\beta\gamma(\chi)+1)}{\beta},$$

where $i \in \{1,2,\ldots,n\}$, $\omega > 0$ and $k_i \in \mathbb{Q}_{>0}$ with $k_i \neq k_j$ for $i \neq j$, where the term $\sqrt{2\rho_x\omega k_i}\cos(\mu k_i \omega t + L(x,\lambda))$ in the $\dot{x}_i$ dynamics is in the form (119) with $\omega_i = \mu k_i \omega$ and $\alpha = \frac{2\rho_x}{\mu}$. The Lie bracket averaged system associated with the bounded ES dynamics above is then shown to be system (263), (264). The authors demonstrate their approach with an example in which the following constrained optimization problem is solved

$$\min_{x\in\mathbb{R}^2} h(x) = (x_1-4)^2 + x_2^2 + 96,$$
$$\text{s.t. } g(x) = x_1^2 + x_2^2 - 4 \leq 0, \tag{265}$$

and the results of using $\beta = 0.5$ and $\beta = 2.0$ are compared for guaranteeing $g(x(t)) \leq \beta^{-1}$. The simulation results are shown in Fig. 36.

Another safe ES approach for systems with analytically unknown safety constraints was recently proposed in Williams, Krstić, and Scheinker (2022). The authors consider a minimization problem in which one has direct control over a variable $\theta(t)$ with the goal of minimizing an analytically unknown cost function $J(\theta)$ while respecting an unknown safety constraint boundary defined by the level set of an unknown barrier function $h(\theta) = 0$.

The overall dynamic optimization problem is then defined as

$$\dot{\theta} = u, \quad y = J(\theta), \tag{266}$$

and the safe ES algorithm is defined as

$$\begin{aligned}
\dot{\theta} &= -kG_J + \Gamma \max_\delta\{kG_J^T G_h - c\eta_h\}G_h + S(t),\\
\dot{G}_J &= -\omega_l G_J + \omega_l(J(\theta)-\eta_J)M(t),\\
\dot{\eta}_J &= -\omega_h \eta_J + \omega_h J(\theta),\\
\dot{G}_h &= -\omega_l G_h + \omega_l(h(\theta)-\eta_h)M(t),\\
\dot{\eta}_h &= -\omega_h \eta_h + \omega_h h(\theta),\\
\dot{\Gamma} &= \omega_l \Gamma(1-\Gamma\|G_h\|^2),
\end{aligned} \tag{267}$$

where the perturbation signal S and demodulation signal M are given by

$$\begin{aligned}
S(t) &= [a\omega_i\cos(\omega_i t), \ \ldots\ , a\omega_i\cos(\omega_n t)]^T,\\
M(t) &= \left[\frac{2}{a}\sin(\omega_i t), \ \ldots\ , \frac{2}{a}\sin(\omega_n t)\right]^T,
\end{aligned} \tag{268}$$

as summarized in Fig. 37.

The functions $J, h : \mathbb{R}^n \mapsto \mathbb{R}$ are scalar valued functions defined over the optimization variables θ. The terms $(J(\theta)-\eta_J)M(t)$ and $(h(\theta)-\eta_h)M(t)$ are the estimates of the gradients of J and h, respectively, performed with the aid of the filter variables η_J and η_h. The variables G_J and G_h are *filtered* versions of these estimates. The estimates of the gradients of J and h are not directly fed into the $\max_\delta$ approximation because averaging

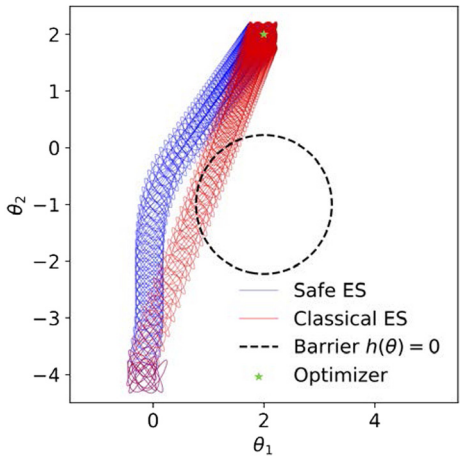


Fig. 38. A safe ES trajectory compared to the classical ES approach to optimizing the same problem, using the method developed in Williams et al. (2022).

in this case would be nearly impossible—with time varying terms appearing inside the argument of the non-linearity $\max_\delta$. The (nonlinear) Riccati filter Γ performs a dynamic computation of $1/G_h^2 > 0$. The standard maximum operation is non-smooth and is therefore approximated by the function $\max_\delta : \mathbb{R} \mapsto \mathbb{R}_{>0}$ given by

$$\max_\delta\{x\} := \frac{1}{2}\left(x + \sqrt{x^2 + \delta}\right) \approx \max\{x, 0\}, \tag{269}$$

for a small parameter $\delta > 0$. Note that $\max_\delta\{x\} > \max\{x, 0\}$ for all $x \in \mathbb{R}$. An example of the scheme avoiding an unsafe region, compared to classical ES without safety is shown in Fig. 38.

This safe ES method has recently been extended to provide semi-global practical stability (Williams, Krstić, & Scheinker, 2023), and has been applied for the safe optimization of high energy charged particle beams (Williams, Scheinker, Huang, Taylor, & Krstić, 2023).

16. Adaptive latent space tuning: ES for enabling machine learning for time-varying systems

Machine learning (ML) techniques, especially deep convolutional neural networks, have been growing in popularity in all areas of science and engineering. However, one major challenge faced by standard ML methods is that of time varying systems, known in the ML community as systems with distribution shift. If a system or its initial conditions change over time, then the data that was previously collected to train an ML model such as a deep convolutional neural network no longer exactly represents the dynamic system that is supposed to be modeled. The typical solution to this phenomenon is brute force re-training, in which once a big enough error is noticed an entire new set of data is collected to update the network weights. This solution is problematic for most complex systems of interest for which repeatedly gathering new data and network re-training is time consuming and requires measurements which interrupt regular system operation.

For example, at large particle accelerator facilities detailed beam profile measurements use destructive processes such as inserting screens which intercept the charged particle beams and interrupt all downstream operations. Such measurements are typically performed only during rare dedicated experimental times. For such systems repeated re-training is not an option.

Recently, a control theory-based dynamic solution was introduced to the problem of initial condition distribution shift (Scheinker, 2021b). The authors consider a problem in which generative encoder–decoder convolutional neural networks (CNN)

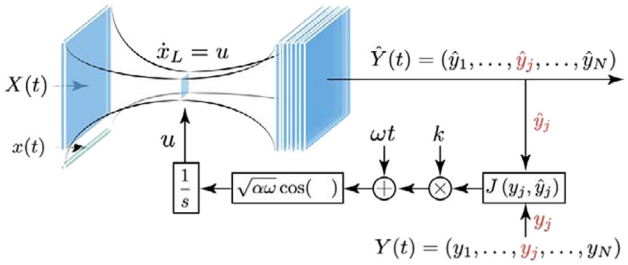


Fig. 39. Adaptive machine learning setup using ES as feedback to tune a low-dimensional latent space representation $x_L(t)$ of a combined set of image and vector inputs $\{x(t), X(t)\}$ in order to minimize a cost function J which compares a subset of an encoder–decoder generative convolutional neural network's outputs $\hat{y}_j(t)$ to measurements $y_j(t)$.

create maps between extremely high dimensional objects such as input–output image pairs which may represent complex distributions at the input and output of a dynamic system

$$\{X(t), x(t)\} \underbrace{\longrightarrow}_{CNN} \hat{Y}(t), \tag{270}$$

where $x(t) \in \mathbb{R}^n$ may be a vector of time-varying parameters, $X(t) \in \mathbb{R}^{N_x \times N_x}$ are time-varying initial conditions represented as images which are incredibly high dimensional objects. For example, given a 256×256 pixel image, $N_x \times N_x = 256^2 = 65536$. The outputs are

$$\hat{Y}(t) = \left(\hat{y}_1(t), \ldots, \hat{y}_N(t)\right) \in \mathbb{R}^{N_y \times N_y \times N}, \tag{271}$$

stacks of N images $\hat{y}_n(t) \in \mathbb{R}^{N_y \times N_y}$, which are estimates being generated by the CNN of some unknown $Y(t)$.

In Scheinker (2021b), the authors consider the case where following initial training access to the time varying inputs $\{X(t), x(t)\}$ is lost and the goal is to track the entire unknown $Y(t)$ of which a partial measurement, a single projection $y_n(t)$ is available for online measurement. This measurement is compared to the CNN's generated version via some cost function such as

$$J\left(y_n(t), \hat{y}_n(t), t\right) = \frac{1}{N_y^2}\sum_i \sum_j \left|y_n(i, j, t) - \hat{y}_n(i, j, t)\right|, \tag{272}$$

The idea is to use the incredible ability of encoder–decoder CNNs to compress images down to a very low dimensional latent space, in this example, the middle of the encoder–decoder network pinches down to a 2D latent representation $x_L(t) \in \mathbb{R}^2$ from which the outputs $\hat{Y}(t)$ are then generated. ES is utilized to adaptively move in this low-dimensional latent space in real time to minimize (272) for unknown and time-varying inputs $\{X(t), x(t)\}$ according to the bounded stabilizing ES feedback

$$\hat{Y}(t) = F_{\text{CNN}}\left(x_L(t)\right), \tag{273}$$

$$\dot{x}_{L,i}(t) = \sqrt{\alpha\omega_i}\cos\left(\omega_i t + kJ\left(y_n(t), \hat{y}_n(t), t\right)\right), \tag{274}$$

as shown in Fig. 39.

Recently, a study of this adaptive latent space tuning approach demonstrated that incorporating ES-based feedback within the autoencoder architecture greatly improves the robustness of neural network-based predictions far beyond the span of the training data (Scheinker, Cropp, & Filippetto, 2023).

This method was also extended to develop a robust framework combining machine learning, an online physics model, and adaptive tuning via ES in order to accurately predict and track 2D phase space distributions of charged particle beams despite time-varying initial beam conditions as well as time-varying particle accelerator components (Scheinker, Cropp, Paiagua, & Filippetto, 2021). ES has also been studied for enabling more general ML

Table 1
Comparison of the various ES methods shown in Fig. 40 relative to the assumptions made about the systems to which they are applied. In each column, the most general/capable ES approaches are highlighted in bold.

Comparison of ES methods and required system assumptions				
ES method	Open loop stability assumed	Hybrid systems allowed	Time-varying systems allowed	Periodic time variation assumed
Classical ES (Krstić & Wang, 2000)	YES	NO	NO	NA
ES for games (Stanković et al., 2010)	YES	NO	NO	NA
Stochastic ES (Liu & Krstić, 2012)	YES	NO	NO	NA
Newton ES (Nesić et al., 2010)	YES	NO	NO	NA
Newton ES delays (Oliveira et al., 2016)	YES	NO	NO	NA
Newton ES PDEs (Oliveira et al., 2020)	YES	NO	NO	NA
Finite time Newton ES (Poveda & Krstić, 2021)	YES	NO	NO	NA
Safe ES (Williams et al., 2022)	**NO**	NO	**YES**	YES
Hybrid ES (Poveda et al., 2018)	YES	**YES**	NO	NA
Stabilizing ES (Scheinker & Krstić, 2012a)	**NO**	NO	**YES**	**NO**
Bounded ES (Scheinker, 2013a)	**NO**	NO	**YES**	**NO**
PI ES (Guay & Dochain, 2017)	**NO**	NO	**YES**	YES
Adaptive latent space ES (Scheinker, 2021b)	**NO**	NO	**YES**	**NO**

for time-varying problems (Scheinker, 2021a), and has been applied for 3D coherent diffraction imaging problems (Scheinker & Pokharel, 2020).

17. ES methods summary

A large family of various ES-based controllers and optimization methods has been presented for dynamic systems. In Fig. 40 an incomplete summary of various ES methods is presented showing the signal pathways and setups for the various ES versions with the various methods broadly categorized in two families: (1). Classical ES approaches and their modifications for the optimization of an analytically unknown output function of an a priori stabilized dynamic system by tuning a control parameter. (2). ES-based feedback stabilization and optimization approaches in which the dynamic system is not assumed to be controlled and ES plays the dual role of optimizing an analytically unknown output function and of forcing the optimal point to be a stable equilibrium of the dynamic system (see Table 1).

Starting at the top of the left column of Fig. 40, we have the classical ES algorithm for which the first analytic proofs were presented in Krstić and Wang (2000). As we continue down the left column we see the various modifications and additions that have been made to the ES method over the years including a stochastic version, a multi-agent version for games, a Newton-based ES setup, the extension of the Newton approach to systems with delays and systems governed by a class of PDEs, finite-time Newton ES, a hybrid ES scheme for systems whose dynamics are a combination of continuous movement and discrete jumps, and safe ES in which analytically unknown boundary constraints must be satisfied by the ES dynamics while minimizing an analytically unknown output function.

Starting at the top of the right column of Fig. 40, we have the ES-based feedback control algorithm first proposed in Scheinker (2012). As we continue down the right column we see various modifications of the ES-based feedback approach including the bounded form of ES, the PI-based ES feedback, and ES coupled with convolutional neural networks for adaptive latent space tuning.

Both of the ES approaches have proven to be useful for an extremely large number of applications, as described below, and for an extremely large family of dynamic systems, as described above. Each family of ES approaches also has its own benefits. The family of classical ES-based optimizers is older, larger, and has many more variants. On the other hand the smaller family of ES-based feedbacks has its own advantage in the increased flexibility of allowing the ES to act directly as the feedback controller

$$\dot{x} = f(x,u,t), \quad u = u_{ES}(x,t), \tag{275}$$

rather than relying on a specific controller which constrains the ES to modify a parameter within the control function

$$\dot{x} = f(x,u,t), \quad u = \alpha(x,\theta_{ES},t). \tag{276}$$

One possible advantage of utilizing an ES-based feedback as a controller directly is the added flexibility of finding and tracking unknown optimal points which may exist for analytically unknown systems.

For example, if we consider the typical ES setup there is usually a pre-existing controller $\alpha(x,\theta,t)$ which guarantees the existence of a smooth output steady state manifold l such that the system equilibrium $f(x,\alpha(x,\theta)) = 0$ is achieved at $x = l(\theta)$ and the ES dynamics evolve along this manifold as they find the optimal value of some $J(x)$. For a large and complex system with unknowns, which is the type of system for which ES is most useful, it is possible that some other controller may achieve a more optimal setting than what is constrained to $l(\theta)$ by having the freedom to explore the space more completely. Such an example is illustrated in Fig. 41 in which the classic ES scheme quickly converges to and evolves along $l(\theta)$ to the global maximum relative to the $l(\theta)$ manifold as enforced by a pre-existing controller $\alpha(x,\theta)$. For this same system when using the approach where ES is the direct feedback itself, from the same starting conditions it is able to find the global maximum that is outside of the $l(\theta)$ constraints.

18. Patents

As described above, the various versions of ES, which continue to grow as it is an active field of research, are applicable to an incredibly wide range of dynamic systems. Furthermore, because ES is by design useful for uncertain systems, it has been widely applied outside of academic studies for real world problems, including scientific studies and industrial settings. It is worth noting the significant impact of ES on industry which can be somewhat quantified by the large volume and diversity of ES-related patents. First of all, there is of course the original 1922 paper of Leblanc which reads more like a patent/design than a typical academic article and now has 277 citations (Leblanc, 1922). Moving on, a simple public search of the US Patent and Trademark Office for the two terms "extremum" and "seeking" returns 1919 records. To give a few concrete examples, a search for "extremum seeking patent" was performed in Google Scholar, which returns 9750 results (many of which are articles rather than patents). These results are ranked by "relevance" as determined by Google's algorithms. If one starts to browse through the patents they include ES applications for regulating the airflow into a building (Seem, 2010), a method for predicting and

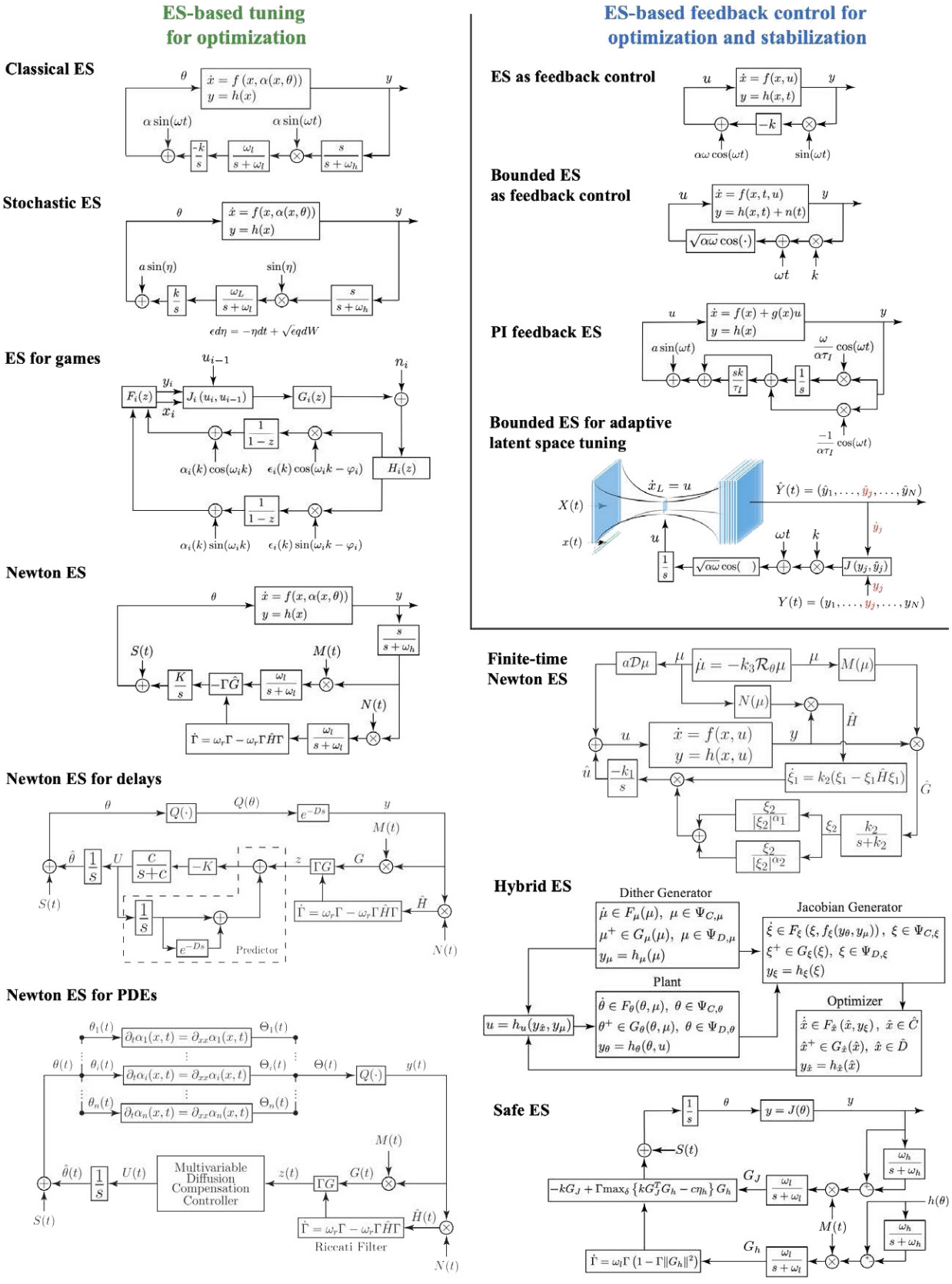


Fig. 40. Extremum Seeking Zoo: Several major forms of the ES algorithm discussed in this survey.

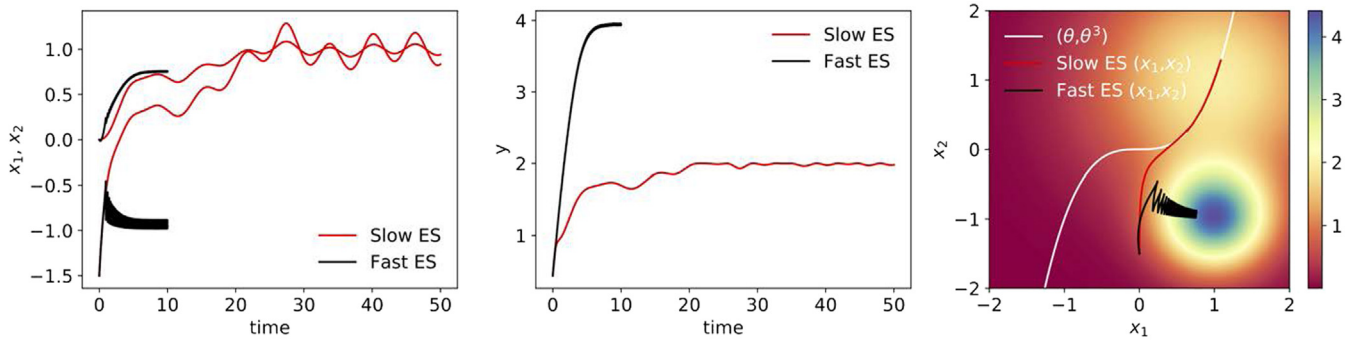


Fig. 41. ES for finding a maximum of an unknown function $y(x_1, x_2)$. Classical ES (red) slowly converges to the local maximum equilibrium dictated by a predefined control structure whereas stabilizing ES (black) quickly finds the global maximum. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

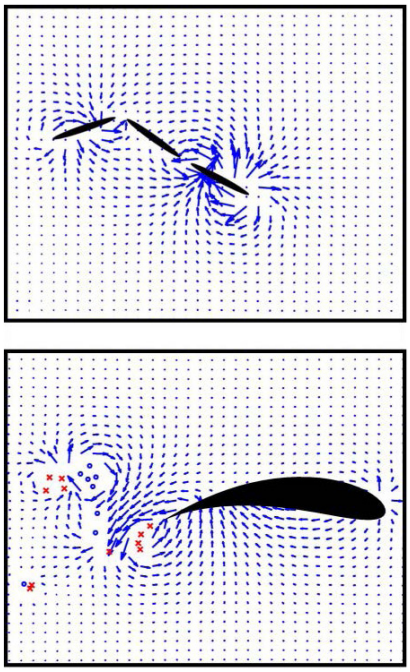


Fig. 42. Fish model setup for a three-linked fish (top) and a Joukowski foil (bottom).
Source: Figure adapted from Cochran, Kanso, Kelly, Xiong, and Krstić (2009).

stabilizing compressor stall (Krener & Krstić, 2000), for chilled water plant control (Salsbury & House, 2019), for vapor compression (Burns, Guay, & Weiss, 2017), for RF impedance matching (Burry, Radomski, Aung, & Klein, 2020), and for flight formation (Ryan, 2018). In the following sections, we present several detailed overviews of ES applications.

19. Autonomous vehicle control

Autonomous underwater vehicles, which do not need input from an operator, are being utilized for scientific and commercial exploration including seafloor mapping and chemical concentration mapping, and for various military applications. Extremum seeking methods have been developed for such systems, for steering vehicles towards a source in GPS-denied environments. In Cochran et al. (2009) and Cochran and Krstić (2009), the authors modeled the mechanics of a fish in two different ways, as a three linked fish and as a Joukowski foil, as shown in Fig. 42.

For both of the fish models considered in Cochran et al. (2009) the equations governing locomotion are variations of Kirchhoff's equations for the motion of a rigid body in an ideal fluid

$$\frac{d\mathbf{L}}{dt} + \Omega \mathbf{k} \times \mathbf{L} = 0, \quad \frac{dA}{dt} = \mathbf{k}([U\ V]^T \times \mathbf{L}) = 0, \tag{277}$$

where $\mathbf{L}$ and A are the linear and angular momenta of the body fluid system and U, V, and Ω are the translational and rotational velocities.

For the three-linked fish model shown in Fig. 42 the ES control law used to maximize the analytically unknown cost function J is given by

$$\dot{\theta}_1 = a\omega \sin(\omega t) - c\eta \cos(\omega t), \tag{278}$$

$$\dot{\theta}_2 = a\omega \cos(\omega t) + c\eta \cos(\omega t), \tag{279}$$

$$\eta = \frac{s}{s+h}[J]. \tag{280}$$

For the Joukowski foil model, the ES controller used is

$$\dot{\eta} = a\omega \cos(\omega t) + c\eta \sin(\omega t), \tag{281}$$

$$\eta = H(s)[J], \quad H(s) = \left(\frac{s}{s+h}\right)^2. \tag{282}$$

It was shown that both of the simulated fish models could utilized ES algorithms to autonomously track source $J(x, y, t)$ whose signal strengths fall off with distance, as shown in Fig. 43.

20. Active flow control

Active flow control is a method which exploits flow instabilities in order to enhance the region of operation of an aircraft over a wide range of operating conditions, including providing extra lift at very a high angle of attack. Airflow is a highly stochastic and time-varying process with various environmental disturbances and requires a very robust and fast online feedback, which makes ES a very strong candidate. ES has been utilized for active flow control (Becker, King, Petz, & Nitsche, 2007; Brunn, Henning, Nitsche, & King, 2008; Henning et al., 2008; Kim, Kasnakoglu, Serrani, & Samimy, 2009; King et al., 2006), and for aeropropulsion (Moeck, Bothien, Paschereit, Gelbert, & King, 2007; Wiederhold et al., 2009). In Becker et al. (2007) the authors developed a Multi-input multi-output (MIMO) ES method for maximizing the lift of a wing by actively controlling pulsating jets in order to control flow separation, as shown in Fig. 44. For the MIMO ES setup the three ES controllers used, (ES_1, ES_2, ES_3) were each of the form

$$u_i = u_0 + a \sin(\omega t + \theta_i) + \hat{u}, \tag{283}$$

$$\hat{u} = K \int_0^t \eta(\tau) d\tau, \tag{284}$$

$$\eta(t) = \frac{\omega_l}{\omega_l + s} \xi \sin(\omega t + \theta_i), \tag{285}$$

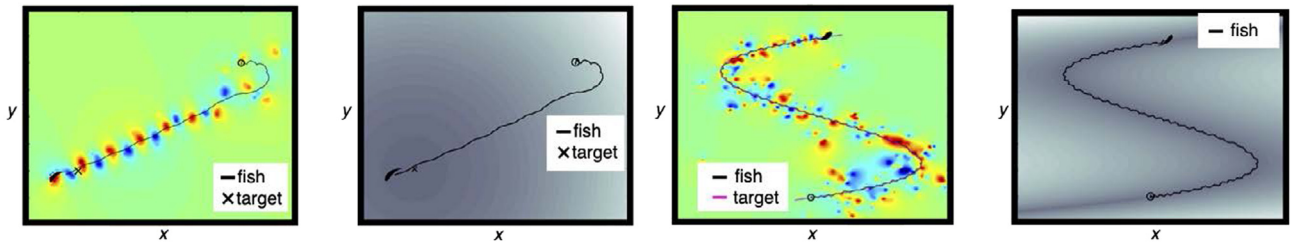


Fig. 43. A fish model utilizing ES to track a stationary and a moving source.
Source: Figure adapted from Cochran et al. (2009).

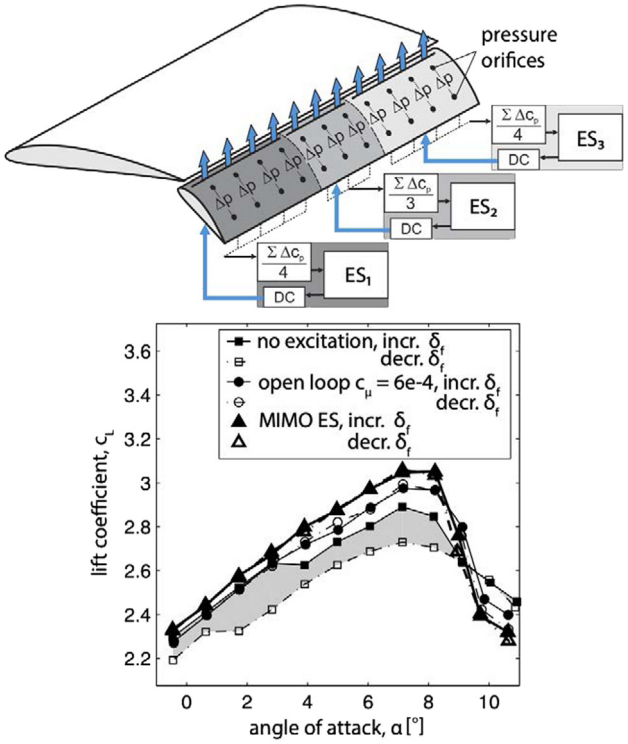


Fig. 44. MIMO ES-based airfoil control.
Source: Figure adapted from King et al. (2006).

$$\xi(t) = \frac{s}{s+\omega_h} y \tag{286}$$

as shown in Fig. 44, where each ES_i utilized the same frequency ES dithers, but with different phase shifts θ_i for the 3 different controllers.

21. Plasma control

High energy plasmas, such as those created inside of Tokomak reactors, are extremely complex dynamic systems which are very difficult to control, for which various ES methods have been developed (Centioli et al., 2008; Luo & Schuster, 2009; Ou et al., 2008). Recently, an innovative Kalman filter-based ES controller was designed for plasma control in Dubbioso et al. (2022) and Tommasi, Dubbioso, Mele, and Pironti (2021).

The Kalman-based ES setup is shown in Fig. 45. The tokamak plasma model used was based on the Grad–Shafranov PDE which, in cylindrical coordinates, is given by

$$\Delta^* \psi = -f\frac{df}{d\psi} - \mu_0 r^2 \frac{dp}{d\psi}, \quad \text{in plasma region,} \tag{287}$$

$$\Delta^* \psi = -\mu_0 r j_{xt}(r,z,t), \quad \text{in the conductors,} \tag{288}$$

$$\Delta^* \psi = 0, \quad \text{elsewhere,} \tag{289}$$

where $\psi(r,z)$ is the poloidal flux per radian, μ_0 is the vacuum magnetic permeability, j_{ext} is the toroidal current density in the external conductors, p is the kinetic pressure profile, f is the poloidal current function profile, and the operator Δ^* is defined as

$$\Delta^* = r\frac{\partial}{\partial r}\left(\frac{1}{\mu_r r}\frac{\partial \psi}{\partial r}\right) + \frac{\partial}{\partial z}\left(\frac{1}{\mu_r}\frac{\partial \psi}{\partial z}\right). \tag{290}$$

It can be shown that

$$j_{ext} = -\frac{\sigma}{r}\dot{\psi} + \frac{\sigma}{2\pi r}u, \tag{291}$$

where u is the voltage applied to the coils and σ is the electric conductivity, with the equation integrated over the conductor regions. The measured outputs were defined as

$$y = \left(\delta I_A^T \ \delta I_p \ \delta Z_c\right)^T, \tag{292}$$

where δI_A are the deviations of the circuit currents from equilibrium values, δI_p is the deviation of the plasma current from equilibrium, and δZ_c is the deviation of the vertical position of the plasma centroid from equilibrium. This input was passed into a Kalman filter to generate a Lyapunov-like function $V = \hat{x}^2$ of the unstable plasma state for minimization by using the stabilizing feedback-based ES method of the form

$$u = \alpha\sqrt{\omega}\cos(\omega t) - k\sqrt{\omega}\sin(\omega t)V(\hat{x}), \tag{293}$$

as shown in Fig. 45(C). Various examples of plasma stabilization were demonstrated. In Fig. 45 we show just the response of the plasma to two different disturbances: VDE of 5 cm and MD, where a VDE is an instantaneous motion of the system's state along an unstable mode resulting in vertical displacement of the plasma centroid and MD represents a loss of plasma thermal energy. The ES method was shown to be incredibly robust over a wide range of conditions with the various Equation numbers in the figure referring to plasma characteristics as described in the paper.

22. Particle accelerator control

Particle accelerators are some of the most powerful scientific tools for studying fundamental physics, biology, and material science. Large accelerators are kilometers in length and are composed of thousands of coupled components such as resonant radio frequency (RF) electromagnetic accelerating cavities and magnets and their associated power sources and amplifiers. The performance of large collections of electromagnetic components is constantly being perturbed by environmental disturbances such as temperature drifts, high voltage ripple, and vibrations, all components face uncertainty in terms of precise alignment at the sub μm scale, and components such as magnets suffer from hysteresis effects which introduces in an uncertainty in the mapping between a control parameter such as a power supply current and

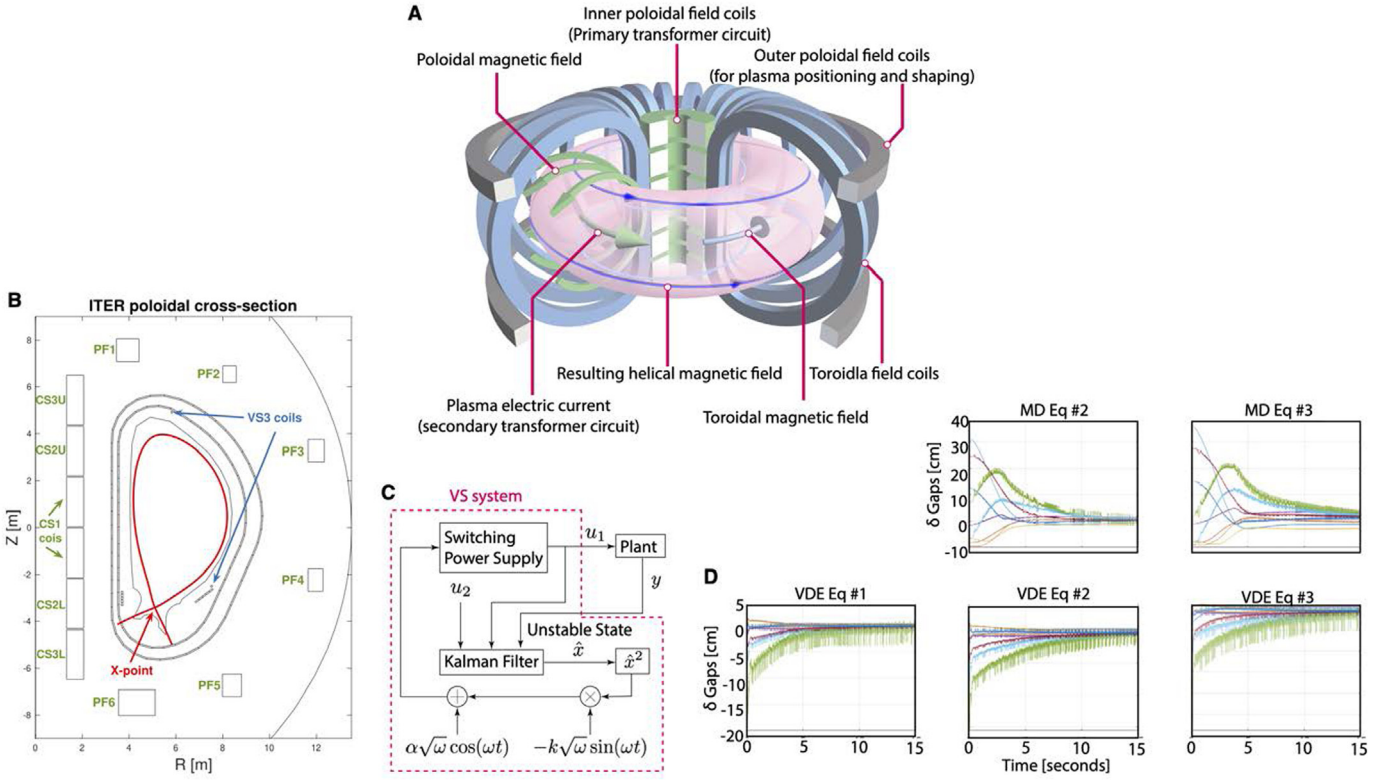


Fig. 45. Tokomak Kalman-based ES controller.
Source: Figure adapted from Dubbioso et al. (2022).

the controlled variable which may be the magnetic field which focuses charged particles. Furthermore, charged particle beams themselves are time-varying with typical beams composed of $10^9 - 10^{14}$ individual particles whose dynamics are governed by complex collective effects such as space charge forces. The result is that a measurable output function of an accelerator, such as a beam current monitor at a particular location, is an analytically unknown function of a control parameter such as a magnet power supply setting, and is governed by an uncertain and time-varying dynamic system.

Two ES feedbacks running at multiple time-scales have been utilized for online multi-objective optimization at the plasma wakefield accelerator (AWAKE) at CERN for simultaneous emittance and orbit control (Scheinker et al., 2020) as shown in Fig. 46(A). In Scheinker et al. (2020) one set of 5 ES parameters that were slowly tuned were 2 solenoid magnets and 3 quadrupole magnets which had an influence over the electron beam size at electron–proton overlap region of the AWAKE beamline, denoted as

$$\mathbf{p}_1 = (p_{11}, p_{12}, \ldots, p_{15}). \tag{294}$$

However, because the beam is not perfectly aligned, changing the solenoid and quadruople magnets not only affected the beam shape, but also its trajectory. Therefore, another set of 10 ES parameters running at a 3× faster rate controlled 10 (x, y) downstream steering magnets, denoted as

$$\mathbf{p}_2 = (p_{21}, p_{22}, \ldots, p_{210}), \tag{295}$$

to keep the beam centered on a screen during the beam shape optimization. Considering all 15 tuned ES parameters as $\mathbf{p} = (\mathbf{p}_1, \mathbf{p}_2)$, the multi-objective ES problem was of the form:

$$\min_{\mathbf{p}} \{F(\mathbf{p}, t)\} = \min_{\mathbf{p}} \{f_1(\mathbf{p}, t), f_2(\mathbf{p}, t)\}, \tag{296}$$

$$f_1(\mathbf{p}, t) = w_1 X_{\text{RMSE}}(\mathbf{p}, t), \quad f_2(\mathbf{p}, t) = w_2 \sigma_{xy}(\mathbf{p}, t), \tag{297}$$

where X_{RMSE} denotes RMS error of the beam trajectory and σ_{xy} is a measure of the (x, y) beam size on the screen.

The two time-scale ES controllers were defined as

$$\dot{\mathbf{p}}_{1i} = \sqrt{\alpha_1 \omega_{1i}} \cos(\omega_{1i} t + kF(\mathbf{p}, t)), \tag{298}$$

$$\dot{\mathbf{p}}_{2i} = \sqrt{\alpha_2 \omega_{2i}} \cos(\omega_{2i} t + kF(\mathbf{p}, t)), \tag{299}$$

where the two time-scale ES dithering frequencies were defined as

$$\omega_{1i} = \omega_0 r_{1i}, \quad r_{1i} \in \{1.0, \ldots, 1.75\}, \quad i = 1, \ldots, 5$$
$$\omega_{2i} = 3 \times \omega_0 r_{2i}, \quad r_{2i} \in \{1.0, \ldots, 1.75\}, \quad i = 1, \ldots, 10,$$

with the initial beam shaping magnets moving much slower than the trajectory-controlling magnets further down stream.

ES has been applied for maximizing free electron laser (FEL) pulse energy at two of the most advanced FEL facilities in the world: the Linac Coherent Light Source (LCLS) at SLAC National Accelerator Laboratory and the European X-ray FEL (EuXFEL) at DESY laboratory, as shown with ES automatically more than doubling the average light output energy of the EuXFEL within 4 minutes in Fig. 46(B) (Scheinker et al., 2019).

ES has also been combined with machine learning-based neural network tools for automatically controlling the (z, E) longitudinal phase space of electron beams at the LCLS FEL as shown in Fig. 46 (C) (Scheinker, Edelen, Bohler, Emma, & Lutman, 2018).

Various other ES applications for accelerator tuning include accelerator magnet tuning (Scheinker, Pang, & Rybarcyk, 2013), adaptive electron beam profile prediction (Scheinker & Gessner, 2015), time varying resonance control (Scheinker, 2016), for min-

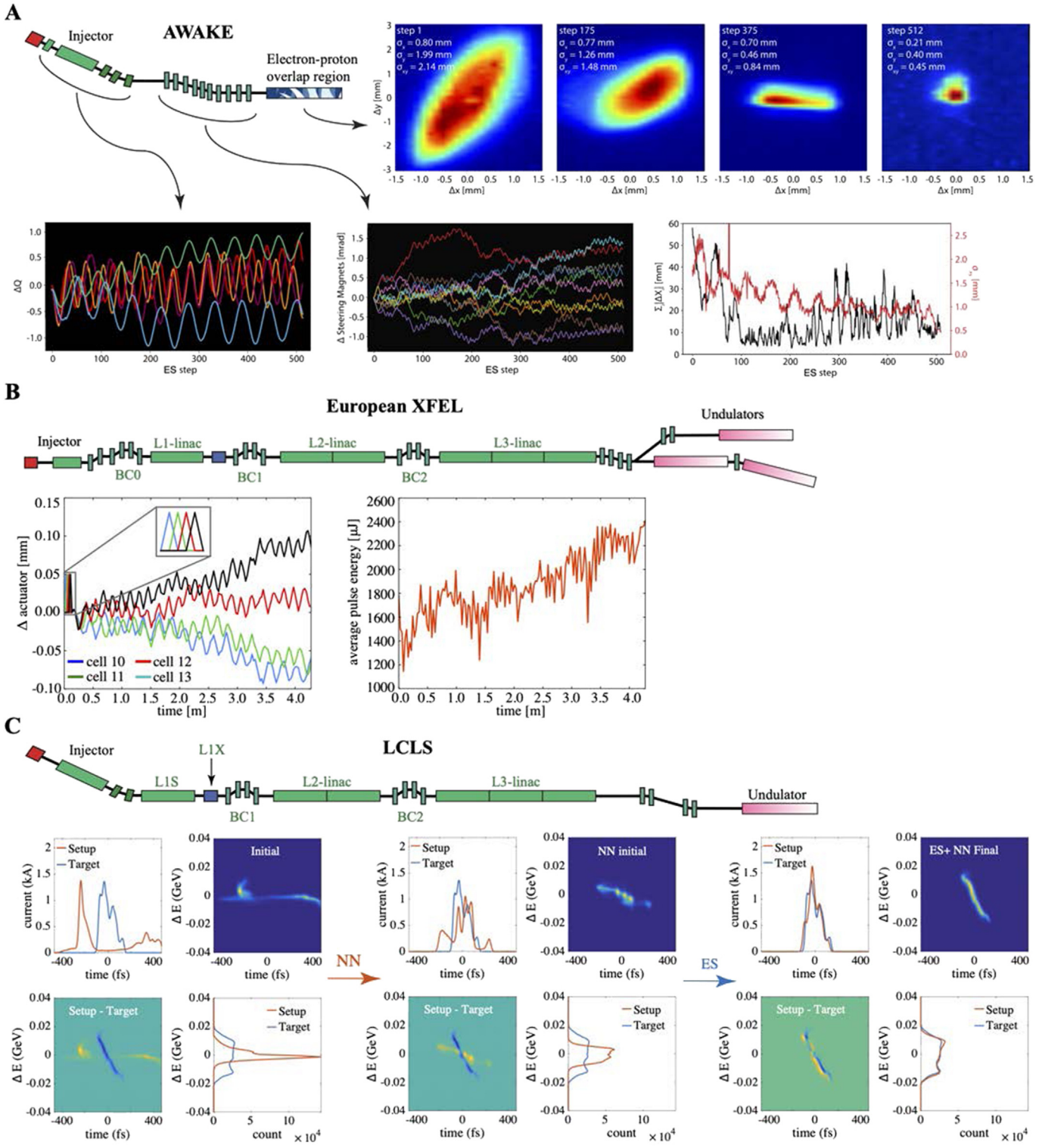


Fig. 46. (A) Application of two ES loops running simultaneously at two different time scales for real-time multi-objective optimization (beam trajectory control and emittance growth minimization) at the AWAKE accelerator at CERN. (B) Application of ES for automatic pulse energy maximization at the European X-ray free electron laser at DESY. (C) Combination of ES and a neural network for direct tuning of the 2D (z, E) longitudinal phase space distribution of the electron beam at the LCLS free electron laser at SLAC National Accelerator Laboratory.

imizing beam oscillations in a time-varying magnetic lattice at SPEAR3 at SLAC (Scheinker, Huang, & Wu, 2017), and is now regularly used as a beam tuning tool at the Los Alamos National Laboratory linear particle accelerator LANSCE for beam loss minimization (Scheinker, Huang, & Taylor, 2021).

23. Maximum power point tracking

Photovoltaic generators which convert solar energy into electricity have current–voltage and power–voltage characteristics which strongly depend on temperature as shown in Fig. 47(A)

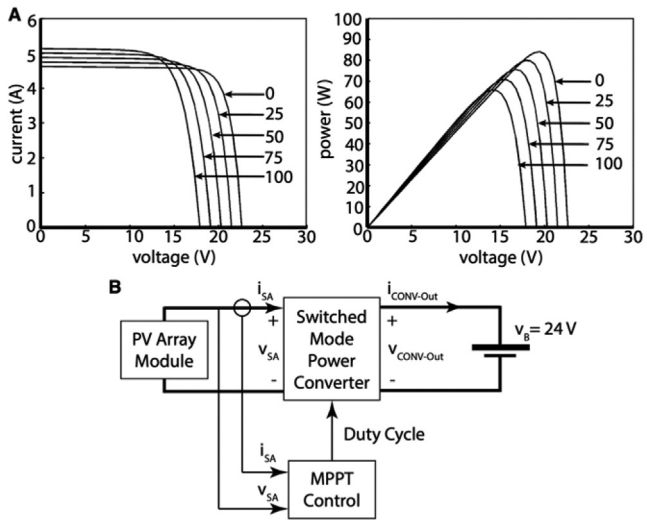


Fig. 47. Voltage, current, power relationships as a function of temperature (0 to 75 degrees C).
Source: Figure adapted from Leyva et al. (2006).

and vary in response to atmospheric conditions. Therefore, the problem of maximum power point tracking (MPPT), becomes an adaptive optimization problem for which ES methods were first demonstrated in Leyva et al. (2006, 2011) and Leyva, Artillan, Cabal, Estibals, and Alonso (2011). The authors present an ES-based MPPT algorithm for efficient extraction of PV array energy by automatic adjustment of the array voltage via modulation of the duty cycle of a switched mode power converter while ensuring the stability of the adjusting process for sudden or fast variations of the external conditions. The ES MPPT setup is shown in Fig. 47(B).

A multivariable Newton ES-based method for MPPT has also been developed for photo-voltaic power optimization (Ghaffari et al., 2014). In that approach the power to be maximized, $P(D, t)$, is considered as a function of switching pulse duration $D = [D_1, D_2, \ldots, D_n]^T$ and is assumed to have a Taylor series expansion at its optimal pulse duration point $D^* = [D_1^*, D_2^*, \ldots, D_n^*]^T$ of the form

$$P = P^* + \frac{1}{2}(\tilde{D} + Q(t))^T H(\tilde{D} + Q(t)) + R(\tilde{D} + Q(t)), \tag{300}$$

where H is the unknown Hessian matrix, $D = D^* + \tilde{D} + Q(t)$, with $\tilde{D} = \hat{D} - D$ an estimation error, R represents higher order terms, and $Q(t)$ is a vector of ES dither signals of the form

$$Q(t) = a[\sin(\omega_1 t), \ldots, \sin(\omega_n t)]^T, \tag{301}$$

where $\omega_i \neq \omega_j$. A matrix W of perturbation signals of the form

$$W_{jj} = \frac{16}{a^2}\left(\sin^2(\omega_j t) - \frac{1}{2}\right),$$
$$W_{jk} = \frac{4}{a^2}\sin(\omega_j t)\sin(\omega_k t), \quad j \neq k, \tag{302}$$

is used to extract an estimate of the Hessian via the first term of the product

$$W(t)P_{ac} = \frac{1}{2}WQ^T HQ + WQ^T H\tilde{D} + \mathcal{O}(a), \tag{303}$$

where P_{ac} is the alternating component of the measured output power and

$$\dot{\hat{D}} = -K_n \Gamma \hat{G}, \tag{304}$$
$$\dot{\Gamma} = \rho\Gamma - \rho\Gamma\hat{H}\Gamma, \tag{305}$$
$$\dot{\hat{\Gamma}} = -\omega_l \hat{G} + \omega_l M(t)P_{ac}(t), \tag{306}$$

where $M(t)$ is a matrix of ES perturbation signals. The performance and advantages of the Newton ES-based MPPT approach are shown in Fig. 48. MPPT is an important problem well suited for ES-based methods and has been studied with numerous additional ES-based approaches (Bazzi & Krein, 2011; Bizon, 2016; Brunton, Rowley, Kulkarni, & Clarkson, 2009, 2010; Heydari-doostabad, Keypour, Khalghani, & Khooban, 2013; Li, Li, Seem, & Lei, 2011, 2015; Reisi, Moradi, & Jamasb, 2013; Yau & Wu, 2011).

24. Bioreactor growth rate optimization

An innovative combination of ES with neural networks has also been applied for the control of complex nonlinear bioreactor systems with unknown growth kinetics for biomass production rate maximization (Guay et al., 2004). The authors consider microbial growth dynamics of the form

$$\dot{x} = \mu(s, d)x - ux, \tag{307}$$
$$\dot{s} = -k_1\mu(s, x)x + u(s_0 - s), \tag{308}$$
$$y = k_2\mu(x, s)x, \tag{309}$$

where $x, s \in \mathbb{R}^+$ denote biomass and substrate concentrations, respectively, $u \geq 0$ is the dilution rate, y is the production rate of the reaction product, s_0 is the concentration of the substrate in the feed, and $k_1, k_2 > 0$ are yield coefficients. The authors consider a case in which only x and s are measurable and y is not available for measurement and control. The goal of the ES controller is to maximize the steady-state production rate y^* by finding the optimal dilution rate u^*.

In this work, the substrate-dependent growth rate expression $\mu(x, s)$ is considered to be unknown and is approximated by a radial basis function (RBF)-based neural network giving an estimate of the form

$$\varphi(z) = W^{*T}S(z) + \mu_l(t), \tag{310}$$

where W^* is unknown, $\mu_l(t)$ is the neural network error, and $S(z)$ is an RBF basis function vector

$$S(z) = [s_1(z), s_2(z), \ldots, s_l(z)]^T,$$
$$s_i(z) = \exp\left[\frac{-(z - \phi_i)^T(z - \phi_i)}{\sigma_i^2}\right], \quad i = 1, 2, \ldots, l,$$

with centers ϕ_i and standard deviations σ_i. The method is demonstrated to perform incredibly well with and without Gaussian measurement noise added to the biomass and substrate concentration measurements, as shown in Fig. 49.

25. Conclusions

The general ES methods described above have proven to be useful for many real-world applications and there are many approaches and applications which were beyond the scope of this survey, some of which we mention here briefly. Beyond what was described, ES has been implemented for cooling systems (Li, Li, & Seem, 2009; Li, Rotea, Chiu, Mongeau, & Paek, 2005), wind energy (Creaby, Li, & Seem, 2009), photovoltaics (Lei, Li, Chen, & Seem, 2010), human exercise machines (Zhang, Dawson, Dixon, & Xian, 2006), optimizing the control of nonisothermal continuously stirred tank reactors (Guay, Dochain, & Perrier, 2005), reducing the impact velocity of an electromechanical valve actuator (Peterson & Stefanopoulou, 2004), Newton-like ES for thermoacousic instability control (Moase et al., 2010), torque maximization of motors (Antonello, Carraro, & Zigliotto, 2013), optimization of variable CAM timing engine operation (Popovic, Jankovic, Magner, & Teel, 2006), ES for cascaded Raman optical

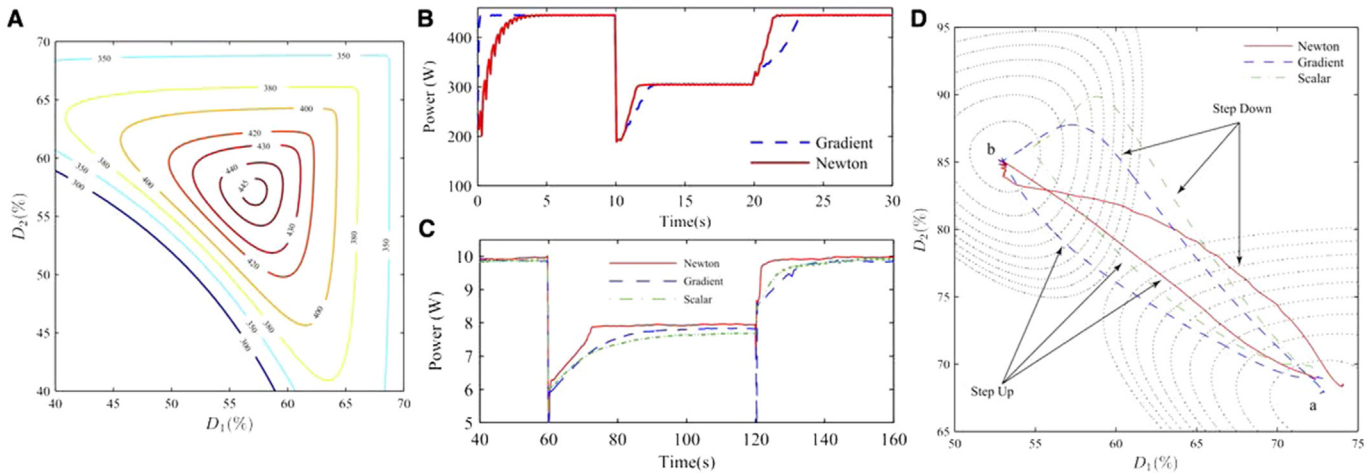


Fig. 48. MPPT with Newton-based ES. Power output as a function of pulse duration for a two-pulse setup $D = [D_1, D_2]$, is shown (A). Following an initial transient time, the Newton-based ES approach has faster settling time than a gradient method alone (B). The Newton-based ES shows superior response to fast power level changes (C, D).
Source: Figure adapted from Ghaffari, Krstić, and Seshagiri (2014).

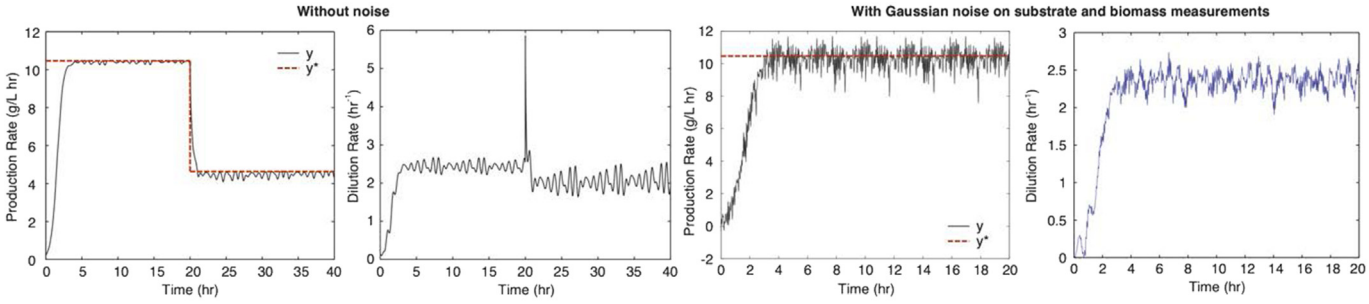


Fig. 49. ES for bioreactor optimization with and without Gaussian measurement noise.
Source: Figure adapted from Guay, Dochain, and Perrier (2004).

amplifiers (Dower, Farrell, & Nesic, 2008), ES for traffic congestion control (Yu, Koga, Oliveira, & Krstić, 2021), GPS-denied source localization (Ghadiri-Modarres et al., 2016a), ES for maximizing RF heating (Carnevale et al., 2009), electromagnetic actuator control (Benosman & Atınç, 2015), for high voltage converter modulator optimization (Scheinker, Bland, Krstić, & Audia, 2013), multi-objective ES has been implemented for mineral grinding (Lu, Krstić, Chai, & Fu, 2020), and sampled-data ES for constrained optimization (Hazeleger, Nešić, & van de Wouw, 2022). The field of ES research is active and continuously growing.

Various extensions and modifications of the ES method continue to generalize to larger classes of systems and to find new applications for real-time control, stabilization, and optimization. Some of the particularly active topics in ES these days include further work on finite time convergence, ES for PDEs, and the combination of ES with machine learning in order to increase the robustness of standard ML techniques for time-varying systems and for input distribution shift. This survey has only brushed the surface of the many published results. As clearly seen from just the few topics in Fig. 2, ES methods continue to branch out and also to overlap. Existing techniques are constantly being generalized for larger class of systems while new techniques are also being developed. ES is a very active field of research with a growing number of approaches and applications.

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